

Finite Observations, Infinite Behaviour: bicategorical semantics for stateful monoidal processes.

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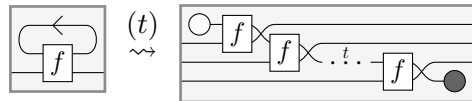
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Abstract

Time-dependent processes are often described by machines with an internal state which is updated as time evolves. An external observer cannot see this state and learns about a process only through finite observations of its inputs and outputs, each of which imposes a constraint on the trajectories the process can exhibit. We introduce a semantic construction in which two stateful processes have the same *behaviour* when they have the same constraints, as determined by finite observations, independent of their internal state. Behaviour in this sense propagates inconsistency, so that impossible processes are identified, and is invariant under time delay, so that the behaviour observed does not depend on when the observation is made. The construction is defined over any preorder-enriched monoidal category with a compatible notion of discarding, which we call a *discard bicategory*, capturing partial, non-deterministic, probabilistic, and quantum processes. The resulting category of behaviours provides a functorial semantics for stateful processes in which the internal state can be reparametrised dinaturally. When the base is compact closed, our construction is a feedback category in the sense of Katis, Sabadini, and Walters. For non-deterministic systems, we prove a categorified compactness theorem: every compatible family of finite observations between compact Hausdorff spaces extends uniquely and functorially to an infinite closed relation. Restricted to affine relations over finite fields, the compactness theorem recovers Willems’ notion of behaviour for linear time-invariant systems.

1 Introduction

Many systems in science and engineering run indefinitely: signal flow graphs [60], stochastic processes [26], and quantum channels with memory [40] continuously transform inputs into outputs. A natural model for such systems is a *stateful process*: one that maintains internal memory coupling past inputs to future outputs. A common specification is a *Mealy machine*: a transition function $f : S \times X \rightarrow S \times Y$ which at each time step consumes an input in X , updates an internal state in S , and produces an output in Y [44]. The time evolution of a Mealy machine is obtained by initialising the state, repeatedly feeding the internal state back into the transition function, and discarding the final state:


(1)

A Mealy machine f together with an initial state $s_0 \in S$ determines a stream transducer $\text{run}(f, s_0) : X^{\mathbb{N}} \rightarrow Y^{\mathbb{N}}$, defined as the final fixed point of the coinductive equation

$$\text{run}(f, s_0)(x_0, x_1, \dots) = (y_0, \text{run}(f, s_1)(x_1, x_2, \dots)) \quad \text{where} \quad (s_1, y_0) = f(s_0, x_0).$$

Two Mealy machines f and g , with initial states s_0 and t_0 , have the same *behaviour* when $\text{run}(f, s_0) = \text{run}(g, t_0)$. This coinductive characterisation is the standard definition of behavioural equivalence in the total deterministic setting. However, as soon as the transition function is allowed to be partial, nondeterministic, probabilistic, or quantum, the coinductive fixed point need not exist, and this notion of behaviour breaks down. This raises the fundamental question that motivates the present paper:

When do two stateful processes produce the same infinite behaviour?

We seek a *denotational* answer to this question, in the tradition initiated by Scott [58]. Our approach is process-theoretic [18]: we model processes as morphisms in symmetric monoidal categories, where different choices of base category accommodate partial [17], nondeterministic [13], probabilistic [27], and quantum [2] processes. Following Katis, Sabadini and Walters [37], we construct feedback categories of stateful processes over these theories and seek a functorial semantics that identifies two such processes whenever they have the same observable behaviour.

What is behaviour? In Willems' behavioural approach to systems theory, the behaviour of a time-dependent system is defined to be the set of infinite trajectories or signals satisfying some global constraints [64]. *Global* constraints, however, are not directly observable as any observation must be made in finite time. The central goal of this paper is therefore to define an analogous notion of behaviour from finite, local observations alone, with no a priori appeal to global constraints.

What is an observation? We model observations as morphisms in an underlying monoidal category, on which we require two further pieces of structure. We first must be able to *compare* observations: for this we require a preorder enrichment, which records when the constraints imposed by one observation x approximate those imposed by another y , denoted $x \subseteq y$. We must also be able to *restrict* the constraints imposed by observations to smaller contexts: for this we require morphisms which discard information. A monoidal category equipped with both structures, suitably compatible, we call a *discard bicategory* (Definition 2.5). Observing a process over time then yields a *compatible family of observations*: in which each observation f_j , restricted to a smaller context $i \subseteq j$, refines the observation f_i in the smaller context, so that $f_j|_i \subseteq f_i$.

Infinite behaviour from finite observations. We define a notion of behaviour similar to that of Willems, except where constraints are determined *locally* by families of finite observations. We say that two compatible families of observations have the same behaviour when any observation of one can be verified by an observation of the other, possibly in a larger context. We formalise this in our central construction (Section 4): given a discard bicategory and a set of finite contexts, we define a discard category of behaviours, whose morphisms are equivalence classes of compatible families of finite observations (Definition 4.8 and Theorem 4.9).

Properties of our construction. Our construction possesses several desirable properties:

- i. *Uniformity across process theories.* (Theorem 4.9 and Proposition 4.10) Our construction applies uniformly to partial, nondeterministic, probabilistic, and quantum processes. Moreover, it is functorial with respect to base change.
- ii. *Dinaturality of state.* (Theorems 4.14 and 4.18) The internal state can be reparametrised dinaturally without changing the behaviour, and when the base is compact closed the construction forms a feedback category in the sense of Katis, Sabadini, and Walters [37].

- iii. *Naturality of the time-delay.* (Proposition 4.15 and Theorem 4.18) Observing a constraint at different times does not change the behaviour: the process that delays time is a natural transformation, which is moreover invertible when the base is compact closed.
- iv. *Propagation of inconsistency.* (Theorem 4.14) Any two families of observations imposing inconsistent constraints have the same behaviour.
- v. *Hiding of internal state.* By construction, internal constraints on the state which do not affect the input or output of a process are discarded.
- vi. *Generality of contexts.* (Definition 4.8) Our construction is agnostic to the index over which a process evolves: although we mostly focus on processes in discrete time, the same construction applies to arbitrary directed families of finite contexts. Therefore, it provides a semantics for stateful processes observed over more general spaces rather than just over time.

The first and last properties reflect the generality of our construction; whereas the other four reflect the systems-theoretic interpretation of behaviour which we generalise.

Recovering the global view of behaviours. In some cases, it is possible to recover a Willems-style global notion of behaviour from our local notion of behaviour. Focusing on the nondeterministic setting, we prove a *compactness theorem* for closed relations between compact Hausdorff spaces (Theorem 5.5). We show that every compatible family of closed relations functorially extends to a unique infinite closed relation between compact Hausdorff spaces. When restricted to affine relations between finite dimensional vector spaces over finite fields, this recovers a categorified notion of Willems’ global behaviour of linear discrete time invariant systems (Theorem 6.5).

Related work. The semantics of stateful processes in monoidal categories was first studied by Katis, Sabadini, and Walters [37], and has since been developed by several authors, including the differentiable causal functions of Sprunger and Katsumata [62], the monoidal streams of Di Lavore et al. [21], the effectful Mealy machines of Bonchi et al. [6], and the delayed-trace calculus of Carette et al. [14]. We argue in Section 3 that none of these satisfies all of desiderata i.–vi. above. We relate our construction to existing work in Sections 4.3 and 4.4.

Similar notions of behaviour and observational equivalence arise in the semantics of programming languages [46, 54, 1] and in automata theory [55, 57]. There, the basic observable is termination: traditionally, two programs are observationally equivalent when no context can distinguish them by halting, so that every observation is registered by a terminating computation. In our setting, by contrast, the processes of interest run indefinitely, and their behaviour consists of the constraints accumulated along the way. In other words, we observe not whether a process halts, but how it continues. Nonetheless, the preorder enrichment at the heart of our construction is inspired by Scott’s domain-theoretic notion of approximation [58], with finite observations playing the role of finite elements.

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2 Discard bicategories

In this section, we introduce the notion of *discard bicategory*, which forms the structural basis for our construction of behaviours in Section 4. A discard bicategory is a process theory in which we can both *discard* the information held by a subsystem and compare processes by how *informative* they are. After giving the abstract definition, we go through our examples of interest. In every case the approximation structure arises from a common discard bicategory structure, which we show subsumes restriction orders on partial functions, subset inclusion of relations, conditional order between probability kernels and the Löwner order between quantum processes.

2.1 Discard bicategories

Recall the definition of symmetric monoidal category.

Definition 2.1. A **strict symmetric monoidal category** (SMC) is a category $\mathcal{C}$ equipped with a functor $\otimes : \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$, called the **monoidal product**, and an object $I \in \mathcal{C}$, called the **monoidal unit**. For all objects X, Y, Z , we impose that:

$$(X \otimes Y) \otimes Z = X \otimes (Y \otimes Z), \quad \text{and} \quad I \otimes X = X = X \otimes I,$$

in addition to a natural isomorphism, the **symmetry**:

$$\sigma_{X,Y} : X \otimes Y \rightarrow Y \otimes X \quad \text{such that} \quad \sigma_{X,Y} = \sigma_{Y,X}^{-1}.$$

Non-strict SMCs are defined by replacing the equalities above with coherent natural isomorphisms. Since every SMC is symmetric monoidally equivalent to a strict SMC [33], we can work in the strict setting without loss of generality.

SMCs have a process-theoretic interpretation. An object of $\mathcal{C}$ is a *system*, and a morphism $f : X \rightarrow Y$ is a *process* taking the system X to the system Y . The monoidal structure equips this with a notion of composition: processes may be composed *in sequence* using $\circ$, and *in parallel* using $\otimes$. Processes of the form $e : X \rightarrow I$ into the monoidal unit are called **effects**; we interpret these as the *observations* that can be made on the system X . Among all observations on the system, there is a distinguished effect which simply forgets all information about X .

Definition 2.2. A **discard category** is a SMC where every object X is equipped with an effect $\top_X : X \rightarrow I$, called the **discard**, such that $\top_{X \otimes Y} = \top_X \otimes \top_Y$ and $\top_I = 1_I$. A **discard functor** is a symmetric monoidal functor preserving the discard morphisms. Moreover, a morphism $f : X \rightarrow Y$ is **total** in case $\top_Y \circ f = \top_X$. A discard category in which all morphisms are total is called an **affine symmetric monoidal category**.

The compatibility conditions say that discarding a composite system is the same as discarding its parts independently, so that the discard is precisely what makes X decompose into *subsystems*.

To compare processes by how informative they are, we enrich in a preorder.

Definition 2.3. A **preorder** on a set X is a binary relation $\subseteq$ on X which is reflexive so that for all $x \in X$, $x \subseteq x$; and transitive so that for all $x, y, z \in X$ such that $x \subseteq y$ and $y \subseteq z$ then $x \subseteq z$. When a preorder is moreover antisymmetric, so that $x \subseteq y$ and $y \subseteq x$ implies $x = y$, then it is called a **partial order**, and X equipped with this order is called a **poset**. Every set X is canonically equipped with the **discrete partial order** where $x \subseteq y$ if and only if $x = y$.

Definition 2.4. A **preorder enrichment** for a SMC $\mathcal{C}$ is a preorder structure $\subseteq$ on every hom-set $\mathcal{C}(X, Y)$ such that for all morphisms f, f', g , and g' of the appropriate type:

$$\text{if } f \subseteq f' \text{ and } g \subseteq g' \text{ then } f \otimes g \subseteq f' \otimes g', \quad \text{moreover} \quad \text{if } f \subseteq f' \text{ and } g \subseteq g' \text{ then } f \circ g \subseteq f' \circ g'.$$

A **symmetric monoidal preorder-enriched functor** is a symmetric monoidal functor between preorder-enriched categories which is monotone on hom-sets. A **preorder-enriched discard category** is a discard category equipped with a preorder enrichment.

We interpret $f \subseteq g$ as denoting that g is an *approximation* of f : the two processes are compatible, but g is possibly less informative than f . The two enrichment conditions say exactly that this notion of approximation is respected by composing processes in sequence and in parallel.

Combining these two structures, a discard bicategory is a preorder-enriched discard category in which the discard is compatible with approximation, in the following sense:

Definition 2.5. A **discard bicategory** is a preorder-enriched discard category with a *laxly natural discard*, so that for any map $f : X \rightarrow Y$, we have that $\top_Y \circ f \subseteq \top_X$. A **discard bicategory functor** is a preorder-enriched discard functor between discard bicategories.

The lax naturality axiom says that running a process and then discarding is approximated by discarding the input directly. In other words, *the discard is the least informative effect*, it approximates any other process. For example, given processes $f : X \rightarrow Y \otimes S$ and $g : X \rightarrow Y \otimes T$ such that $(1_Y \otimes \top_T) \circ g \subseteq (1_Y \otimes \top_S) \circ f$, we interpret g as giving more information about the system Y than f .

Many process theories have a bottom element in the approximation order called *zero morphism* and corresponding to the most informative effect: denoting that a particular event is impossible or *inconsistent*.

Definition 2.6. A SMC $\mathcal{C}$ has **monoidal zero morphisms** if for all objects X, Y there exists a morphism $0_{X,Y} : X \rightarrow Y$ such that for all morphisms $f : X' \rightarrow X$ and $g : Y \rightarrow Y'$,

$$0_{X,Y} \circ f = 0_{X',Y}, \quad g \circ 0_{X,Y} = 0_{X,Y'}, \quad \text{and} \quad 1_Z \otimes 0_{X,Y} = 0_{Z \otimes X, Z \otimes Y}.$$

Lemma 2.7. *If a discard bicategory has zero morphisms then zero is the bottom element in every hom-set.*

See proof on page 34.

The notion of discard bicategory can be seen as a generalisation of Carboni and Walters' notion of a Cartesian bicategory of relations [13]. In this setting, *compact closed* structure induces a form of nondeterminism in the computation. We recall the definition.

Definition 2.8. A **compact-closed category** (CCC) is a SMC where every object X has a *dual* X^* , such that for all objects X and Y , $(X \otimes Y)^* = Y^* \otimes X^*$; for all objects X , there are morphisms $\cup_X : I \rightarrow X \otimes X^*$ and $\cap_X : X^* \otimes X \rightarrow I$, such that:

$$(1_X \otimes \cap_X) \circ (\cup_X \otimes 1_X) = 1_X, \quad \text{and} \quad (\cap_X \otimes 1_{X^*}) \circ (1_{X^*} \otimes \cup_X) = 1_{X^*}.$$

Given any object X in a compact-closed discard bicategory, there is a canonical morphism $\perp_X : I \rightarrow X$, which we interpret as the *state of no knowledge* about the system X . Moreover, the two maps $\top_X$ and $\perp_X$ form an adjoint pair (in the monoidal bicategory sense):

Lemma 2.9. *If $\mathcal{C}$ is a compact-closed discard bicategory, then for any object X of $\mathcal{C}$, denoting $\perp_X := (1_X \otimes \top_{X^*}) \circ \cup_X$, we have that $\top_X \circ \perp_X \subseteq 1_I$ and $1_X \subseteq \perp_X \circ \top_X$.*

See proof on page 34.

Therefore, compact-closed discard bicategories are very close to being Cartesian bicategories, except that we do not ask for diagonal nor codiagonal maps. This generalisation is needed to capture the partial quantum examples, which have neither diagonals nor codiagonals. In the remainder of this section, we verify that all of our examples are discard bicategories.

2.2 Classical process theories

The prototypical example of a process theory is the category of sets and functions with the cartesian product, which can be seen as a model of deterministic computation. It is often useful to consider computations that can fail, in which case we obtain partial structure.

Example 2.1 (Partial deterministic processes). The category $\mathbf{Par}$ of sets and partial functions has sets as objects. Morphisms $f : X \rightarrow Y$ consist of a subset $X_f \subseteq X$, the *domain of definition*, together with a function $f : X_f \rightarrow Y$. Given an input $x \in X \setminus X_f$ not in the domain of definition, we say that f is *undefined* at x , i.e. it returns no output. The monoidal product is the Cartesian product and the monoidal unit is the singleton set $\{*\}$.

The discard $\top_X : X \rightarrow \{*\}$ is the total function sending every element of X to $*$. The approximation order is the **restriction order**: $f \subseteq g$ when the domain of definition of f is contained in that of g , and the two agree on X_f . Thus a process is approximated by any of its restrictions to a smaller domain, and the discard is the unique total effect.

$\mathbf{Par}$ is a canonical example of Robinson and Rosolini's [56] notion of a *p-category*, also known as a *Cartesian restriction category* by Cockett and Lack [17, P. 21].

Partial functions are instances of relations which can be seen as a model of nondeterministic computation.

Example 2.2 (Nondeterministic processes). The category $\mathbf{Rel}$ of sets and relations has sets as objects and subsets $R \subseteq X \times Y$ as morphisms $R : X \rightarrow Y$. Composition and identities are given by $R \circ S := \{(x, z) \mid \exists y : (x, y) \in S, (y, z) \in R\}$ and $1_X := \{(x, x) \in X^2\}$. The monoidal structure is again given by the Cartesian product and the singleton set.

The discard $\top_X : X \rightarrow \{*\}$ is the total relation $\{(x, *) \mid x \in X\}$. The approximation order is *subset inclusion*: $S \subseteq T$ when S is a subset of T . A relation is therefore approximated by any relation containing it, and the empty relation is the most informative process of its type, while the discard relates everything to the point $*$.

More generally, given a regular category $\mathcal{C}$ one can form the category of *internal relations* $\mathbf{Rel}(\mathcal{C})$; in particular $\mathbf{Rel}(\mathbf{Set}) \simeq \mathbf{Rel}$. Relations internal to finite-dimensional vector spaces are a categorical semantics for linear control theory [11, 8, 9, 4, 7, 10, 19, 12], as we investigate further in Section 6; linear relations also model mechanical networks such as electrical circuits [3, 5]. Categories of internal relations are canonical examples of Carboni and Walters' notion of a **Cartesian bicategory** [13, § 1.], which are compact-closed with dual $X^* = X$.

Several systems feature randomness and uncertainty. We recall the concrete categories of interest in categorical probability theory.

Example 2.3 (Probabilistic processes). Functions with random outcomes are commonly modelled in the category of measurable spaces and probability kernels [29]. We mostly consider the subcategory $\mathbf{BorelStoch}$ whose objects are standard Borel spaces (X, Σ_X) , where Σ_X is the Borel σ -algebra on a Polish space X . The morphisms $f : (X, \Sigma_X) \rightarrow (Y, \Sigma_Y)$ are Markov kernels; that is, functions $f : X \times \Sigma_Y \rightarrow [0, 1]$ measurable in X and a probability measure over Σ_Y . The monoidal structure is the product of Borel spaces. For discrete probability, one may also consider the full subcategory $\mathbf{FinStoch} \hookrightarrow \mathbf{BorelStoch}$ of finite sets and stochastic matrices.

The discard $\top_X(x, \{*\}) = 1$ is the Markov kernel sending every state to the unique probability measure on $\{*\}$. Since every Markov kernel is total, $\mathbf{BorelStoch}$ is affine symmetric monoidal, and is equipped with the discrete partial order. The category $\mathbf{BorelStoch}$ is the canonical example of a **Markov category with conditionals** [27, Ex. 11.3.].

To obtain a notion of approximation between probabilistic processes, we add effects to our theory corresponding to observations or conditions that the distribution satisfies. When conditioning a distribution on some effect we obtain a new *unnormalised* distribution.

Example 2.4 (Partial probabilistic processes). Panangaden’s category of subprobability kernels [52, Def. 3.1] is an established model of partial probabilistic systems. The well-behaved subcategory $\mathbf{BorelStoch}_{\leq 1}$ is defined exactly as $\mathbf{BorelStoch}$, except that $f : X \times \Sigma_Y \rightarrow [0, 1]$ is allowed to be a *subprobability* measure over Σ_Y , i.e. $f(x, Y) \leq 1$ for all $x \in X$; the discard is again the kernel to the unique measure on $\{*\}$, but is no longer total. The approximation order is the conditional preorder of Proposition 2.11 below.

$\mathbf{BorelStoch}_{\leq 1}$ is the canonical example of Di Lavore et al.’s notion of a *partial Markov category* (with conditionals) [24, Cor. 3.13.], which models updating and conditioning in substochastic processes.

In fact, all of the examples seen so far are instances of a single structure, which abstracts the copy-and-discard structure of probability theory together with the existence of conditionals.

Definition 2.10. A **partial Markov category** is a symmetric monoidal category where each object X is equipped with a *copy map* $\Lambda_X : X \rightarrow X \otimes X$ and a *discarding map* $\top_X : X \rightarrow I$ such that:

- $(\Lambda_X, \top_X)$ is a cocommutative comonoid;
- for all objects X and Y , the comonoid structure is compatible with the monoidal structure:

$$\Lambda_{X \otimes Y} = (1_X \otimes \sigma_{X,Y} \otimes 1_Y) \circ (\Lambda_X \otimes \Lambda_Y), \quad \Lambda_I = 1_I, \quad \top_{X \otimes Y} = \top_X \otimes \top_Y, \quad \text{and} \quad \top_I = 1_I;$$

- for every morphism $f : X \rightarrow Y \otimes Z$ there exists some (possibly non-unique) morphism $f|_X : Y \otimes X \rightarrow Z$ called a *conditional* of f with respect to X , such that

$$f = (1_Y \otimes f|_X) \circ (\Lambda_Y \otimes 1_X) \circ (((1_Y \otimes \top_Z) \circ f) \otimes 1_X) \circ \Lambda_X.$$

It was shown by Di Lavore et al. [25] that partial Markov categories carry a canonical approximation order:

Proposition 2.11. [25, Thm. 3.3] *Partial Markov categories are enriched in the **conditional partial order**, where given two maps $f, g : X \rightarrow Y$, $f \subseteq g$ in case there exists a map $r : Y \otimes X \rightarrow I$ such that $f = (1_Y \otimes r) \circ (\Lambda_Y \otimes 1_X) \circ (g \otimes 1_X) \circ \Lambda_X$.*

Moreover, it follows immediately from [25, Prop. 3.5] that:

Proposition 2.12. *Every partial Markov category equipped with the conditional order, is a discard bicategory.*

Therefore, all our examples of classical processes form discard bicategories. Indeed, Cartesian restriction categories are partial Markov categories [25, Props. 2.13, 3.11]; and Cartesian bicategories are partial Markov categories [25, Props. 2.16, 3.12].

2.3 Quantum process theories

We consider three notions of quantum process, capturing unnormalised quantum processes, quantum channels, and quantum channels with post-selection (see Nielsen and Chuang [47] for reference).

Example 2.5 (Quantum processes). Given a finite-dimensional Hilbert space $\mathcal{H}$, let $\mathcal{B}(\mathcal{H})$ be the space of linear operators on $\mathcal{H}$. Write $0 \preceq a$ to mean that a is *positive semidefinite*, i.e. $a = b^\dagger b$ for some $b \in \mathcal{B}(\mathcal{H})$.

A linear map $\Phi : \mathcal{B}(\mathcal{H}) \rightarrow \mathcal{B}(\mathcal{K})$ is:

- *positive* if for every $a \in \mathcal{B}(\mathcal{H})$, $0 \preceq a$ implies $0 \preceq \Phi(a)$;

- *completely positive* if for all finite-dimensional Hilbert spaces $\mathcal{E}$, $\Phi \otimes \text{id}_{\mathcal{B}(\mathcal{E})}$ is positive;
- *trace-preserving* if for all $a \in \mathcal{B}(\mathcal{H})$, $\text{Tr}(\Phi(a)) = \text{Tr}(a)$;
- *trace-nonincreasing* if for all $a \in \mathcal{B}(\mathcal{H})$ with $0 \preceq a$, $\text{Tr}(\Phi(a)) \leq \text{Tr}(a)$.

Here Tr denotes the usual linear algebraic trace. Define the categories CPM , CPTP and CPTNI whose objects are of the form $\mathcal{B}(\mathcal{H})$, for finite-dimensional Hilbert spaces $\mathcal{H}$; and whose morphisms are respectively completely positive; both completely positive and trace-preserving; and both completely positive and trace-nonincreasing maps. In all cases, the monoidal structure is the bilinear tensor product, and the discard is the partial trace, regarded as a map $\text{Tr} : \mathcal{B}(\mathcal{H}) \rightarrow \mathbb{C}$. Note that CPM is compact-closed, whereas CPTP and CPTNI are not, and that CPTP is affine.

Interpreting $\mathcal{B}(\mathcal{H})$ as the space of unnormalised quantum states on $\mathcal{H}$, CPM captures the unnormalised quantum processes; CPTP captures the physically implementable processes, also known as *quantum channels*; and CPTNI captures the post-selected processes, also known as *quantum operations*.

Our quantum examples also carry natural preorder enrichments, which can be divided into two families given by the Löwner order and the purification order.

First, recall the following partial order on completely positive maps (see [47] for reference):

Definition 2.13. Two parallel completely positive maps $\Phi, \Psi : \mathcal{B}(\mathcal{H}) \rightarrow \mathcal{B}(\mathcal{K})$ are related by the (completely positive) **Löwner order**, denoted $\Phi \preceq \Psi$, in case $\Psi - \Phi$ is completely positive.

It is easy to see that this gives us the following poset enrichments:

Lemma 2.14. *The Löwner order is a symmetric monoidal poset enrichment for CPTNI and is a discrete poset enrichment for CPTP .*

See proof on page 34.

Second, recall the following order introduced by Carrette et al. [14, Def. 5]:

Definition 2.15. Two parallel morphisms $f, g : X \rightarrow Y$ in a discard category are related by the **purification order** $f \subseteq g$ in case there exists some $g_0 : X \rightarrow S \otimes Y$ and $f_0 : S \rightarrow I$ such that

$$f = (f_0 \otimes 1_Y) \circ g_0, \quad \text{and} \quad g = (\top_S \otimes 1_Y) \circ g_0.$$

It is a direct consequence of [14, Lem. 6] that:

Lemma 2.16. *The purification order is a symmetric monoidal preorder enrichment for CPM and a discrete poset enrichment for CPTP .*

Carrette et al. do not formally compare the purification order with the Löwner order, which we now do:

Proposition 2.17. *The Löwner and purification orders coincide in CPTP and CPTNI but differ in CPM .*

See proof on page 34.

Equipping the quantum categories with the purification order makes each a discard bicategory, completing our list of examples.

Proposition 2.18. *When equipped with the purification order, CPM , CPTP and CPTNI are discard bicategories.*

See proof on page 35.

Note that CPM equipped with the Löwner order is a poset enrichment but *not* a discard bicategory structure, since lax naturality fails. Apart from this exception, all the process theories reviewed in this section are discard bicategories. They are in fact discard bicategories equipped with the purification order, as the following proposition shows.

Proposition 2.19. *In any partial Markov category, the purification order and the conditional order coincide.*

See proof on page 35.

3 Stateful processes

Having seen how process theories have the structure of discard bicategories, we now ask how to perform *stateful computation* inside them. This section is a detailed review of the existing categorical semantics for stateful processes, examining in each case what equivalence it induces and where it fails on the examples we care about. We will see that no existing approach simultaneously interprets state functorially, lets the time delay be natural, and propagates zero morphisms globally. These shortcomings are precisely what motivate the notion of *behaviour* developed in Section 4. Throughout this section we work in a plain symmetric monoidal category, the approximation structure becoming relevant only later, when we use it to compare partial observations.

3.1 Mealy machines in monoidal categories

The most basic specification of a stateful process is a *Mealy machine* [44]: a transition function $S \times X \rightarrow S \times Y$ which consumes an input in X , updates an internal state in S , and produces an output in Y . In Cartesian closed categories, such processes are often packaged using the *state monad* $T_S(X) := (X \times S)^S$ [45]: a Kleisli arrow for the state monad is exactly a Mealy machine, and iterating the Kleisli composition gives the n -step time evolution illustrated in Equation (1).

We are interested in a generalised notion of Mealy machine, in which the underlying transition $S \otimes X \rightarrow S \otimes Y$ need not be a function, but can instead live in any symmetric monoidal category.

Definition 3.1. Let $\mathcal{C}$ be a symmetric monoidal category and let S be an object of $\mathcal{C}$. A **symmetric monoidal Mealy machine** is a morphism $f: S \otimes X \rightarrow S \otimes Y$, where X is interpreted as the input, Y as the output and S as the internal state. For $n \geq 1$, its n -step evolution is the canonical composite $f^{(n)}: S \otimes X^{\otimes n} \rightarrow Y^{\otimes n} \otimes S$ obtained by wiring together n copies of f so that the output state of the k -th copy is fed back to the input state of the $(k+1)$ -st copy, repeatedly using σ , as in Equation (1).

Given an **initial state** $s_0: I \rightarrow S$, the **evaluation** at time step n is given by

$$f^{(n)} \circ (s_0 \otimes 1_{X^{\otimes n}}): X^{\otimes n} \rightarrow Y^{\otimes n} \otimes S.$$

Instantiating Definition 3.1 in each of our guiding examples recovers a well-studied class of stateful process:

- In **Par**, a Mealy machine is a **partial Mealy machine**: at each step the transition may be undefined, halting the computation. Partial Mealy machines are a standard example in the coalgebraic theory of state-based systems, where their behaviour is obtained through the (generalised) powerset/determinization construction [61].
- In **Rel**, a Mealy machine is a **nondeterministic Mealy machine**: each input may lead to several output-and-state pairs. These coincide with the *observable nondeterministic finite-state machines* studied in model-based testing and automata learning [39].
- In **BorelStoch** or **FinStoch**, a Mealy machine has a random transition function, and its time evolution generates a **controlled stochastic process** [21, Def. 7.1]: a Markov chain whose dynamics are driven by the input stream.

- In CPTP, a Mealy machine is a **quantum channel with finite memory** [40], where the state object S is the memory carried between uses of the channel. Such channels are widely studied in the contexts of spin chains, quantum coding, and quantum communication [53, 15, 49]. Going beyond this trace-preserving literature, our framework also accommodates the non-trace-preserving setting: working over CPTNI or CPM additionally captures conditioning and post-selection.

Running any of these machines from an initial state should, intuitively, produce a process that transforms input streams into output streams; but when should two such machines be deemed equivalent? This depends on what one takes the behaviour of a process to be. For the partial and nondeterministic examples, the cited works adopt a computational view, in which the behaviour of a machine is the language it accepts or the computation it performs. We instead regard the behaviour of a process as the family of constraints it imposes, in the systems-theoretic tradition of Willems [64]. In the remainder of this section we review several categorical semantics for stateful processes that formalise the computational view, and argue that none of them accommodates this constraint-based notion of behaviour, which we develop in Section 4.

3.2 The state construction

It is natural to ask: *what is the categorical semantics of Mealy machines over an arbitrary symmetric monoidal category?* At a minimum, such a semantics should provide a principled way to *hide* the internal state S , turning a stateful description into an external process from X to Y as in Equation (1). Following the approach of Katis, Sabadini, and Walters [37, 35] (and later by various others such as [62, 31]), we axiomatise this state-hiding operation with the notion of a *feedback category*. Our formulation is a mild variant of that of Katis, Sabadini, and Walters [37], adapted to the guarded/delayed setting needed later:¹

Definition 3.2. A **feedback category** is a tuple $(\mathcal{C}, \mathcal{D}, \Gamma, \Delta, \text{fbk})$ where:

- $\Gamma: \mathcal{D} \rightarrow \mathcal{C}$ is a symmetric monoidal functor, called the **guard functor**;
- $\Delta: \mathcal{D} \rightarrow \mathcal{D}$ is a symmetric monoidal endofunctor, called the **delay endofunctor**;
- $\text{fbk}_S: \mathcal{C}(\Gamma(\Delta S) \otimes X, \Gamma(S) \otimes Y) \rightarrow \mathcal{C}(X, Y)$ is a family of functions, called the **feedback operator**.

Moreover for all $f \in \mathcal{C}(\Gamma(\Delta S) \otimes X, \Gamma(S) \otimes Y)$, $u \in \mathcal{C}(X', X)$, $v \in \mathcal{C}(Y, Y')$, and $h \in \mathcal{D}(S, T)$, we ask that the following equations hold:

- **Tightening:** $v \circ \text{fbk}_S(f) \circ u = \text{fbk}_S((1_{\Gamma S} \otimes v) \circ f \circ (1_{\Gamma(\Delta S)} \otimes u))$;
- **Vanishing:** $\text{fbk}_I(f) = f$;
- **Joining:** $\text{fbk}_T(\text{fbk}_S(f)) = \text{fbk}_{S \otimes T}(f)$;
- **Strength:** $\text{fbk}_S(f) \otimes g = \text{fbk}_S(f \otimes g)$;
- **Sliding:** $\text{fbk}_T((\Gamma(h) \otimes 1_Y) \circ f) = \text{fbk}_S(f \circ (\Gamma(\Delta h) \otimes 1_X))$.

In case Γ is faithful call $\mathcal{D}$ the **guarded subcategory**.

A **feedback functor** $(F, F|_{\mathcal{D}}): (\mathcal{C}, \mathcal{D}, \Gamma, \Delta, \text{fbk}) \rightarrow (\mathcal{C}', \mathcal{D}', \Gamma', \Delta', \text{fbk}')$ between feedback categories consists of symmetric monoidal functors $F: \mathcal{C} \rightarrow \mathcal{C}'$ and $F|_{\mathcal{D}}: \mathcal{D} \rightarrow \mathcal{D}'$, satisfying $\Gamma' \circ F|_{\mathcal{D}} = F \circ \Gamma$, $\Delta' \circ F|_{\mathcal{D}} = F|_{\mathcal{D}} \circ \Delta$, and $F(\text{fbk}_S(f)) = \text{fbk}'_{F(S)}(F(f))$.

¹More generally one could replace Γ, Δ and $\mathcal{D}$ with a monoidal endoprofunctor on $\mathcal{C}$, modifying the sliding equation appropriately.

We interpret $\Gamma: \mathcal{D} \rightarrow \mathcal{C}$ as parametrising what is allowed to be passed through the state at successive timesteps, and $\Delta: \mathcal{D} \rightarrow \mathcal{D}$ as an effect which is applied when something is passed through the state; graphically:

$$(2)$$

In particular, when $\Gamma = \Delta = 1_{\mathcal{C}}$, a morphism $f: S \otimes X \rightarrow S \otimes Y$ is a symmetric monoidal Mealy machine in the sense of Definition 3.1, where $\text{fbk}_S(f): X \rightarrow Y$ is interpreted as the process given by hiding the memory of f as an internal state.

Note that a *traced symmetric monoidal structure* [34] on a SMC $\mathcal{C}$ is precisely a feedback structure with $\Delta = \Gamma = 1_{\mathcal{C}}$ satisfying the additional *yanking identity*: $\text{Tr}_X(\sigma_{X,X}) = 1_X$, which trivialises the time dependency.

Generalising the “Circ” construction of Katis et al. [37], now more commonly referred to in the theoretical computer science literature as the *state construction* (see for example [21]), we can *freely* construct a feedback category to give a categorical semantics for stateful processes over any base symmetric monoidal category:

Definition 3.3 (State construction). Given SMCs $\mathcal{C}$ and $\mathcal{D}$ with symmetric monoidal functors $\Gamma: \mathcal{D} \rightarrow \mathcal{C}$ and $\Delta: \mathcal{D} \rightarrow \mathcal{D}$, the SMC $\text{St}_{\mathcal{D}}(\mathcal{C}, \Delta)$ has the same objects as $\mathcal{C}$, where the morphisms given by elements of the set:

$$\text{St}_{\mathcal{D}}(\mathcal{C}, \Delta)(X, Y) := \int^{S \in \mathcal{D}} \mathcal{C}(\Gamma(\Delta S) \otimes X, \Gamma(S) \otimes Y).$$

The rest of the structure defining a symmetric monoidal category is given by the evident natural transformations.

A morphism in $\text{St}_{\mathcal{D}}(\mathcal{C}, \Delta)$ is called a **stateful morphism**. In case, $\Gamma = 1_{\mathcal{C}}$ or $\Delta = 1_{\mathcal{D}}$, we omit the labels $\mathcal{D}$ or Δ .

Remark 3.4. This integral symbol denotes a coend [41], so that elements of $\text{St}_{\mathcal{D}}(\mathcal{C}, \Delta)(X, Y)$ are pairs

$$(S \in \text{ob}(\mathcal{D}), f \in \mathcal{C}(\Gamma(\Delta S) \otimes X, \Gamma(S) \otimes Y)),$$

modulo the largest equivalence relation such that for any $r: S \rightarrow S'$ and any $h: S' \otimes X \rightarrow S \otimes Y$:

$$[S', (\Gamma(s) \otimes 1_Y) \circ h] = [S, h \circ (\Gamma(\Delta(s)) \otimes 1_X)].$$

The state construction is the *free* feedback category built from $\mathcal{C}$, $\mathcal{D}$, Γ , and Δ , see Proposition D.2 for a detailed proof. Given $f \in \text{St}_{\mathcal{D}}(\mathcal{C}, \Delta)(\Gamma(\Delta S) \otimes X, \Gamma(S) \otimes Y)$, the process $\text{fbk}(f): X \rightarrow Y$ is obtained by adjoining S to the memory. By the definition of the coend, two stateful processes are identified precisely when they are related by finitely many applications of the sliding rule (2).

We call the map $\delta_X := \text{fbk}_X(\sigma_{\Gamma(\Delta X), \Gamma X})$, the **delay morphism** at X . Generalising the work of Katis et al. [35, P. 5], we find that given some weak assumptions on the guard functor, feedback structures in compact-closed categories are determined by the delay morphisms $\{\delta_X\}_{X \in \text{ob}(\mathcal{D})}$:

Proposition 3.5. *Take a compact-closed category $\mathcal{C}$, a faithful symmetric monoidal functor $\Gamma: \mathcal{D} \rightarrow \mathcal{C}$ such that for all $X \in \text{ob}(\mathcal{D})$, $\Gamma^{-1}(\delta_X) \neq \emptyset$, and a symmetric monoidal endofunctor $\Delta: \mathcal{D} \rightarrow \mathcal{D}$. The existence of a feedback operator is equivalent to a monoidal natural transformation $\delta: 1_{\mathcal{D}} \Rightarrow \Delta$, called the **delay natural transformation**. Moreover, such a natural transformation is necessarily a natural isomorphism.*

See proof on page 38.

3.3 Stateful morphism sequences

Applying the state construction to $\mathcal{C}$ identifies stateful processes built from $\mathcal{C}$ that differ only by a reparametrisation of the internal state. However, this semantics is purely formal: it does not specify how to iterate a process through time. To model iteration explicitly, we pass from $\mathcal{C}$ to categories of time-indexed families in $\mathcal{C}$. The key extra ingredient is a canonical endofunctor expressing a one timestep delay:

Definition 3.6. Let $\mathbb{Z}$ and $\mathbb{N}$ denote the discrete symmetric monoidal categories whose objects are respectively the integers and the natural numbers, where the monoidal product and unit are given by addition and 0.

Given a symmetric monoidal category $\mathcal{C}$, the functor categories $[\mathbb{Z}, \mathcal{C}]$ and $[\mathbb{N}, \mathcal{C}]$ have objects and morphisms given by (bi)infinite sequences of objects and morphisms in $\mathcal{C}$. Equipping these functor categories with the *pointwise* symmetric monoidal structure, we find that there are symmetric monoidal functors:

$$\begin{aligned} \text{Shift}: [\mathbb{Z}, \mathcal{C}] &\rightarrow [\mathbb{Z}, \mathcal{C}]; & (\text{Shift}X)_{j+1} &:= X_j, & (\text{Shift}f)_{j+1} &:= f_j; \\ \text{Delay}: [\mathbb{N}, \mathcal{C}] &\rightarrow [\mathbb{N}, \mathcal{C}]; & (\text{Delay}X)_0 &:= I, & (\text{Delay}X)_{j+1} &:= X_j, \\ & & (\text{Delay}f)_0 &:= 1_I, & (\text{Delay}f)_{j+1} &:= f_j. \end{aligned}$$

With this data, we can build feedback categories of **extensional monoidal streams** [21, § 3.2]. Call a morphism in $\text{St}([\mathbb{N}, \mathcal{C}], \text{Delay})$ an **initialised stateful morphism sequence** in $\mathcal{C}$; and a morphism in $\text{St}([\mathbb{Z}, \mathcal{C}], \text{Shift})$ an **uninitialised stateful morphism sequence**.

A guard functor $\Gamma: \mathcal{D} \rightarrow \mathcal{C}$ induces functors $\Gamma: [\mathbb{N}, \mathcal{D}] \rightarrow [\mathbb{N}, \mathcal{C}]$ and $\Gamma: [\mathbb{Z}, \mathcal{D}] \rightarrow [\mathbb{Z}, \mathcal{C}]$ which can be used to restrict the sliding rule in stateful morphism sequences. For instance, initialised stateful morphism sequences with sliding guarded by Γ live in $\text{St}_{[\mathbb{N}, \mathcal{D}]}([\mathbb{N}, \mathcal{C}], \text{Delay})$. We however do not require this restriction in our results.

In the uninitialised setting, we can unroll stateful morphisms from the free feedback category into stateful morphism sequences analogously to Equation (1):

Lemma 3.7. *For every SMC $\mathcal{C}$, there is a unique feedback functor $U: \text{St}(\mathcal{C}) \rightarrow \text{St}([\mathbb{Z}, \mathcal{C}], \text{Shift})$ which is the identity on objects and acts on morphisms by:*

$$([S, f]: X \rightarrow Y) \quad \mapsto \quad ([(f: S \otimes X \rightarrow S \otimes Y)_{j \in \mathbb{Z}}]: X^{\mathbb{Z}} \rightarrow Y^{\mathbb{Z}}).$$

See proof on page 39.

Remark 3.8. On the other hand, unless $\mathcal{C}$ is a *monoid*, there is no feedback functor $F: \text{St}(\mathcal{C}) \rightarrow \text{St}([\mathbb{N}, \mathcal{C}], \text{Delay})$ acting on objects by $X \mapsto (X, X, \dots)$, since we would then have that $(X, X, \dots) = F(1_{\mathcal{C}}(X)) = \text{Delay}(F(X)) = (I, X, X, \dots)$, and thus $I = X$ for all objects X .

Nevertheless, one may still obtain an $\mathbb{N}$ -indexed interpretation of stateful morphisms by equipping morphisms in St with an initial state as in the work of Bonchi et al. [6, § III.E.], but this requires changing the definition of the feedback operation [6, Def. III.15]. Here, we will use $\text{St}([\mathbb{N}, \mathcal{C}], \text{Delay})$ directly as a syntax for initialised automata for convenience.

Extensional streams provide a categorical semantics for Mealy machines, though in a different sense from the state construction over the base category: they identify processes whose state can be propagated finitely far into the future or the past. The functor of Lemma 3.7 may therefore be seen as equipping the more primitive semantics of the previous subsection with a notion of discrete-time dynamics.

This finite propagation, however, is also a defect. If a process is found to be inconsistent at some point in time, then the process as a whole should be interpreted as inconsistent. Because extensional streams permit morphisms to slide only *finitely* far into the past or future, an inconsistency propagates only finitely far, and it follows immediately that:

Lemma 3.9. *Take a symmetric monoidal category $\mathcal{C}$ with monoidal zero morphisms. Consider a representative dinatural stream $[\mathbf{f}]$ in $\text{St}([\mathbb{Z}, \mathcal{C}], \text{Shift})$ or $\text{St}([\mathbb{N}, \mathcal{C}], \text{Delay})$ with some $f_j = 0$. Then in general $[\mathbf{f}]$ will not be a monoidal zero morphism.*

This is only one example of the many shortcomings of extensional streams. For example, using the feedback operator, any endomorphism can be turned into a scalar that never interfaces with the inputs or outputs of the stream. Consequently, extensionally indistinguishable processes need not lie in the same equivalence class. However many of these problems are resolved via taking a more sophisticated coalgebraic approach.

3.4 Coalgebraic streams

Stateful morphism sequences only allow for the internal state at one time to be reparametrised finitely far into the past or the future. Therefore, in a certain sense, the induced equivalence relation is very coarse as it does not account for truly infinite behaviour. In order to give a properly infinite stream semantics, in this subsection, we recall the coalgebraic notion of monoidal streams first developed by Di Lavore et al. [21, Def. 4.1] and later generalised by Bonchi et al. [6, Def. IV.4.].

Given a symmetric monoidal category $\mathcal{C}$ and a symmetric monoidal subcategory $\Gamma: \mathcal{D} \rightarrow \mathcal{C}$, Bonchi et al. [6] define a feedback category, which we denote by $\text{Stream}_{\mathcal{D}}(\mathcal{C})$, whose objects are infinite lists of objects in $\mathcal{C}$, where the hom-sets are given by the final fixpoint of profunctors:²

$$\text{Stream}_{\mathcal{D}}(\mathcal{C})(\mathbf{X}, \mathbf{Y}) \cong \int^{M \in \mathcal{D}} \mathcal{C}(X_0, (\Gamma M) \otimes Y_0) \times \text{Stream}_{\mathcal{D}}(\mathcal{C})((\Gamma M) \cdot \mathbf{X}^+, \mathbf{Y}^+), \quad (3)$$

where given $\mathbf{X}, \mathbf{Y} \in \text{ob}(\mathcal{C})^{\mathbb{N}}$ and $N \in \text{ob}(\mathcal{C})$, we denote

$$N \cdot \mathbf{X} := (N \otimes X_0, X_1, X_2, \dots), \quad \text{and} \quad \mathbf{X}^+ := (X_1, X_2, \dots).$$

In case $\Gamma = 1_{\mathcal{C}}$ denote this category by $\text{Stream}(\mathcal{C})$ as in [21]. The category $\text{Stream}(\mathcal{C})$ elegantly captures total stateful processes; for example, stateful processes built from Cartesian monoidal categories or Markov categories with ranges and conditionals [21, Rem. 4.10]. However, we argue that in the presence of compact-closure or monoidal zero morphisms, which are often present in nondeterministic partial and quantum settings, coalgebraic monoidal streams are poorly behaved.

In the presence of compact-closure, coalgebraic monoidal streams without a guarded subcategory are degenerate:

Lemma 3.10. *$\text{Stream}(\mathcal{C})$ is indiscrete for compact-closed $\mathcal{C}$.*

See proof on page 40.

As remarked by Bonchi et al. [6, Rem. V.12] this problem can be sidestepped by working in $\text{Stream}_{\mathcal{D}}(\mathcal{C})$ for some non-compact-closed symmetric monoidal subcategory $\mathcal{D}$ of $\mathcal{C}$. When $\mathcal{C}$ is a partial Markov category, they choose $\mathcal{D}$ to be the category of total maps $\text{Tot}(\mathcal{C})$.

Without a guard, in the presence of monoidal zero morphisms, coalgebraic monoidal streams are also degenerate:

Lemma 3.11. *If $\mathcal{C}$ is a SMC with monoidal zero morphisms, then $\text{Stream}(\mathcal{C})$ is indiscrete.*

See proof on page 40. This problem is also addressed for partial Markov categories by guarding over the category of total maps. However, guarding over the total maps disallows the delay from being a natural isomorphism. Similarly, when one takes this approach of guarding over the total maps, just as is the case for extensional streams in Lemma 3.9, the presence of a zero morphism at some time in the stream will not make the stream itself zero, and thus will not detect inconsistencies in a satisfactory manner.

²In fact they give a more general definition for premonoidal categories; however, we are only interested in monoidal setting in this paper.

3.5 Causal and monotone sequences

Another approach to categorical semantics for stateful processes is to represent a stream *concretely* as a family of morphisms in the base category of *increasingly larger* or *increasingly informative* approximations.

First, consider the following notion due to Raney [55], which constructs stream transducers so that the value of the stream at every time step does not depend on the value of the stream in the future. Let A and B be sets. A function $f: A^{\mathbb{N}} \rightarrow B^{\mathbb{N}}$ is *causal* (called sequential by Raney) if for all $x, y \in A^{\mathbb{N}}$ and $n \in \mathbb{N}$, $(\forall i \leq n: x_i = y_i) \implies (\forall i \leq n: f(x)_i = f(y)_i)$.

Categorifying this concept we obtain:

Definition 3.12. Let $\mathcal{C}$ be a discard category. The affine symmetric monoidal category of **causal sequences** $\text{Caus}_{\mathbb{N}}(\mathcal{C})$ has:

- **Objects:** infinite sequences $\mathbf{X} \in \text{ob}(\mathcal{C})^{\mathbb{N}}$;
- **Morphisms:** $\mathbf{f}: \mathbf{X} \rightarrow \mathbf{Y}$ are given by $\mathbb{N}$ -indexed sequences of morphisms in $\mathcal{C}$ with components $\{f_t: X_0 \otimes \cdots \otimes X_t \rightarrow Y_0 \otimes \cdots \otimes Y_t\}_{t \in \mathbb{N}}$ such that for all $t \in \mathbb{N}$,

$$(1_{Y_0 \otimes \cdots \otimes Y_t} \otimes \top_{Y_{t+1}}) \circ f_{t+1} = f_t \otimes \top_{X_{t+1}}; \quad (4)$$

where the rest of the affine symmetric monoidal category structure is defined pointwise.

This notion of causality is well-behaved in categories where every morphism is total. For example, it recovers the causal functions of Katsumata and Sprunger [62, Thm. 14] in Cartesian categories, and controlled stochastic processes in FinStoch [21, Def. 7.1].

However, this is not the correct notion in categories of partial processes. For example, a process f in $\text{Caus}_{\mathbb{N}}(\text{Par})$ such that f_0 is total, must remain total for all $n > 0$. Similarly, for any constant sequence $\{f^{\otimes n}\} \in \text{Caus}_{\mathbb{N}}(\mathcal{C})$ we must have $\top_Y \circ f = \top_X$, i.e. f must be total. Therefore, causal streams do not accommodate inconsistencies, which suggests that the equality from Definition 3.12 needs to be relaxed to an *inequality*

$$(1_{Y_0 \otimes \cdots \otimes Y_t} \otimes \top_{Y_{t+1}}) \circ f_{t+1} \subseteq f_t \otimes \top_{X_{t+1}}. \quad (5)$$

Finally, with all of these examples in mind, we can relax Definition 3.12 to the preorder-enriched setting:

Definition 3.13. Let $\mathcal{C}$ be a preorder-enriched discard category. The discard category of **monotone sequences** $\text{Mon}_{\mathbb{N}}(\mathcal{C})$ has the same objects as $\text{Caus}_{\mathbb{N}}(\mathcal{C})$. A morphism $f: \mathbf{X} \rightarrow \mathbf{Y}$ is a monotone sequence, i.e. a family of morphisms $\{f_t: X_0 \otimes \cdots \otimes X_t \rightarrow Y_0 \otimes \cdots \otimes Y_t\}_{t \in \mathbb{N}}$ satisfying the inequality of Equation (5), where the rest of the discard category structure is defined pointwise.

Bonchi et al. show [6, Prop. 6.7] that, in a Markov category with conditionals, their notion of “causal sequences” (which conflicts with the terminology that we use in this paper) are exactly monotone sequences. Carette et al. define a similar construction for discard categories with pseudo-purification, where they impose moreover that every sequence satisfies the very strong assumption of regularity [14, Def. 8].

Monotone sequences accommodate inconsistency: they can contain a component $f_t = 0$ without forcing the whole family $\{f_t\}_{t \in \mathbb{N}}$ to be the pointwise-zero morphism in $\text{Mon}_{\mathbb{N}}(\mathcal{C})$. However, inconsistencies are still not detected at the level of the whole stream: an inconsistency that appears only at a later time step changes only later components in the family.

This motivates imposing an equivalence relation which takes the information theoretic interpretation of preorder enrichment more seriously, identifying two monotone sequences in case they are compatible and neither is more informative than the other.

4 Observational semantics

In the previous section, we reviewed various semantics for stateful processes, and saw that each makes a different compromise. Stateful morphism sequences are purely local: a constraint enforced at one time step only propagates finitely far into the future or the past. Coalgebraic streams sit at the opposite, where a fixed-point equation constrains the entire infinite behaviour at once, but collapses in the presence of partiality and nondeterminism. Causal and monotone sequences are concrete and well-behaved, but they fail to identify dinaturally equivalent streams.

In this section, we build a categorical semantics in which the *global behaviour* of a process is determined by the totality of its *local observations*, glued together by a consistency condition: observations made in a larger context may only *refine* those made in a smaller one. Two processes are then identified when they are *observationally indistinguishable*, that is, when every observation of one can be verified by observing the other, possibly in a larger context. We show that this notion satisfies the desiderata set out in Section 1. A further advantage over existing stream semantics is that the contexts need not be discrete intervals of time. We can therefore interpret processes evolving through more general spaces.

4.1 The bicategory of behaviours

Our semantic construction is parametrised by an arbitrary **index set** $\mathcal{I}$ and a collection $\mathcal{J} \subseteq \mathcal{P}_f(\mathcal{I})$ of *finite* subsets of $\mathcal{I}$ called **contexts**, which define the local finite regions from which a global system can be observed. Given two contexts $j \subseteq j'$, we denote the set-theoretic difference by $j' \setminus j$. We assume a total order on $\mathcal{I}$ throughout so that the indexed monoidal product $\otimes_{i \in j}$ is well-defined for any subset $j \in \mathcal{J}$. We often omit the symmetry morphisms for simplicity and readability, or explicitly use the **shuffle morphisms**, defined for arbitrary $\mathbf{X}, \mathbf{Y} \in (\text{ob}(\mathcal{C}))^{\mathcal{I}}$ and contexts $j \in \mathcal{J}$ as the following isomorphisms:

$$(\zeta_{\mathbf{X}, \mathbf{Y}})_j : \left(\bigotimes_{k \in j} X_k \right) \otimes \left(\bigotimes_{k \in j} Y_k \right) \cong \bigotimes_{k \in j} (X_k \otimes Y_k).$$

We represent consistent families of observations using the following structure:

Definition 4.1. Take a discard bicategory $\mathcal{C}$, index set $\mathcal{I}$, a set of contexts $\mathcal{J} \subseteq \mathcal{P}_f(\mathcal{I})$, and a $\mathcal{I}$ -indexed family of objects $\mathbf{X}$ and $\mathbf{Y}$ in $\mathcal{C}$.

A **monotone family** $f : \mathbf{X} \rightarrow \mathbf{Y}$ in $\mathcal{C}$ is a $\mathcal{J}$ -indexed family of morphisms

$$\{f_j : \bigotimes_{i \in j} X_i \rightarrow \bigotimes_{i \in j} Y_i\}_{j \in \mathcal{J}},$$

where $Y_j := \bigotimes_{k \in j} Y_k$ and $X_j := \bigotimes_{k \in j} X_k$, such that for all $j \subseteq j' \in \mathcal{J}$:

$$(1_{Y_j} \otimes \top_{Y_{j' \setminus j}}) \circ f_{j'} \subseteq f_j \otimes \top_{X_{j' \setminus j}}.$$

A monotone family is **causal** when all inclusions are equalities.

Note that monotone families generalise the notion of monotone sequences. A monotone sequence is a monotone family with $\mathcal{I} := \mathbb{N}$ and $\mathcal{J} := \{[0, n] \mid \forall n \in \mathbb{N}\}$.

We interpret a context as a finite region in which properties of some global system can be observed (for monotone sequences, this finite region is interpreted as an interval of time). Under this interpretation, the condition of monotonicity imposes that the observations made in any context *approximate* the observations which can be made in a larger context, so that all observations are consistent with each other.

We introduce a preorder between monotone families that captures when one family of observations approximates another:

Definition 4.2. Given two monotone families $\mathbf{f}, \mathbf{g}: \mathbf{X} \rightarrow \mathbf{Y}$, say that $\mathbf{g}$ **approximates** $\mathbf{f}$, written $\mathbf{f} \subseteq \mathbf{g}$, if for every $j \in \mathcal{J}$ there exists some $j' \in \mathcal{J}$ with $j \subseteq j'$ such that

$$(1_{Y_j} \otimes \top_{Y_{j' \setminus j}}) \circ f_{j'} \subseteq g_j \otimes \top_{X_{j' \setminus j}}.$$

Lemma 4.3. *Approximation is a preorder.*

See proof on page 41.

As the name suggests, we interpret $\mathbf{f} \subseteq \mathbf{g}$ to denote that the observations made in $\mathbf{f}$ refine those made in $\mathbf{g}$, so that $\mathbf{f}$ is a more accurate description of the underlying process which is being observed.

We define a discard bicategory of monotone families ordered by approximation:

Definition 4.4. Given a set of contexts $\mathcal{J} \subseteq \mathcal{P}_f(\mathcal{I})$ and a discard bicategory $\mathcal{C}$, the discard bicategory of **monotone families**, $\text{Mon}_{\mathcal{I}, \mathcal{J}}(\mathcal{C})$, has:

- **objects:** families $\mathbf{X} = \{X_i\}_{i \in \mathcal{I}}$;
- **morphisms:** monotone families $\mathbf{f}: \mathbf{X} \rightarrow \mathbf{Y}$;
- **poset-enrichment:** given by approximation;
- **discard category structure:** contextwise.

Proposition 4.5. $\text{Mon}_{\mathcal{I}, \mathcal{J}}(\mathcal{C})$ is a discard bicategory.

See proof on page 41.

In particular, we have that $\text{Mon}_{\{[0, n] \mid \forall n \in \mathbb{N}\}, \mathbb{N}}(\mathcal{C}) \cong \text{Mon}_{\mathbb{N}}(\mathcal{C})$; however, by changing the set of contexts we can model processes evolving through more general spaces, rather than merely time.

Consider the equivalence relation on monotone families generated by mutual approximation:

Definition 4.6. We say that $\mathbf{f}$ and $\mathbf{g}$ are **observationally equivalent**, in case $\mathbf{f}$ and $\mathbf{g}$ approximate each other, i.e. $\mathbf{f} \subseteq \mathbf{g}$ and $\mathbf{g} \subseteq \mathbf{f}$.

We interpret two processes to be observationally equivalent in case their observations never contradict each other; so that every observation of one process can be shown to approximate an observation in the other and vice versa.

For monotone families indexed by arbitrary sets of contexts, observational equivalence is not necessarily a congruence with respect to composition. To ensure that this holds, we ask that the set of contexts possesses the following property:

Definition 4.7. $\mathcal{J} \subseteq \mathcal{P}_f(\mathcal{I})$ is **upward-directed**, in case for any pair of contexts $j, j' \in \mathcal{J}$ there exists some $j'' \supseteq j \cup j'$ in $\mathcal{J}$. Call a monotone family a **monotone net**³ if it is indexed by an upward-directed set of contexts.

Quotienting the category of monotone *nets* by observational equivalence yields the following category, which is the central construction of this article:

Definition 4.8. Given an upward-directed set of contexts $\mathcal{J} \subseteq \mathcal{P}_f(\mathcal{I})$ and a discard bicategory $\mathcal{C}$, the (poset-enriched) discard bicategory of **behaviours**, $\text{Obs}_{\mathcal{I}, \mathcal{J}}(\mathcal{C})$, is given by quotienting $\text{Mon}_{\mathcal{I}, \mathcal{J}}(\mathcal{C})$ by observational equivalence.

³We use the term “monotone net” as a categorification of the notion of a “net” coined by Kelley, which denotes a function whose domain is a directed set [38].

Theorem 4.9. $\text{Obs}_{\mathcal{I}, \mathcal{J}}(\mathcal{C})$ is a poset-enriched discard bicategory.

See proof on page 42.

The construction of behaviours is functorial:

Proposition 4.10. Let $F: \mathcal{C} \rightarrow \mathcal{D}$ be a discard bicategory functor. There is an induced discard bicategory functor $\text{Obs}_{\mathcal{I}, \mathcal{J}}(F): \text{Obs}_{\mathcal{I}, \mathcal{J}}(\mathcal{C}) \rightarrow \text{Obs}_{\mathcal{I}, \mathcal{J}}(\mathcal{D})$ defined contextwise.

Moreover, if F reflects the preorder then $\text{Obs}_{\mathcal{I}, \mathcal{J}}(F)$ is faithful. If F reflects the preorder and is full, then $\text{Obs}_{\mathcal{I}, \mathcal{J}}(F)$ is full and faithful.

See proof on page 42.

4.2 Propagation of inconsistency

Recall from Section 3 that the existence of symmetric monoidal zero morphisms makes the category of coalgebraic streams satisfying Equation (3) codiscrete. On the other hand, in extensional streams a zero morphism occurring at some time step cannot propagate infinitely far into the future, so the global process is not a monoidal zero morphism; and for monotone sequences, zero morphisms propagate only into the future, never the past. By contrast, behaviours *detect inconsistencies* globally: if a process is inconsistent in any single finite context, then its whole behaviour is the zero morphism.

Proposition 4.11. Take a discard bicategory $\mathcal{C}$ with monoidal zero morphisms, and upward-directed $\mathcal{J} \subseteq \mathcal{P}_f(\mathcal{I})$. Then $\text{Obs}_{\mathcal{I}, \mathcal{J}}(\mathcal{C})$ has monoidal zero morphisms. Moreover, if a monotone net $\mathbf{f}: \mathbf{X} \rightarrow \mathbf{Y}$ satisfies $f_{j_0} = 0$ for some $j_0 \in \mathcal{J}$, then $[\mathbf{f}] = [\{0\}_{j \in \mathcal{J}}]$ is a zero morphism.

See proof on page 43.

4.3 Dinaturality of state

We now turn to the question: *under which conditions can stateful processes be interpreted as behaviours?* We answer it by exhibiting functors from *stateful morphism sequences* into categories of behaviours. Crucially, these functors equate machines when they are related by dinatural reparametrisation of the internal state. Note that the same mapping onto monotone sequences would be ill-defined as dinaturally equivalent machines can give rise to distinct monotone sequences. We first study the case of stateful processes with a specified initial state. Their behaviours are then indexed by the natural numbers.

Definition 4.12. Let $\mathbb{N}_{\leq}^2 := \{[0, t] \mid t \in \mathbb{N}\}$ denote the set of finite intervals in $\mathbb{N}$ containing 0. Given a discard bicategory $\mathcal{C}$, denote the category of **initialised behaviours** over $\mathcal{C}$ by $\text{Obs}_{\mathbb{N}}(\mathcal{C}) := \text{Obs}_{\mathbb{N}, \mathbb{N}_{\leq}^2}(\mathcal{C})$.

For convenience, we index morphisms in $\text{Obs}_{\mathbb{N}}(\mathcal{C})$ by $t \in \mathbb{N}$, writing f_t for the component at context $[0, t]$; note that X_t denotes the single object at time t , whereas f_t has domain $X_0 \otimes \cdots \otimes X_t$.

Definition 4.13. Define the delay endofunctor $\text{Delay}: \text{Obs}_{\mathbb{N}}(\mathcal{C}) \rightarrow \text{Obs}_{\mathbb{N}}(\mathcal{C})$ on objects by

$$\text{Delay}(\mathbf{X}) := \{X_{t-1}\}_{t \in \mathbb{N}}, \quad \text{with } X_{-1} := I,$$

and on morphisms by

$$([\mathbf{f}]: \mathbf{X} \rightarrow \mathbf{Y}) \quad \longmapsto \quad ([\{1_I \otimes f_{t-1}\}_{t \in \mathbb{N}}]: \text{Delay}(\mathbf{X}) \rightarrow \text{Delay}(\mathbf{Y})).$$

The following theorem justifies our claim that behaviours provide a categorical semantics for Mealy machines:

Theorem 4.14. *If $\mathcal{C}$ is a discard bicategory then there is a symmetric monoidal functor $\text{St}([\mathbb{N}, \mathcal{C}], \text{Delay}) \rightarrow \text{Obs}_{\mathbb{N}}(\mathcal{C})$.*

See proof on page 44.

In general, $\text{Obs}_{\mathbb{N}}(\mathcal{C})$ is not known to form a feedback category for monoidal $\mathcal{C}$, although the existence of a functor from $\text{St}([\mathbb{N}, \mathcal{C}])$ ensures that dinaturally equivalent stateful morphisms will have the same interpretation. Even so, the delay is natural: delaying an observation by one time step does not change the behaviour.

Proposition 4.15. *For any discard bicategory $\mathcal{C}$, there is a monoidal natural transformation*

$$\delta: 1_{\text{Obs}_{\mathbb{N}}(\mathcal{C})} \Rightarrow \text{Delay}; \quad \delta_{\mathbf{X}} := \left[\left\{ \left(\bigotimes_{k=0}^{t-1} 1_{X_k} \right) \otimes \top_{X_t} \right\}_{t \in \mathbb{N}} \right].$$

See proof on page 44.

Example 4.1 (Universality of stateful processes). A natural question to ask is whether all behaviours can be built as stateful morphism sequences. Categorically, this corresponds to showing that the functor $\text{St}([\mathbb{N}, \mathcal{C}]) \rightarrow \text{Obs}_{\mathbb{N}}(\mathcal{C})$ is full. This holds for all our examples of interest.

Classical processes. For a partial Markov category $\mathcal{C}$, the fullness of the functor $U : \text{St}([\mathbb{N}, \mathcal{C}]) \rightarrow \text{Obs}_{\mathbb{N}}(\mathcal{C})$ can be shown using the property of conditionals, encoding *Bayes rule*. Given a representative monotone sequence, the conditional preorder supplies an effect witnessing each monotonicity inequality [21, § 7]. Note however that this functor does not factor through the category of monotone sequences $\text{Mon}_{\mathbb{N}}(\mathcal{C})$, because it doesn't have a natural delay. Still, the quotient functor $\text{Mon}_{\mathbb{N}}(\mathcal{C}) \rightarrow \text{Obs}_{\mathbb{N}}(\mathcal{C})$ is full and it was shown by Bonchi et al. [6, Thm. 7.19] that, if $\mathcal{C}$ has *ranges*, monotone sequences can be identified with coalgebraic streams guarded over the total maps, $\text{Stream}_{\text{Tot}(\mathcal{C})}(\mathcal{C}) \simeq \text{Mon}_{\mathbb{N}}(\mathcal{C})$. Composing with the quotient, the functor $\text{Stream}_{\text{Tot}(\mathcal{C})}(\mathcal{C}) \rightarrow \text{Obs}_{\mathbb{N}}(\mathcal{C})$ is full. This means we can interpret coinductive equations on string diagrams [22], as long as the rule is restricted to total maps. A *complete* coalgebraic calculus of initialised behaviours would have to combine the two: finite sliding of arbitrary morphisms together with coinductive rules restricted to a subclass.

Quantum processes. A similar story should hold in the quantum setting. Here fullness of U arises instead from the *Stinespring dilation theorem* for completely positive maps, which plays the role that conditionals play classically. This has been axiomatised by Carette et al. [14] via the notion of a *pseudo-purifiable* discard category equipped with the purification order of Definition 2.15, which can be seen as a strengthening of the notion discard bicategory where every morphism is uniquely expressed by a pure morphism with a minimal discarded environment. Following the same argument as [14, Prop. 2], U is full for all CPTP, CPTNI, and CPM with the purification order; by Proposition 2.17, the same holds for CPTNI with the Löwner order.

4.4 Compact-closed behaviours

When working in compact-closed categories, the simplest way to construct a feedback category of stateful processes is to freely add a *delay* natural transformation, see Proposition 3.5. This is convenient because, if we have an axiomatisation of a compact-closed category $\mathcal{C}$ in terms of generators and relations, we directly obtain an axiomatisation of $\text{St}(\mathcal{C})$ by adding one generator for delay and imposing that it is natural with respect to all other generators.

However, it is not clear what the time evolution of these processes should be, as there cannot be a functor from $\text{St}(\mathcal{C})$ to initialised streams or behaviours. For example, signal flow graphs add a natural delay transformation to the category of affine relations [8, 9], but their time-domain interpretation has so far not been functorial.

As we proceed to show, stateful morphisms over compact-closed discard bicategories can be interpreted functorially as behaviours indexed by the integers rather than the natural numbers.

Definition 4.16. Let $\mathbb{Z}_{\leq}^2 := \{[\ell, r] \mid \ell, r \in \mathbb{Z}: \ell \leq r\}$ denote the set of finite closed intervals in $\mathbb{Z}$. Given a discard bicategory $\mathcal{C}$, denote the category of **uninitialised behaviours** over $\mathcal{C}$ by $\text{Obs}_{\mathbb{Z}}(\mathcal{C}) := \text{Obs}_{\mathbb{Z}, \mathbb{Z}_{\leq}^2}(\mathcal{C})$.

Definition 4.17. Define the delay endofunctor $\text{Shift}: \text{Obs}_{\mathbb{Z}}(\mathcal{C}) \rightarrow \text{Obs}_{\mathbb{Z}}(\mathcal{C})$ on objects by

$$\text{Shift}(\mathbf{X}) := \{X_{k-1}\}_{k \in \mathbb{Z}},$$

and on morphisms by:

$$([f]: \mathbf{X} \rightarrow \mathbf{Y}) \longmapsto ([\{f_{[\ell-1, r-1]}\}_{[\ell, r] \in \mathbb{Z}_{\leq}^2}]: \text{Shift}(\mathbf{X}) \rightarrow \text{Shift}(\mathbf{Y})).$$

It follows from Lemma 2.9 that:

Theorem 4.18. *Given a compact-closed discard bicategory $\mathcal{C}$ with $\perp_X := (1_X \otimes \top_{X^*}) \circ \cup_X$, there is a monoidal natural isomorphism*

$$\delta: 1_{\text{Obs}_{\mathbb{Z}}(\mathcal{C})} \Rightarrow \text{Shift}; \quad \delta_{\mathbf{X}} := [\{\perp_{X_{\ell-1}} \otimes (\bigotimes_{k=\ell}^{r-1} 1_{X_k}) \otimes \top_{X_r}\}_{[\ell, r] \in \mathbb{Z}_{\leq}^2}],$$

making $\text{Obs}_{\mathbb{Z}}(\mathcal{C})$ into a compact-closed feedback category.

See proof on page 45. Using the universal property of the state construction and lax naturality, we obtain a functor witnessing that uninitialised behaviours are a feedback-preserving categorical semantics for Mealy machines in compact-closed categories:

Corollary 4.19. *Given a compact-closed discard bicategory $\mathcal{C}$, there are feedback functors $\text{St}(\mathcal{C}) \rightarrow \text{St}([\mathbb{Z}, \mathcal{C}], \text{Shift}) \rightarrow \text{Obs}_{\mathbb{Z}}(\mathcal{C})$.*

See proof on page 46.

Example 4.2. Among our examples, these results apply to (i) non-deterministic processes in any Cartesian bicategory $\text{Rel}(\mathcal{C})$ and (ii) unnormalised quantum processes in CPM with the purification order. Indeed both are compact-closed discard bicategories. The resulting behaviours are very well-behaved in both cases, having the structure of a feedback compact closed discard bicategory. $\text{Obs}_{\mathbb{Z}}(\text{CPM})$ can be interpreted as a semantics for infinite dimensional quantum systems such as spin chains and quantum channels with memory.

The *structure theorem* of Kretschman and Werner [40] corresponds to the fullness of the functor $U: \text{St}(\text{CPM}) \rightarrow \text{Obs}_{\mathbb{Z}}(\text{CPM})$. Note that this functor is not full if CPM contains only *finite dimensional* systems. This is because, differently from the $\mathbb{N}$ -indexed setting, behaviours could in principle retain memory from infinitely far into the past, and these cannot be expressed as time-evolutions of machines with finite-dimensional memory.

The full structure theorem could possibly be recovered by allowing infinite-dimensional quantum systems in the process theory, as in [59, 16]. The advantage of a *finite memory* assumption in this case, is that the resulting equational theory is more tractable and we gain access to truly infinite behaviour while retaining compact closure and a natural delay.

We leave the analysis of stateful quantum processes to future work and, in the remainder of this paper, we investigate the compact-closed relational setting in more detail.

5 Compactness theorem

When do local approximations completely and uniquely determine a global process? Just as series need not converge, neither do behaviours. However, when the category is sufficiently well-behaved, we may be able to glue the local approximations into a coherent infinite process.

In this section, we show that this is the case for the category of closed relations between compact Hausdorff spaces, exhibiting a functor $\text{Obs}(\text{Rel}(\mathbf{KHaus})) \rightarrow \text{Rel}(\mathbf{KHaus})$.

In first-order logic, the *compactness theorem* states that a set of first-order sentences has a model if and only if every finite subset of it has a model (see Gödel [30, Satz X] and later Tarski [63, Thm. 13]). The construction of this functor can be interpreted as a categorified compactness theorem for relational models.

Definition 5.1. Let $\mathbf{KHaus}$ denote the category of **compact Hausdorff topological spaces** and **continuous functions**.

Lemma 5.2 ([32, prop 15.]). $\mathbf{KHaus}$ is a regular category, such that the morphisms $X \rightarrow Y$ in $\text{Rel}(\mathbf{KHaus})$ are exactly the closed subspaces of $X \times Y$ with respect to the product topology.

Hausdorffness is needed for the identity to be closed; whereas compactness is needed for composition to preserve closedness.

We exhibit a functor $\text{Obs}(\text{Rel}(\mathbf{KHaus})) \rightarrow \text{Rel}(\mathbf{KHaus})$ which glues together observational equivalence classes of monotone nets on $\{X_j\}_{j \in \mathcal{J}}$ into a closed subspace of $\prod_{j \in \mathcal{J}} X_j$, subject to a covering condition on the set of contexts. We moreover show that this restricts to a *faithful* functor $\text{Obs}(\text{Rel}(\mathbf{FSet})) \rightarrow \text{Rel}(\mathbf{KHaus})$.

For an index set $\mathcal{I}$ and a set of finite contexts $j, j' \in \mathcal{J}$ such that $j \subseteq j'$, we use the following notations for projections:

$$\pi_j^\infty : \prod_{k \in \mathcal{I}} X_k \rightarrow \prod_{k \in j} X_k, \quad \text{and} \quad \pi_j^{j'} : \prod_{k \in j'} X_k \rightarrow \prod_{k \in j} X_k.$$

Using the inverse image of these projections and intersections, we glue together monotone nets using the following construction:

Lemma 5.3. Given an $\mathcal{I}$ -indexed set of compact Hausdorff spaces $\mathbf{X} = \{X_k \in \text{ob}(\mathbf{KHaus})\}_{k \in \mathcal{I}}$, and a monotone net of closed subspaces $\mathbf{C} := \{C_j \subseteq \prod_{k \in j} X_k\}_{j \in \mathcal{J}}$, then there is a compact, closed subspace

$$\lim(\mathbf{C}) := \bigcap_{j \in \mathcal{J}} (\pi_j^\infty)^{-1}(C_j) \subseteq \prod_{k \in \mathcal{I}} X_k.$$

See proof on page 46.

The construction of Lemma 5.3 lifts to a functor when the set of contexts possesses the following property:

Definition 5.4. $\mathcal{J} \subseteq \mathcal{P}_f(\mathcal{I})$ is a **cover**, in case for any $i \in \mathcal{I}$ there exists a subset $j \in \mathcal{J}$ containing $i \in j$. $\mathcal{J}$ is **cofinal**, in case for each finite subset $s \in \mathcal{P}_f(\mathcal{I})$ there is some $j \supseteq s$ in $\mathcal{J}$. Note that $\mathcal{J}$ is cofinal if and only if it is an upward-directed cover.

Theorem 5.5 (Compactness). Given a set $\mathcal{I}$ with cofinal $\mathcal{J} \subseteq \mathcal{P}_f(\mathcal{I})$, then $\lim$ induces a poset-enriched discard functor $\text{Lim} : \text{Obs}_{\mathcal{I}, \mathcal{J}}(\text{Rel}(\mathbf{KHaus})) \rightarrow \text{Rel}(\mathbf{KHaus})$.

See proof on page 46.

Note that the assumption that $\mathcal{J}$ is a cover is only used for preservation of the identity, while the condition of directedness is only used for preservation of composition.

Moreover, when each finite context admits only finitely many possible observations, the global behaviour *uniquely determines* which local observations are possible:

Corollary 5.6 (Faithfulness). Given a set $\mathcal{I}$ with cofinal $\mathcal{J} \subseteq \mathcal{P}_f(\mathcal{I})$, Lim restricts to a faithful poset-enriched discard functor $\text{Lim} : \text{Obs}_{\mathcal{I}, \mathcal{J}}(\text{Rel}(\mathbf{FSet})) \rightarrow \text{Rel}(\mathbf{KHaus})$.

This follows because all descending chains of subsets of finite sets terminate (see Lemma F.2 and Lemma F.3 for a detailed argument). However, because every infinite compact Hausdorff space has an infinite strictly descending chain of subsets, $\mathbf{FSet}$ is the largest *full* subcategory of $\mathbf{KHaus}$ which admits this faithfulness property.

6 Linear time-invariant systems

Linear time-invariant (LTI) systems are central objects of study in signal processing [50], control theory [64], and electrical circuits [48, § 3.1.3]. They have received increased attention in the context of categorical semantics, thanks to the work of Bonchi, Zanasi et al. [8, 9, 11, 65] as well as Baez et al. [4, 5, 3].

While their categorical semantics is well-understood from the perspective of the frequency domain (which we will call the z -domain), particularly with the use of affine relations over fields of polynomials [11, 4], their time-domain interpretation has long eluded categorical treatment. Different approaches exist to capture the “executions” of LTI systems, from the behaviours of Willems [64] and Oberst et al. [48, § 3.1.1] to the trajectories of Bonchi et al. [11], but none is compositional.

In this section, we show that $\text{Obs}_{\mathbb{Z}}$ and $\text{Obs}_{\mathbb{N}}$ give a categorical semantics for uninitialised and initialised LTI systems in the time-domain. We relate the z -domain interpretation of [11, 4] to our time-domain semantics via an oplax feedback functor capturing the inverse z -transform. We moreover show that our notion of uninitialised behaviour, instantiated in this setting, precisely captures the behaviours of Willems [64]. We end by discussing the syntactic implications for the theory of signal flow graphs.

6.1 The z -domain

Fix a field $\mathbb{K}$. Let $\text{FAff}_{\mathbb{K}}$ denote the SMC of affine transformations between finite-dimensional vector spaces, and let $\text{AR}_{\mathbb{K}}^{\text{fd}}$ denote the full subcategory of $\text{Rel}(\text{FAff}_{\mathbb{K}})$ whose objects are finite-dimensional vector spaces. The category of finite-dimensional affine relations, $\text{AR}_{\mathbb{K}}^{\text{fd}}$, is a categorical semantics for time-independent open $\mathbb{K}$ -linear systems [4, 7]. The linear relations are interpreted as global constraints; whereas elements of $\mathbb{K}^n$, regarded as affine displacements, are interpreted as signals propagating through the system according to these constraints.

Let $\mathbb{K}[[z]]$ denote the ring of formal power series, and $\mathbb{K}((z))$ its field of fractions, the formal Laurent series (see Appendix B for a review of the theory of polynomial-type rings). Following Oberst et al. [48, Eq. (8.2) & Rem. 8.1.1], we identify initialised discrete-time signals in the z -domain⁴ with formal power series $a(z) = \sum_{t=0}^{\infty} a_t z^t$, so that the signal at time t is given by the coefficient a_t , where multiplication by z corresponds to a discrete time-delay.

By extending scalars along $\mathbb{K}((z)) \supset \mathbb{K}[[z]]$, the delay becomes invertible, so that by forgetting initial conditions, LTI systems in the z -domain may be conveniently represented in terms of $\mathbb{K}((z))$ -linear subspaces of the vector space $\mathbb{K}((z))^{\text{in}} \times \mathbb{K}((z))^{\text{out}}$ [48, Cor. & Def. 8.1.4].

Just as linear time-independent systems can be composed in $\text{AR}_{\mathbb{K}}^{\text{fd}}$, following Baez and Erbele [4] as well as Bonchi et al. [9, § 6.2], uninitialised LTI systems in the z -domain can be composed in $\text{LR}_{\mathbb{K}((z))}^{\text{fd}}$. In order to also account for the signals, we moreover need to include affine displacements, which altogether yields the following feedback category:

Lemma 6.1. *There is faithful symmetric monoidal functor $\text{LR}_{\mathbb{K}((z))}^{\text{fd}} \rightarrow \text{AR}_{\mathbb{K}((z))}^{\text{fd}}$ and a monoidal natural transformation $\partial: 1_{\text{LR}_{\mathbb{K}((z))}^{\text{fd}}} \Rightarrow 1_{\text{AR}_{\mathbb{K}((z))}^{\text{fd}}}$; $\partial_V := \{(\mathbf{v}, z \cdot \mathbf{v}) \mid \mathbf{v} \in V\}$, making $\text{AR}_{\mathbb{K}((z))}^{\text{fd}}$ into a feedback category.*

Note that in order to compose LTI systems in the z -domain using relational composition we are forced to use formal Laurent series, rather than formal power series, since $\text{AR}_{\mathbb{K}[[z]]}^{\text{fd}} \simeq \text{AR}_{\mathbb{K}((z))}^{\text{fd}}$ [65, Thm. 4.14]. Therefore, this relational categorical semantics for LTI systems in the z -domain cannot accommodate for initial conditions.

⁴Oberst et al. refer to this as the algebraic domain [48, P. 3].

6.2 The time-domain

In the *time-domain*, initialised discrete-time signals are represented by $\mathbb{N}$ -indexed sequences in $\mathbb{K}$ [48, § 5.2.2]. Accordingly, the *behaviours* or *trajectories* of initialised LTI systems can be represented by systems of linear constant coefficient *difference equations*, the discrete analogue of differential equations [48, § 3.1.1]. The solutions to these difference equations are certain $\mathbb{K}[z]$ -linear subspaces of $(\mathbb{K}^{\mathbb{N}})^{\text{in}} \times (\mathbb{K}^{\mathbb{N}})^{\text{out}}$.

The **inverse z -transform** is the $\mathbb{K}[z]$ -linear transformation $\mathcal{Z}_n^\circ: \mathbb{K}[[z]]^n \rightarrow (\mathbb{K}^{\mathbb{N}})^n$ which extracts coefficients, transforming signals in the z -domain into the time-domain [48, Rem. 8.1.1].

In order to remain compatible with the categorical semantics for the z -domain, we abandon initial conditions, and define the delay $\Delta: \mathbb{K}^{\mathbb{Z}} \rightarrow \mathbb{K}^{\mathbb{Z}}$ by $(\Delta x)_{k+1} := x_k$. In this setting we can now define inverse z -transform as the $\mathbb{K}[z, 1/z]$ -linear transformation $\mathcal{Z}_n^\circ: \mathbb{K}((z))^n \rightarrow (\mathbb{K}^{\mathbb{Z}})^n$ which extracts coefficients.

Note that the inverse z -transform preserves the time-delay so that $\mathcal{Z}_n^\circ(z \cdot f(z)) = \Delta(\mathcal{Z}_n^\circ(f(z)))$.

For each context $[\ell, r] \in \mathbb{Z}_{\leq}^2$ postcomposing this $\mathcal{Z}_n^\circ$ with the projection $(\mathbb{K}^{\mathbb{Z}})^n \rightarrow \prod_{k=\ell}^r \mathbb{K}^n$, yields the $\mathbb{K}$ -linear map $\zeta_{[\ell, r]}^n: \mathbb{K}((z))^n \rightarrow \prod_{k=\ell}^r \mathbb{K}^n$. This allows us to categorify the inverse z -transform, making $\text{Obs}_{\mathbb{Z}}(\text{AR}_{\mathbb{K}}^{\text{fd}})$ a categorical semantics for uninitialised linear time-dependent systems in the time-domain.

Theorem 6.2. *There is an oplax normal feedback functor $\mathcal{Z}^*: \text{AR}_{\mathbb{K}((z))}^{\text{fd}} \rightarrow \text{Obs}_{\mathbb{Z}}(\text{AR}_{\mathbb{K}}^{\text{fd}})$ given on objects by $n \mapsto n^{\mathbb{Z}}$ and on morphisms $S: n \rightarrow m$ by*

$$\mathcal{Z}^*(S) := \left[\left\{ \left\{ (\zeta_j^n(\mathbf{x}), \zeta_j^m(\mathbf{y})) \mid (\mathbf{x}, \mathbf{y}) \in S \right\} \right\}_{j \in \mathbb{Z}_{\leq}^2} \right];$$

where oplax means that $\mathcal{Z}^*(T \circ S) \subseteq \mathcal{Z}^*(T) \circ \mathcal{Z}^*(S)$ and normal means that $\mathcal{Z}^*(1_n) = 1_{n^{\mathbb{Z}}}$.

See proof on page 50.

By replacing $\mathbb{Z}$ with $\mathbb{N}$ we can therefore interpret $\text{Obs}_{\mathbb{N}}(\text{AR}_{\mathbb{K}}^{\text{fd}})$ as a categorical semantics for *initialised* linear time-dependent systems in the time-domain.

Compactness. Given a field $\mathbb{K}$, let $\text{Aff}_{\mathbb{K}}$ denote the SMC of affine transformations between (possibly infinite) vector spaces and the empty set; moreover, let $\text{AR}_{\mathbb{K}}$ denote the full subcategory of $\text{Rel}(\text{Aff}_{\mathbb{K}})$ whose objects are vector spaces. Using Theorem 5.5, and observing that only the *finite* fields are compact, it follows that:

Proposition 6.3. *Given a finite field $\mathbb{F}_q$, and a cofinal set $\mathcal{J} \subseteq \mathcal{P}_f(\mathbb{I})$ there is a faithful poset-enriched discard functor $\text{Lim}: \text{Obs}_{\mathcal{I}, \mathcal{J}}(\text{AR}_{\mathbb{F}_q}^{\text{fd}}) \rightarrow \text{AR}_{\mathbb{F}_q}$.*

See proof on page 50.

Therefore, observing that $\mathbb{Z}_{\leq}^2 \subseteq \mathcal{P}_f(\mathbb{Z})$ and $\{[0, t] \mid t \in \mathbb{N}\} \subseteq \mathcal{P}_f(\mathbb{N})$ are both cofinal it follows that for finite fields $\mathbb{F}_q$, there are faithful poset-enriched discard functors

$$\text{Obs}_{\mathbb{N}}(\text{AR}_{\mathbb{F}_q}^{\text{fd}}) \rightarrow \text{AR}_{\mathbb{F}_q} \quad \text{and} \quad \text{Obs}_{\mathbb{Z}}(\text{AR}_{\mathbb{F}_q}^{\text{fd}}) \rightarrow \text{AR}_{\mathbb{F}_q},$$

witnessing a “global” behavioural semantics both for initialised and uninitialised linear time-dependent systems. Moreover, the induced oplax functor $\text{Lim} \circ \mathcal{Z}^*$ is given on morphisms by the $\mathbb{F}_q$ -linear extension of $\mathcal{Z}_n^\circ: \mathbb{F}_q((z))^n \rightarrow (\mathbb{F}_q^{\mathbb{Z}})^n$.

Following Willems, the behaviour of an LTI system is a $\mathbb{K}$ -subspace of trajectories cut out by a finite system of constant-coefficient difference equations [64]. Note that the work of Willems was one of the original motivations of Zanasi for developing $\text{AR}_{\mathbb{F}_q((z))}^{\text{fd}}$ as a categorical semantics for LTI systems [65, P. 12].

The name “behaviour” is no coincidence. Over a finite field, which is the setting where the compactness theorem holds, the induced oplax functor $\text{AR}_{\mathbb{F}_q((z))}^{\text{fd}} \rightarrow \text{AR}_{\mathbb{F}_q}$ can be seen to categorify this notion in our setting.

We take the formal definition of behaviours from the book of Oberst et al. [48, § 3.1.1].

Definition 6.4 (Affine behaviour). Let $S \subseteq \mathbb{K}(z)^\ell$ be a finite-dimensional $\mathbb{K}(z)$ -linear subspace. Choose $r \in \mathbb{N}$ and a $\mathbb{K}(z)$ -linear map $A: \mathbb{K}(z)^\ell \rightarrow \mathbb{K}(z)^r$ with $S = \ker A$. Pick some nonzero $d(z) \in \mathbb{K}[z, 1/z]$ clearing denominators and set $B(z) := d(z)A \in \mathbb{K}[z, 1/z]^{r \times \ell}$. Evaluate entrywise to obtain the $\mathbb{K}[z, 1/z]$ -linear map $B^{\rho^\ell}: ((\mathbb{K}^\mathbb{Z})^\ell, \rho^\ell) \rightarrow ((\mathbb{K}^\mathbb{Z})^r, \rho^r)$ given by $B^{\rho^\ell}(u) := B(\Delta)u$. Moreover, define $\text{beh}(S) := \ker(B^{\rho^\ell}) \subseteq (\mathbb{K}^\mathbb{Z})^\ell$.

Let $\gamma: \mathbb{K}(z) \rightarrow \mathbb{K}((z))$ denote the geometric series expansion and define $\mathcal{Z}_\ell^\gamma(-) := \mathcal{Z}_\ell^\circ(\gamma(-))$. We extend beh to $\mathbb{K}(z)$ -affine subspaces $S \subseteq \mathbb{K}(z)^\ell$ by: for $S = \emptyset$ define $\text{beh}(S) := \emptyset$; otherwise choosing some $\mathbf{a} \in S$, with $R := S - \mathbf{a}$: $\text{beh}(S) := \text{beh}(R) + \mathcal{Z}_\ell^\gamma(\mathbf{a}) \subseteq (\mathbb{K}^\mathbb{Z})^\ell$.

Note that Oberst et al. [48, § 3.1.1] define this only over linear subspaces, but the extension to affine relations as above is immediately obvious. Indeed the oplax functor $\text{Lim} \circ \mathcal{Z}^*$ can be seen to categorify this notion of behaviour, albeit only after restricting to the trajectories which are rational.

Theorem 6.5. Let $\mathbb{F}_q$ be a finite field and $S \subseteq \mathbb{F}_q(z)^\ell$ be a finite-dimensional $\mathbb{F}_q(z)$ -affine subspace. Then $(\text{Lim} \circ \mathcal{Z}^*)(S) = \text{beh}(S) \cap \mathcal{Z}_\ell^\gamma(\mathbb{F}_q(z)^\ell)$.

See proof on page 51.

6.3 Signal flow graphs

Linear *signal flow graphs* are a syntax for linear time-invariant systems; see for example, the seminal work Shannon [60], and later of Mason [42, 43]. In this section, we discuss how the z -domain and time-domain semantics for uninitialised LTI systems produce different equational theories for signal flow graphs; followed briefly by a discussion concerning the syntactic implications of our initialised time-domain semantics.

First we recall the following equational theory for linear *time-independent* systems:

Theorem 6.6 (Bonchi et al. [7]). Given a field $\mathbb{K}$, the SMC $\text{AR}_\mathbb{K}^{\text{fd}}$ is presented by the following generators, for all $a \in \mathbb{K}$:

$$\begin{aligned} \left[\begin{array}{c} \text{---} \bullet \text{---} \\ \text{---} \bullet \text{---} \end{array} \right] &:= \{(\mathbf{v}, \mathbf{w}) \mid \forall j, k: v_j = w_k\}, \\ \left[\begin{array}{c} a \\ \text{---} \circ \text{---} \\ \text{---} \bullet \text{---} \end{array} \right] &:= \{(\mathbf{v}, \mathbf{w}) \mid a + \Sigma v_j = \Sigma w_j\}, \\ \left[\begin{array}{c} \text{---} \text{---} a \text{---} \end{array} \right] &:= \{(b, ab) \mid \forall b \in \mathbb{K}\}. \end{aligned}$$

modulo the equations given in Appendix A.

These generators impose linear constraints on systems; where the label a on the white node is interpreted as a signal propagating through this linear system.

The question of finding an equational theory for LTI systems, and axiomatizing signal flow graphs, can therefore be reduced to the question of how to add a notion of *time* to the equational theory of $\text{AR}_\mathbb{K}^{\text{fd}}$.

The simplest way to answer this question is to freely add a discrete-time delay generator which commutes only with the constraints, which by Proposition 3.5 is captured by the feedback category $\text{St}_{\text{LR}_\mathbb{K}^{\text{fd}}}(\text{AR}_\mathbb{K}^{\text{fd}})$. Concretely, this is presented by adding the generator $\boxed{\overline{\partial}}$ to the equational theory of $\text{AR}_\mathbb{K}^{\text{fd}}$ modulo the equations:

$$\begin{array}{c} \boxed{\overline{\partial}} \bullet \\ \text{---} \bullet \text{---} \\ \text{---} \bullet \text{---} \end{array} = \begin{array}{c} \text{---} \bullet \text{---} \\ \text{---} \bullet \text{---} \\ \boxed{\overline{\partial}} \bullet \end{array}, \quad \begin{array}{c} \boxed{\overline{\partial}} \circ \\ \text{---} \bullet \text{---} \\ \text{---} \bullet \text{---} \end{array} = \begin{array}{c} \text{---} \bullet \text{---} \\ \text{---} \bullet \text{---} \\ \text{---} \circ \text{---} \end{array} \boxed{\overline{\partial}}, \quad \boxed{\overline{\partial}} \text{---} a = \text{---} a \text{---} \boxed{\overline{\partial}}.$$

Where the white spider with no label denotes a white spider with label 0.

Let $\mathbb{K}(z)$ denote the field of fractions of the polynomial ring $\mathbb{K}[z]$ (i.e. the rational functions over $\mathbb{K}$). Just as $\text{AR}_{\mathbb{K}(z)}^{\text{fd}}$ is a feedback category, so is $\text{AR}_{\mathbb{K}(z)}^{\text{fd}}$. Moreover,

Lemma 6.7. *There is a full feedback functor $\text{St}_{\text{LR}_{\mathbb{K}}}^{\text{fd}}(\text{AR}_{\mathbb{K}}^{\text{fd}}) \rightarrow \text{AR}_{\mathbb{K}(z)}^{\text{fd}}$, given by identifying the delays and extending scalars along $\mathbb{K}(z)/\mathbb{K}$, and extending scalars along the geometric series embedding $\mathbb{K}((z))/\mathbb{K}(z)$ induces a feedback functor $\text{Ext}: \text{St}_{\text{LR}_{\mathbb{K}}}^{\text{fd}}(\text{AR}_{\mathbb{K}}^{\text{fd}}) \rightarrow \text{AR}_{\mathbb{K}((z))}^{\text{fd}}$.*

See proof on page 51.

Identifying the respective delay maps, so that $\boxed{\partial^n} = \boxed{z^n}$, the equational theory of $\text{AR}_{\mathbb{K}(z)}^{\text{fd}}$ is given by quotienting the equational theory of $\text{St}_{\text{LR}_{\mathbb{K}}}^{\text{fd}}(\text{AR}_{\mathbb{K}}^{\text{fd}})$ by the following equations for all nonzero $\mathbf{a} \in \mathbb{K}[z]$:

$$\begin{array}{c} \boxed{\begin{array}{c} \begin{array}{c} \xrightarrow{a_0} \partial \xrightarrow{a_1} \partial \xrightarrow{a_n} \partial^n \end{array} \\ \begin{array}{c} \xleftarrow{a_0} \partial \xleftarrow{a_1} \partial \xleftarrow{a_n} \partial^n \end{array} \end{array}} = \boxed{}, \end{array} \quad (6)$$

$$\begin{array}{c} \boxed{\begin{array}{c} \begin{array}{c} \xleftarrow{a_0} \partial \xleftarrow{a_1} \partial \xleftarrow{a_n} \partial^n \end{array} \\ \begin{array}{c} \xrightarrow{a_0} \partial \xrightarrow{a_1} \partial \xrightarrow{a_n} \partial^n \end{array} \end{array}} = \boxed{}. \end{array} \quad (7)$$

Where the left facing arrows denote the transposes of the right facing arrows. These equations mean that the relation which multiplies its inputs by a nonzero polynomial is inverse to its converse relation.

We can also interpret the “free” syntax for LTI systems in our categorical semantics for the time-domain so that the affine translations are interpreted as impulses at time step 0 and the linear subspaces are invariant with respect to time:

Lemma 6.8. *Let $\eta: \text{AR}_{\mathbb{K}}^{\text{fd}} \rightarrow \text{St}_{\text{LR}_{\mathbb{K}}}^{\text{fd}}(\text{AR}_{\mathbb{K}}^{\text{fd}})$ be the canonical functor. Define $J: \text{AR}_{\mathbb{K}}^{\text{fd}} \rightarrow \text{Obs}_{\mathbb{Z}}(\text{AR}_{\mathbb{K}}^{\text{fd}})$ by $J(V)_t := V$ for all $t \in \mathbb{Z}$, and on morphisms by $J(\emptyset) := [\{\emptyset\}_{j \in \mathbb{Z}_{\leq}^2}]$ and*

$$J(A + \mathbf{a}) := \left[\left\{ \bigoplus_{t=\ell}^r A + \mathbf{a} \quad \text{if } t = 0, \right. \right. \\ \left. \left. A \quad \text{if } t \neq 0 \right\}_{[\ell, r] \in \mathbb{Z}_{\leq}^2} \right].$$

There is a unique feedback functor $\text{Unroll}: \text{St}_{\text{LR}_{\mathbb{K}}}^{\text{fd}}(\text{AR}_{\mathbb{K}}^{\text{fd}}) \rightarrow \text{Obs}_{\mathbb{Z}}(\text{AR}_{\mathbb{K}}^{\text{fd}})$ satisfying $\text{Unroll} \circ \eta = J$.

See proof on page 52. It is natural to ask if invertibility equations Equations (6) and (7) also hold in $\text{Unroll}(\text{St}_{\text{LR}_{\mathbb{K}}}^{\text{fd}}(\text{AR}_{\mathbb{K}}^{\text{fd}}))$; this is only partly true:

Proposition 6.9. *Given any polynomial $z + a \in \mathbb{K}[z]$ for $a \neq 0$:*

$$\text{Unroll} \left(\boxed{\begin{array}{c} \begin{array}{c} \xrightarrow{a_0} \partial \xrightarrow{a_1} \partial \xrightarrow{a_n} \partial^n \end{array} \\ \begin{array}{c} \xleftarrow{a_0} \partial \xleftarrow{a_1} \partial \xleftarrow{a_n} \partial^n \end{array} \end{array}} \right) = \text{Unroll} \left(\boxed{} \right) \subsetneq \text{Unroll} \left(\boxed{\begin{array}{c} \begin{array}{c} \xrightarrow{a_0} \partial \xrightarrow{a_1} \partial \xrightarrow{a_n} \partial^n \end{array} \\ \begin{array}{c} \xleftarrow{a_0} \partial \xleftarrow{a_1} \partial \xleftarrow{a_n} \partial^n \end{array} \end{array}} \right)$$

See proof on page 53.

In particular, this means, at least that for algebraically closed $\mathbb{K}$, all nonzero polynomials in $\mathbb{K}[z]$ have left inverse in $\text{Obs}_{\mathbb{Z}}(\text{AR}_{\mathbb{K}}^{\text{fd}})$. However, only polynomials of the form az^n for nonzero $a \in \mathbb{K}$ and $n \in \mathbb{N}$ have right inverses, where for all other nonzero polynomials the right inverse equation holds opaxly. Nevertheless, in a *relaxed* sense, both semantics are compatible:

Proposition 6.10. *For any map f in $\text{St}_{\text{LR}_{\mathbb{K}}}^{\text{fd}}(\text{AR}_{\mathbb{K}}^{\text{fd}})$, we have that $\mathcal{Z}^*(\text{Ext}(f)) \subseteq \text{Unroll}(f)$.*

See proof on page 54.

The obstruction to invertibility is a consequence of the fact that the solutions to difference equations are determined only up to initial conditions. This can be resolved by working in $\text{Obs}_{\mathbb{N}}(\text{AR}_{\mathbb{K}}^{\text{fd}})$ and initialising the relevant monotone sequences to begin with the zero subspace.

7 Conclusion

Given any discard bicategory $\mathcal{C}$ and an upward-directed set of finite contexts $\mathcal{J} \subseteq \mathcal{P}_f(\mathcal{I})$, we introduced the discard bicategory of observational behaviours $\text{Obs}_{\mathcal{I}, \mathcal{J}}(\mathcal{C})$, in which two processes have the same behaviour when every finite observation of one can be verified by an observation of the other (Definition 4.8 and Theorem 4.9). We showed that our construction gives a functorial interpretation of stateful morphism sequences (Proposition 4.10), propagates inconsistency globally (Theorem 4.14), and carries a natural time delay (Proposition 4.15 and Theorem 4.18). Moreover, in the relational setting, we proved a compactness theorem (Theorem 5.5) gluing each compatible family of finite observations into a single infinite relation, yielding a compositional time-domain semantics for signal flow graphs that recovers Willems’ behaviours over finite fields (Theorem 6.5).

Several questions remain open. A natural follow up would be generalising our compactness theorem from relations to probabilistic and quantum systems. In the total probabilistic setting, this should follow from the existence of Kolmogorov products [28, Def. 4.1]. For partial probabilistic systems, this would yield a generalisation of the Kolmogorov extension theorem to substochastic processes. In the quantum setting, Kretschmann and Werner interpret causal quantum channels with memory as infinite channels between quasi-local algebras [40]. It would be interesting to recover this result and generalise it to post-selected quantum processes via a quantised version of our compactness theorem.

A second direction for future work is the study of graphical languages for behaviours. We have shown that several intuitive axioms hold, but completeness is still an open problem. For uninitialised linear time-invariant behaviours, we expect that some of the axioms of Bonchi et al. [11] need to be relaxed to inequations. In the quantum setting, our framework gives a semantics for the ZX calculus with a *natural* delay. A complete axiomatisation in this setting would complement previous approaches to quantum dataflow programming [14, 20].

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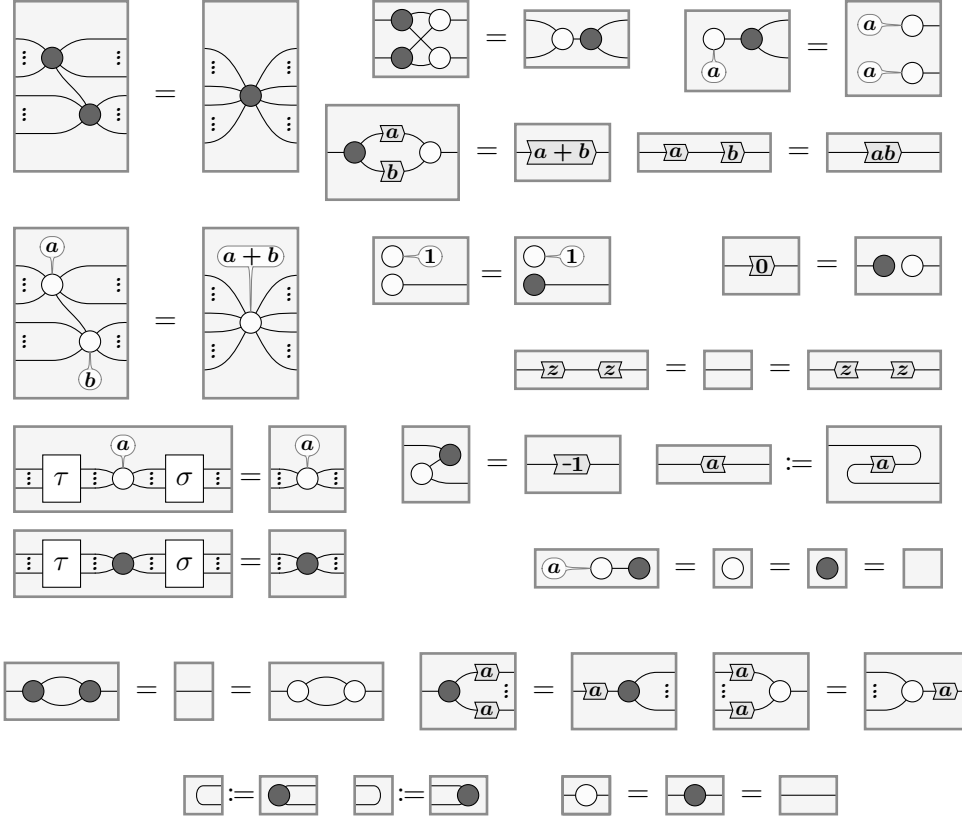
A Graphical affine algebra

The following result is due to Bonchi et al. [7]; however, for convenience, we use the “spider” notation, which is given in [12]:

Theorem 6.6. *Given a field $\mathbb{K}$, the SMC $\text{AR}_{\mathbb{K}}^{\text{fd}}$ is presented by the generators, for all $a \in \mathbb{K}$:*

$$\begin{aligned}
 \left[\begin{array}{c} \text{---} \bullet \text{---} \\ \vdots \\ \text{---} \end{array} \right] &:= \{(\mathbf{v}, \mathbf{w}) \mid \forall j, k : v_j = w_k\}, \\
 \left[\begin{array}{c} a \\ \text{---} \circ \text{---} \\ \vdots \\ \text{---} \end{array} \right] &:= \{(\mathbf{v}, \mathbf{w}) \mid a + \Sigma v_j = \Sigma w_j\}, \\
 \left[\begin{array}{c} \text{---} \overline{a} \text{---} \end{array} \right] &:= \{(b, ab) \mid \forall b \in \mathbb{K}\},
 \end{aligned}$$

modulo the following equations, for all $a, b \in \mathbb{K}$ and nonzero $c \in \mathbb{K}$:



By reflecting the bicategory structure of $\mathbf{AR}_{\mathbb{K}}^{\text{fd}}$ into its diagrammatic presentation, we find that the following equation holds, which we shall use:

Lemma A.1.



This lemma can be checked semantically; moreover, it follows from the bicategorical axiomatisation of $\text{Rel}(\mathbf{FVect})$ of Paixão et al. [51].

B Rings of polynomials

In this Appendix we recall some basic algebraic notions involving rings of polynomials. In order to establish notation, we start with the following definition:

Definition B.1. Given a field $\mathbb{K}$, the **polynomial ring** $\mathbb{K}[z]$ is a principal ideal domain (PID).

This can be extended to a larger PID by localising z

Definition B.2. The PID $\mathbb{K}[z, 1/z]$ given by adding an inverse to the polynomial indeterminate z in $\mathbb{K}[z]$ is called the PID of **Laurent polynomials** over $\mathbb{K}$ in z .

Lemma B.3. *There is an embedding of PIDs*

$$\mathbb{K}[z] \hookrightarrow \mathbb{K}[z, 1/z].$$

The polynomial ring can also be extended to a larger PID:

Definition B.4. The PID of **formal power series** $\mathbb{K}[[z]]$ has elements given by infinite sequences in $\mathbb{K}$, denoted

$$f(z) := \sum_{j=0}^{\infty} f_j \cdot z^j$$

The additive, Abelian group structure is given pointwise as for polynomials:

$$f(z) + g(z) := \sum_{j=0}^{\infty} (f_j + g_j) \cdot z^j,$$

$$-f(z) := \sum_{j=0}^{\infty} (-f_j) \cdot z^j, \quad \text{and} \quad 0 := 0 \cdot z^0.$$

The multiplicative monoid structure is given by:

$$f(z)g(z) := \sum_{j=0}^{\infty} \sum_{k=0}^{\infty} (f_j \cdot g_k) \cdot z^{j+k}, \quad \text{and} \quad 1 := 1 \cdot z^0$$

Lemma B.5. *There is an embedding of PIDs $\mathbb{K}[z] \hookrightarrow \mathbb{K}[[z]]$.*

The polynomial ring can be extended to a field:

Definition B.6. The field of fractions of polynomials or Laurent polynomials in z , $\mathbb{K}(z)$ is called the field of **rational functions** over $\mathbb{K}$ in z .

Lemma B.7. *There is an embedding of PIDs $\mathbb{K}[z, 1/z] \hookrightarrow \mathbb{K}(z)$.*

By taking the z -adic completion, this can be extended to an even bigger field, which is a bit more involved to define:

Definition B.8. The field of **formal Laurent series** $\mathbb{K}((z))$ is the z adic completion of $\mathbb{K}[z]$, $\mathbb{K}[z, 1/z]$, $\mathbb{K}[[z]]$ or $\mathbb{K}(z)$, which is more complicated to describe concretely. $\mathbb{K}((z))$ has elements which are biinfinite sequences $f(z) \in \mathbb{K}^{\mathbb{Z}}$ of elements in $\mathbb{K}$, bounded below by some $m \in \mathbb{Z}$, denoted:

$$f(z) := \sum_{j=m}^{\infty} f_j \cdot z^j$$

The largest j such that for all $k \in \mathbb{N}$ $f_{j-k-1} = 0$, if it exists, is called the **order** of $f(z)$, denoted $\text{ord}(f(z))$; otherwise, set $\text{ord}(f(z)) := 0$.

The additive group structure and the multiplicative monoid structure are essentially the same as for formal power series.

The multiplicative inverse is a bit more complicated. Take some nonzero $f(z)$. Then we can factor out the lowest order term:

$$f(z) = f_{\text{ord}(f(z))} z^{\text{ord}(f(z))} \left(1 + \sum_{j=1}^{\infty} \frac{f_{j+\text{ord}(f(z))}}{f_{\text{ord}(f(z))}} z^j \right)$$

The inverse of $1 + \sum_{j=1}^{\infty} \frac{f_{j+\text{ord}(f(z))}}{f_{\text{ord}(f(z))}} z^j$ is therefore a geometric series $g(z) := \sum_{j=0}^{\infty} g_j \cdot z^j$ where the coefficients are given by the recursive formula:

$$g_0 = 1, \quad \text{and for all } n \geq 0, \quad g_{n+1} = - \sum_{k=1}^{n+1} \frac{f_{k+\text{ord}(f)}}{f_{\text{ord}(f)}} g_{n+1-k}$$

This gives us the inverse $f^{-1}(z) := \frac{z^{-\text{ord}(f(z))}}{f_{\text{ord}(f(z))}} g(z)$.

Lemma B.9. *There is a field extension $\gamma : \mathbb{K}(z) \hookrightarrow \mathbb{K}((z))$, called the **geometric series embedding**, sending $f(z)/g(z) \in \mathbb{K}(z)$ to $f(z) \cdot g^{-1}(z) \in \mathbb{K}((z))$.*

Putting this all together we have the lattice of PIDs:

$$\begin{array}{ccccccc} \mathbb{K} & \twoheadrightarrow & \mathbb{K}[z] & \twoheadrightarrow & \mathbb{K}[z, 1/z] & \twoheadrightarrow & \mathbb{K}(z) \\ & & \downarrow & & & & \downarrow \\ & & \mathbb{K}[[z]] & \longrightarrow & & & \mathbb{K}((z)) \end{array}$$

C Proofs of Section 2

Lemma 2.7. *If a discard bicategory has zero morphisms then zero is the bottom element in every hom-set.*

Proof of Lemma 2.7. By definition of discard bicategory we have $\top_I \circ 0_{I,I} \subseteq \top_I$, then using the definition of zero morphism, for any $f : X \rightarrow Y$

$$0_{X,Y} = f \otimes 0_{I,I} = f \otimes (\top_I \circ 0_{I,I}) \subseteq f \otimes \top_I = f. \quad \square$$

Lemma 2.9. *If $\mathcal{C}$ is a compact-closed discard bicategory, then for any object X of $\mathcal{C}$, denoting $\perp_X := (1_X \otimes \top_{X^*}) \circ \cup_X$, we have that $\top_X \circ \perp_X \subseteq 1_I$ and $1_X \subseteq \perp_X \circ \top_X$.*

Proof of Lemma 2.9. The first inequation follows directly by lax naturality of $\top_X$. The second inequation follows by the snake equation and lax naturality:

$$1_X = (1_X \otimes \cap_X) \circ (\cup_X \otimes 1_X) \subseteq (1_X \otimes \top_X \otimes \top_X) \circ (\cup_X \otimes 1_X) = \perp_X \circ \top_X. \quad \square$$

Lemma 2.14. *The Löwner order is a symmetric monoidal poset enrichment for CPTNI and is a discrete poset enrichment for CPTP.*

Proof of Lemma 2.14. Selinger remarked that the Löwner order is a poset enrichment for CPTNI [59, Lem. 6.4], where DCPO-enrichment is formally proved by Cho [16, Thm. 5.5].

To see that the Löwner order between completely positive trace-preserving maps is discrete, suppose $\Phi \preceq \Psi$, then $\Psi - \Phi$ is completely positive. Then for any normalised state $a \in \mathcal{B}(\mathcal{H})$, $\Psi(a) - \Phi(a)$ is positive. Moreover since these maps preserve the trace $\text{Tr}(\Psi(a)) = \text{Tr}(\Phi(a)) = \text{Tr}(a) = 1$. Therefore $\text{Tr}(\Psi(a) - \Phi(a)) = 0$ by linearity of the trace. The only CP map that sends all normalised states to states of trace 0 is the zero map, therefore $\Psi = \Phi$. $\square$

Proposition 2.17. *The Löwner and purification orders coincide in CPTP and CPTNI but differ in CPM.*

Proof of Proposition 2.17. We prove each claim separately.

First, the orders coincide in CPTP by the uniqueness of the discrete partial order structure.

Next we show that the orders coincide in CPTNI. First, suppose that $\Phi \preceq \Psi$, then $\Xi = \Psi - \Phi$ is completely positive. Consider the qubit space $\mathbb{C}^2$ with an orthonormal basis $|0\rangle, |1\rangle$. Define $g_0 : \mathcal{B}(\mathcal{H}) \rightarrow \mathcal{B}(\mathcal{K} \otimes \mathbb{C}^2)$ and $f_0 : \mathcal{B}(\mathbb{C}^2) \rightarrow \mathbb{C}$ by:

$$g_0 = \Phi \otimes |0\rangle\langle 0| + \Xi \otimes |1\rangle\langle 1| \quad \text{and} \quad f_0(a) = \langle 0| a |0\rangle.$$

for any $a \in \mathcal{B}(\mathbb{C}^2)$. It is easy to check that these maps are trace non-increasing and that $\Phi = (1 \otimes f_0) \circ g_0$ and $\Psi = (1 \otimes \text{Tr}) \circ g_0$. Therefore $\Phi \subseteq \Psi$ in the purification order. Now for the other direction, suppose that $\Phi \subseteq \Psi$. Then there exists trace non-increasing maps g_0, f_0 such that $\Phi = (1 \otimes f_0) \circ g_0$ and $\Psi = (1 \otimes \text{Tr}) \circ g_0$. Therefore, $\Psi - \Phi = (1 \otimes (\text{Tr} - f_0)) \circ g_0$; moreover, since f_0 is trace non-increasing $\text{Tr} - f_0$ is completely positive. Since the composition of completely positive maps is completely positive we obtain $\Psi \subseteq \Phi$.

To see that these two orders do not coincide we can simply look at scalars. Note that the scalars in CPM are precisely the non-negative real numbers $\mathbb{R}^{\geq 0}$. For any pair of scalars $a, b \in \mathbb{R}^{\geq 0}$, taking $g_0 = b$ and $f_0 = \frac{a}{b}$ we have that $a \subseteq b$ in the purification order. However if $b \leq a$ as real numbers, the inequality does not hold in the Löwner order. $\square$

Proposition 2.18. *When equipped with the purification order, CPM, CPTP and CPTNI are discard bicategories.*

Proof of Proposition 2.18. Lax naturality of the discard is immediate from Definition 2.15: for any $f : X \rightarrow Y$, taking $g_0 = f$ and $f_0 = \top_Y$ exhibits $\top_Y \circ f \subseteq \top_X$. The remaining preorder-enrichment axioms hold by [14, Lem. 6]. For CPTP and CPTNI the purification order coincides with the Löwner order by Proposition 2.17. $\square$

Proposition 2.19. *In any partial Markov category, the purification order and the conditional order coincide.*

Proof of Proposition 2.19. Denote by $\subseteq_P$ the purification order and by $\subseteq_C$ the conditional order. Suppose that $f \subseteq_P g$, then there exists f_0 and g_0 such that $f = (f_0 \otimes 1_Y) \circ g_0$ and $g = (\top_S \otimes 1_Y) \circ g_0$. Then, since $f_0 \subseteq_C \top_S$ [25, Prop. 3.5] and $\subseteq_C$ respects the monoidal structure, we have:

$$f = (f_0 \otimes 1_Y) \circ g_0 \subseteq_C (\top_S \otimes 1_Y) \circ g_0 = g$$

For the other direction, suppose that $f \subseteq_C g$, then there exists $r : Y \otimes X \rightarrow I$ such that $f = (1_Y \otimes r) \circ (\Lambda_Y \otimes 1_X) \circ (g \otimes 1_X) \circ \Lambda_X$. Let $E = Y \otimes X$, $f_0 = r : Y \otimes X \rightarrow I$ and $g_0 = (\Lambda_Y \otimes 1_X) \circ (g \otimes 1_X) \circ \Lambda_X$. Then $f = (f_0 \otimes 1_Y) \circ g_0$ and $g = (\top_S \otimes 1_Y) \circ g_0$, therefore $f \subseteq_P g$. $\square$

D Proofs of Section 3

Lemma D.1. *Take two symmetric monoidal functors $\Gamma : \mathcal{D} \rightarrow \mathcal{C}$ and $\Delta : \mathcal{D} \rightarrow \mathcal{D}$, and let $\eta : \mathcal{C} \rightarrow \text{St}_{\mathcal{D}}(\mathcal{C}, \Delta)$ be the canonical symmetric monoidal functor. Then $\text{St}_{\mathcal{D}}(\mathcal{C}, \Delta)$ carries a canonical feedback structure given by $\Delta : \mathcal{D} \rightarrow \mathcal{D}$,*

$$\hat{\Gamma} := \eta \circ \Gamma : \mathcal{D} \rightarrow \text{St}_{\mathcal{D}}(\mathcal{C}, \Delta), \quad \text{and} \quad \text{fbk}_S([M, f]) := [M \otimes S, f].$$

Proof. We need to show that the feedback operator:

- i. is well-typed;
- ii. is well-defined on equivalence classes;
- iii. satisfies the feedback category axioms.

Take objects $S \in \text{ob}(\mathcal{D})$ and $X, Y \in \text{ob}(\mathcal{C})$.

First, because η is the identity on objects, we have that $\hat{\Gamma}(S) = \Gamma(S)$ and $\hat{\Gamma}(\Delta S) = \Gamma(\Delta S)$.

Next, take a representative $[M, f] \in \text{St}_{\mathcal{D}}(\mathcal{C}, \Delta)(\hat{\Gamma}(\Delta S) \otimes X, \hat{\Gamma}(S) \otimes Y)$, where

$$M \in \text{ob}(\mathcal{D}) \quad \text{and} \quad f \in \mathcal{C}(\Gamma(\Delta M) \otimes \Gamma(\Delta S) \otimes X, \Gamma(M) \otimes \Gamma(S) \otimes Y).$$

Define the feedback operator on the representatives $[M, f]$ by $\text{fbk}_S([M, f]) := [M \otimes S, f]$.

i. Feedback operator is well-typed. Because Δ and Γ are strict symmetric monoidal, we have that $\Gamma(\Delta(M \otimes S)) = \Gamma(\Delta M) \otimes \Gamma(\Delta S)$ and $\Gamma(M \otimes S) = \Gamma(M) \otimes \Gamma(S)$, so that f is also an element of $\mathcal{C}(\Gamma(\Delta(M \otimes S)) \otimes X, \Gamma(M \otimes S) \otimes Y)$, and thus, $[M \otimes S, f] \in \text{St}_{\mathcal{D}}(\mathcal{C}, \Delta)(X, Y)$. Therefore, fbk_S is well-typed.

ii. Well-definedness of feedback operator. Fix $t \in \mathcal{D}(M, M')$ and

$$f \in \mathcal{C}(\Gamma(\Delta M) \otimes \Gamma(\Delta S) \otimes X, \Gamma(M') \otimes \Gamma(S) \otimes Y).$$

Moreover, write $A := \widehat{\Gamma}(\Delta S) \otimes X$ and $B := \widehat{\Gamma}(S) \otimes Y$.

Observe that the feedback operation in $\text{St}_{\mathcal{D}}(\mathcal{C}, \Delta)$ satisfies the sliding equation on representatives of the equivalence class:

$$\begin{aligned} \text{fbk}_{M'}(M' \otimes S, (\widehat{\Gamma}(t) \otimes 1_B) \circ f) &= (S, (\widehat{\Gamma}(t) \otimes 1_B) \circ f) = (S, f \circ (\widehat{\Gamma}(\Delta t) \otimes 1_A)) \\ &= \text{fbk}_M(M \otimes S, f \circ (\widehat{\Gamma}(\Delta t) \otimes 1_A)). \end{aligned}$$

Therefore fbk_M is well-defined.

iii. Feedback category axioms. The other feedback category axioms follow from the evident diagram chases. $\square$

This is the *free* feedback category, in the following sense:

Proposition D.2. *Take two symmetric monoidal functors $\Gamma: \mathcal{D} \rightarrow \mathcal{C}$ and $\Delta: \mathcal{D} \rightarrow \mathcal{D}$. Given a feedback category $(\mathcal{C}', \mathcal{D}', \Gamma', \Delta', \text{fbk}')$ and two symmetric monoidal functors $F: \mathcal{C} \rightarrow \mathcal{C}'$ and $F|_{\mathcal{D}}: \mathcal{D} \rightarrow \mathcal{D}'$ satisfying $\Gamma' \circ F|_{\mathcal{D}} = F \circ \Gamma$ and $\Delta' \circ F|_{\mathcal{D}} = F|_{\mathcal{D}} \circ \Delta$, there is a unique feedback functor $(F^\sharp, F|_{\mathcal{D}}): \text{St}_{\mathcal{D}}(\mathcal{C}, \Delta) \rightarrow \mathcal{C}'$ making the diagram commute:*

$$\begin{array}{ccc} \mathcal{C} & \xrightarrow{\eta} & \text{St}_{\mathcal{D}}(\mathcal{C}, \Delta) \\ & \searrow F & \downarrow \check{\gamma}^{(F^\sharp, F|_{\mathcal{D}})} \\ & & \mathcal{C}' \end{array}$$

Proof. Define $F^\sharp: \text{St}_{\mathcal{D}}(\mathcal{C}, \Delta) \rightarrow \mathcal{C}'$ on objects by $F^\sharp(X) := F(X)$. Given some $[S, f] \in \text{St}_{\mathcal{D}}(\mathcal{C}, \Delta)(X, Y)$, define $F^\sharp([S, f]) := \text{fbk}'_{F|_{\mathcal{D}}(S)}(F(f)) \in \mathcal{C}'(F(X), F(Y))$.

We need to show that $F^\sharp$:

- i. is well-typed;
- ii. is well-defined on equivalence classes;
- iii. is functorial and symmetric monoidal;
- iv. satisfies $F^\sharp \circ \eta = F$;
- v. is compatible with Γ and Δ ;
- vi. preserves the feedback operator;
- vii. is the unique such functor.

i. Well-typedness. Since $\Gamma' \circ F|_{\mathcal{D}} = F \circ \Gamma$ and $\Delta' \circ F|_{\mathcal{D}} = F|_{\mathcal{D}} \circ \Delta$, we have that

$$F(\Gamma(S)) = \Gamma'(F|_{\mathcal{D}}(S)), \quad \text{and} \quad F(\Gamma(\Delta S)) = \Gamma'(\Delta'(F|_{\mathcal{D}}(S))).$$

ii. Well-definedness. Take morphisms $s \in \mathcal{D}(S, S')$ and $f \in \mathcal{C}(\Gamma(\Delta S') \otimes X, \Gamma(S) \otimes Y)$. Observe that in $\text{St}_{\mathcal{D}}(\mathcal{C}, \Delta)$ we have $[S', (\Gamma(s) \otimes 1_Y) \circ f] = [S, f \circ (\Gamma(\Delta s) \otimes 1_X)]$. On the other hand in $\mathcal{C}'$, we have

$$\begin{aligned}
& \text{fbk}'_{F|_{\mathcal{D}(S')}}(F((\Gamma(s) \otimes 1_Y) \circ f)) \\
&= \text{fbk}'_{F|_{\mathcal{D}(S')}}((F(\Gamma(s)) \otimes 1_{F(Y)}) \circ F(f)) \\
&= \text{fbk}'_{F|_{\mathcal{D}(S')}}((\Gamma'(F|_{\mathcal{D}(s)}) \otimes 1_{F(Y)}) \circ F(f)) \\
&= \text{fbk}'_{F|_{\mathcal{D}(S)}}(F(f) \circ (\Gamma'(\Delta'(F|_{\mathcal{D}(s)})) \otimes 1_{F(X)})) \\
&= \text{fbk}'_{F|_{\mathcal{D}(S)}}(F(f) \circ (\Gamma'(F|_{\mathcal{D}(\Delta s)}) \otimes 1_{F(X)})) \\
&= \text{fbk}'_{F|_{\mathcal{D}(S)}}(F(f) \circ (F(\Gamma(\Delta s)) \otimes 1_{F(X)})) \\
&= \text{fbk}'_{F|_{\mathcal{D}(S)}}(F(f) \circ F(\Gamma(\Delta s) \otimes 1_X)) \\
&= \text{fbk}'_{F|_{\mathcal{D}(S)}}(F(f \circ (\Gamma(\Delta s) \otimes 1_X))).
\end{aligned}$$

Hence $F^\sharp$ is well-defined.

iii. Functoriality and symmetric monoidality. Follow from the evident diagram chases.

iv. Triangle commutes. Given $f \in \mathcal{C}(X, Y)$ we have

$$F^\sharp(\eta(f)) = F^\sharp([I, f]) = \text{fbk}'_{F|_{\mathcal{D}(I)}}(F(f)) = \text{fbk}'_I(F(f)) = F(f).$$

v. Compatibility with Γ and Δ . By assumption we have $\Delta' \circ F|_{\mathcal{D}} = F|_{\mathcal{D}} \circ \Delta$; moreover,

$$F^\sharp \circ \hat{\Gamma} = F^\sharp \circ \eta \circ \Gamma = F \circ \Gamma = \Gamma' \circ F|_{\mathcal{D}}.$$

vi. Preservation of feedback. Let $S \in \text{ob}(\mathcal{D})$ and take a representative

$$[M, h] \in \text{St}_{\mathcal{D}}(\mathcal{C}, \Delta)(\hat{\Gamma}(\Delta S) \otimes X, \hat{\Gamma}(S) \otimes Y).$$

Then

$$\begin{aligned}
F^\sharp(\text{fbk}_S([M, h])) &= F^\sharp([S \otimes M, h]) = \text{fbk}'_{F|_{\mathcal{D}(S \otimes M)}}(F(h)) = \text{fbk}'_{F|_{\mathcal{D}(S)}}(\text{fbk}'_{F|_{\mathcal{D}(M)}}(F(h))) \\
&= \text{fbk}'_{F|_{\mathcal{D}(S)}}(F^\sharp([M, h])).
\end{aligned}$$

vii. Uniqueness. Take another feedback functor $(H, F|_{\mathcal{D}}): \text{St}_{\mathcal{D}}(\mathcal{C}, \Delta) \rightarrow \mathcal{C}'$, such that $H \circ \eta = F$. Moreover take any representative $(S, f) \in \text{St}_{\mathcal{D}}(\mathcal{C}, \Delta)(\hat{\Gamma}(\Delta S) \otimes X, \hat{\Gamma}(S) \otimes Y)$. Therefore

$$H([S, f]) = H(\text{fbk}_S([I, f])) = H(\text{fbk}_S(\eta f)) = \text{fbk}'_{F|_{\mathcal{D}(S)}}(H(\eta f)) = \text{fbk}'_{F|_{\mathcal{D}(S)}}(F(f)) = F^\sharp([S, f]).$$

Hence $H = F^\sharp$, so $(F^\sharp, F|_{\mathcal{D}})$ is unique. $\square$

Proposition 3.5. Take a compact-closed category $\mathcal{C}$, a faithful symmetric monoidal functor $\Gamma: \mathcal{D} \rightarrow \mathcal{C}$ such that for all $X \in \text{ob}(\mathcal{D})$, $\Gamma^{-1}(\delta_X) \neq \emptyset$, and a symmetric monoidal endofunctor $\Delta: \mathcal{D} \rightarrow \mathcal{D}$. The existence of a feedback operator is equivalent to a monoidal natural transformation $\delta: 1_{\mathcal{D}} \Rightarrow \Delta$, called the **delay natural transformation**. Moreover, such a natural transformation is necessarily a natural isomorphism.

Proof of Proposition 3.5. We need to prove that:

- i. the existence of a feedback operator implies a monoidal natural transformation $\delta: 1_{\mathcal{D}} \Rightarrow \Delta$;
- ii. a monoidal natural transformation $\delta: 1_{\mathcal{D}} \Rightarrow \Delta$ implies a feedback operator;
- iii. the delay δ is necessarily a natural isomorphism.

i. Feedback operator implies delay. Assume that for all objects $S \in \mathcal{D}$, we have a feedback operator $\text{fbk}_S: \mathcal{C}(\Gamma(\Delta S) \otimes X, \Gamma(S) \otimes Y) \rightarrow \mathcal{C}(X, Y)$.

Using the fact that $\Gamma: \mathcal{D} \rightarrow \mathcal{C}$ is a faithful symmetric monoidal functor, for all $S \in \text{ob}(\mathcal{D})$, define $\delta_S \in \mathcal{D}(S, \Delta S)$ by $\delta_S := \Gamma^{-1}(\text{fbk}_S(\sigma_{\Gamma(\Delta S), \Gamma(S)})) \in \mathcal{C}(\Gamma(S), \Gamma(\Delta S))$. Note that δ_S exists by assumption and is well-defined due to the faithfulness of Γ .

Now we prove naturality. Define

$$f := \sigma_{\Gamma(\Delta S), \Gamma(S)} \in \mathcal{C}(\Gamma(\Delta T) \otimes \Gamma(S), \Gamma(S) \otimes \Gamma(\Delta T)).$$

Take any $h \in \mathcal{D}(S, T)$. Using sliding, we have

$$\text{fbk}_T((\Gamma(h) \otimes 1_{\Gamma(\Delta T)}) \circ f) = \text{fbk}_S(f \circ (\Gamma(\Delta h) \otimes 1_{\Gamma(S)})).$$

Using the naturality of σ , we have

$$(\Gamma(h) \otimes 1) \circ \sigma_{\Gamma(\Delta T), \Gamma(S)} = \sigma_{\Gamma(\Delta T), \Gamma(T)} \circ (1 \otimes \Gamma(h)),$$

and similarly

$$\sigma_{\Gamma(\Delta T), \Gamma(S)} \circ (\Gamma(\Delta h) \otimes 1) = (1 \otimes \Gamma(\Delta h)) \circ \sigma_{\Gamma(\Delta S), \Gamma(S)}.$$

Substituting these into the sliding equation gives:

$$\text{fbk}_T(\sigma_{\Gamma(\Delta T), \Gamma(T)} \circ (1 \otimes \Gamma(h))) = \text{fbk}_S((1 \otimes \Gamma(\Delta h)) \circ \sigma_{\Gamma(\Delta S), \Gamma(S)}).$$

Now applying tightening twice yields

$$\text{fbk}_T(\sigma_{\Gamma(\Delta T), \Gamma(T)} \circ (1 \otimes \Gamma(h))) = \text{fbk}_T(\sigma_{\Gamma(\Delta T), \Gamma(T)}) \circ \Gamma(h) = \Gamma(\delta_T) \circ \Gamma(h),$$

and similarly

$$\text{fbk}_S((1 \otimes \Gamma(\Delta h)) \circ \sigma_{\Gamma(\Delta S), \Gamma(S)}) = \Gamma(\Delta h) \circ \text{fbk}_S(\sigma_{\Gamma(\Delta S), \Gamma(S)}) = \Gamma(\Delta h) \circ \Gamma(\delta_S).$$

Hence

$$\Gamma(\Delta h) \circ \Gamma(\delta_S) = \Gamma(\delta_T) \circ \Gamma(h).$$

Since Γ is faithful, this implies that $\Delta h \circ \delta_S = \delta_T \circ h$, therefore $\delta: 1_{\mathcal{D}} \Rightarrow \Delta$ is natural. Monoidality of δ follows from strength, joining, and vanishing together with coherence of σ .

ii. Delay implies feedback operator. Conversely, assume we have a monoidal natural transformation $\delta: 1_{\mathcal{D}} \Rightarrow \Delta$. Take $S \in \text{ob}(\mathcal{D})$ and $f \in \mathcal{C}(\Gamma(\Delta S) \otimes X, \Gamma(S) \otimes Y)$. Define

$$\begin{aligned} \text{fbk}_S(f) &:= (\cap_{\Gamma(S)} \otimes 1_Y) \circ (\sigma_{\Gamma(S), \Gamma(S)^*} \otimes 1_Y) \circ (1_{\Gamma(S)} \otimes \sigma_{Y, \Gamma(S)^*}) \\ &\quad \circ ((f \circ (\Gamma(\delta_S) \otimes 1_X)) \otimes 1_{\Gamma(S)^*}) \\ &\quad \circ (1_{\Gamma(S)} \otimes \sigma_{\Gamma(S)^*, X}) \circ (\cup_{\Gamma(S)} \otimes 1_X). \end{aligned}$$

The feedback axioms follow from compact-closure together with the fact that δ is a monoidal natural transformation, using essentially the same argument as is spelled out in [23].

iii. **The delay is a natural isomorphism.** Consider the delay natural transformation $\delta: 1_{\mathcal{D}} \Rightarrow \Delta$ from Proposition 3.5. Define $\gamma: \Delta \Rightarrow 1_{\mathcal{D}}$ componentwise by

$$\gamma_X := (\cap_{\Delta(X^*)} \otimes 1_X) \circ (1_{\Delta X} \otimes \delta_{X^*} \otimes 1_X) \circ (1_{\Delta X} \otimes \cup_{X^*}).$$

Naturality of γ follows by applying the dual $(-)^*$ to the naturality square for δ .

Now we show that γ is inverse to δ . First note that because Δ is a symmetric monoidal functor and $\mathcal{D}$ is compact-closed, for all objects D in $\mathcal{D}$ we have that $\cup_{\Delta Z} = \Delta(\cup_Z)$ and $\cap_{\Delta Z} = \Delta(\cap_Z)$.

For the one side of the inverse, note that naturality of δ at $\cap_{X^*}: X \otimes X^* \rightarrow I$, together with monoidality of δ , gives $\cap_{\Delta(X^*)} \circ (\delta_X \otimes \delta_{X^*}) = \cap_{X^*}$. Therefore

$$\begin{aligned} \gamma_X \circ \delta_X &= (\cap_{\Delta(X^*)} \otimes 1_X) \circ (1_{\Delta X} \otimes \delta_{X^*} \otimes 1_X) \circ (1_{\Delta X} \otimes \cup_{X^*}) \circ \delta_X \\ &= (\cap_{\Delta(X^*)} \otimes 1_X) \circ (\delta_X \otimes \delta_{X^*} \otimes 1_X) \circ (1_X \otimes \cup_{X^*}) \\ &= (\cap_{X^*} \otimes 1_X) \circ (1_X \otimes \cup_{X^*}) = 1_X. \end{aligned}$$

On the other hand, the naturality of δ at $\cup_{X^*}: I \rightarrow X^* \otimes X$, together with the monoidality of δ , imply that $(\delta_{X^*} \otimes \delta_X) \circ \cup_{X^*} = \cup_{\Delta(X^*)}$. Therefore

$$\begin{aligned} \delta_X \circ \gamma_X &= \delta_X \circ (\cap_{\Delta(X^*)} \otimes 1_X) \circ (1_{\Delta X} \otimes \delta_{X^*} \otimes 1_X) \circ (1_{\Delta X} \otimes \cup_{X^*}) \\ &= (\cap_{\Delta(X^*)} \otimes 1_{\Delta X}) \circ (1_{\Delta X} \otimes \delta_{X^*} \otimes \delta_X) \circ (1_{\Delta X} \otimes \cup_{X^*}) \\ &= (\cap_{\Delta(X^*)} \otimes 1_{\Delta X}) \circ (1_{\Delta X} \otimes \cup_{\Delta(X^*)}) = 1_{\Delta X}. \end{aligned} \quad \square$$

Lemma 3.7. *For every SMC $\mathcal{C}$, there is a unique feedback functor $U: \text{St}(\mathcal{C}) \rightarrow \text{St}([\mathbb{Z}, \mathcal{C}], \text{Shift})$ which is the identity on objects and acts on morphisms by:*

$$([S, f]: X \rightarrow Y) \quad \mapsto \quad ([(f: S \otimes X \rightarrow S \otimes Y)_{j \in \mathbb{Z}}]: X^{\mathbb{Z}} \rightarrow Y^{\mathbb{Z}}).$$

Proof of Lemma 3.7. Consider the symmetric monoidal functor:

$$J: \mathcal{C} \rightarrow \text{St}([\mathbb{Z}, \mathcal{C}], \text{Shift}); \quad (f: X \rightarrow Y) \mapsto [(I \otimes f)_{j \in \mathbb{Z}}].$$

Using the canonical functor $\eta: \mathcal{C} \rightarrow \text{St}(\mathcal{C})$, the universal property of the state construction given in Proposition D.2 induces a unique feedback functor $J^\sharp: \text{St}(\mathcal{C}) \rightarrow \text{St}([\mathbb{Z}, \mathcal{C}], \text{Shift})$, such that for all $f \in \mathcal{C}(X, Y)$ $J^\sharp([I, f]) = [(f)_{j \in \mathbb{Z}}]$. Moreover, for any $[S, f] \in \text{St}(\mathcal{C})(X, Y)$, since $J^\sharp$ is a feedback functor, it follows that $J^\sharp([S, f]) = \text{Shift}(J^\sharp([S, f]))$. $\square$

To further understand coalgebraic streams, we must further expose the work of Di Lavore et al. [21].

Definition D.3 ([21, Def. 4.2]). Fix a symmetric monoidal category $(\mathcal{C}, \otimes, I)$ and $\mathbf{X}, \mathbf{Y} \in \text{ob}(\mathcal{C})^{\mathbb{N}}$. Define $M_{-1} := I$ and for all $n \in \mathbb{N}$ define

$$\text{Stage}_n(\mathbf{X}, \mathbf{Y}) := \int^{\mathbf{M} \in \mathcal{C}^{n-1}} \prod_{i=0}^n \mathcal{C}(M_{i-1} \otimes X_i, M_i \otimes Y_i),$$

denoting representatives by $[\langle f_0 | \cdots | f_n \rangle] \in \text{Stage}_n(\mathbf{X}, \mathbf{Y})$. Define the truncation maps by:

$$\pi_n: \text{Stage}_{n+1}(\mathbf{X}, \mathbf{Y}) \rightarrow \text{Stage}_n(\mathbf{X}, \mathbf{Y}); \quad \langle f_0 | \cdots | f_{n+1} \rangle \mapsto \langle f_0 | \cdots | f_n \rangle.$$

Next we recall the following notion under which coalgebraic streams are “well-behaved”:

Definition D.4 ([21, Def. 9]). A SMC is **productive** if for all $\alpha \in \text{Stage}_0(\mathbf{X}, \mathbf{Y})$ there exists a representative $\alpha_0: X_0 \rightarrow M_0 \otimes Y_0$ such that, for every representative $\alpha': X_0 \rightarrow M \otimes Y_0$ of α , there exists $s: M_0 \rightarrow M$ with $\alpha' = (s \otimes 1_{Y_0}) \circ \alpha_0$.

Lemma D.5. *The following SMCs are productive:*

- Cartesian monoidal categories;
- compact-closed categories;
- SMCs with monoidal zero morphisms.

Proof. The first two claims are proved by Di Lavore et al. [21, Remark 4.10]. Therefore, we only prove the final claim involving monoidal zero morphisms.

Fix $\mathbf{X}, \mathbf{Y} \in \text{ob}(\mathcal{C})^{\mathbb{N}}$ and let $\alpha \in \text{Stage}_0(\mathbf{X}, \mathbf{Y})$. Choose a representative $\alpha = \langle \alpha_0 |$ with $\alpha_0: X_0 \rightarrow M \otimes Y_0$ for some $M \in \text{ob}(\mathcal{C})$. The axioms of monoidal zero morphisms imply that

$$0_{M,I} \otimes 1_{Y_0} = \sigma_{I,Y_0} \circ (1_{Y_0} \otimes 0_{M,I}) \circ \sigma_{M,Y_0} = 0_{M \otimes Y_0, I \otimes Y_0}.$$

Moreover, by dinaturality $\langle \alpha_0 | = \langle (0_{M,I} \otimes 1_{Y_0}) \circ \alpha_0 | = \langle 0_{X_0, I \otimes Y_0} |$. Therefore $\text{Stage}_0(\mathbf{X}, \mathbf{Y})$ is a singleton set. Hence every 0-stage process is terminating so that $\mathcal{C}$ is productive. $\square$

Lemma D.6 ([21, Thm. 4.11.]). *If $\mathcal{C}$ is productive, then for all $\mathbf{X}, \mathbf{Y} \in \text{ob}(\mathcal{C})^{\mathbb{N}}$ there is a canonical bijection $\text{Stream}(\mathcal{C})(\mathbf{X}, \mathbf{Y}) \cong \lim_{n \in \mathbb{N}} \text{Stage}_n(\mathbf{X}, \mathbf{Y})$, where the limit is taken along the truncations π_n .*

Lemma 3.10. *$\text{Stream}(\mathcal{C})$ is indiscrete for compact-closed $\mathcal{C}$.*

Proof of Lemma 3.10. Fix $\mathbf{X}, \mathbf{Y} \in \text{ob}(\mathcal{C})^{\mathbb{N}}$. Take some $\langle f_0 | f_1 | \cdots | f_n | \in \text{Stage}_n(\mathbf{X}, \mathbf{Y})$ with witnesses

$f_i: M_{i-1} \otimes X_i \rightarrow M_i \otimes Y_i$ and $M_{-1} := I$. Define

$$\eta_{M,X,Y} := 1_{M \otimes X} \otimes (\sigma_{Y,Y^*} \circ \cup_Y): M \otimes X \rightarrow (M \otimes X \otimes Y^*) \otimes Y,$$

Moreover for any $f: M \otimes X \rightarrow N \otimes Y$, define

$$s_f := (1_N \otimes \cap_Y) \circ (1_N \otimes \sigma_{Y,Y^*}) \circ (f \otimes 1_{Y^*}): M \otimes X \otimes Y^* \rightarrow N.$$

Therefore it follows that $(s_f \otimes 1_Y) \circ \eta_{M,X,Y} = f$. Write $N_{-1} := I$ and $N_k := N_{k-1} \otimes X_k \otimes Y_k^*$. Then by dinaturality of the coends and the above factorisation, we have that

$$\begin{aligned} \langle f_0 | \cdots | f_{n-1} | f_n | &\approx \langle f_0 | \cdots | f_{n-1} | \eta_{M_{n-1}, X_n, Y_n} | \\ &\approx \langle f_0 | \cdots | f_{n-2} | \eta_{M_{n-2}, X_{n-1}, Y_{n-1}} | \eta_{M_{n-2} \otimes X_{n-1} \otimes Y_{n-1}^*, X_n, Y_n} | \\ &\vdots \\ &= \langle \eta_{N_{-1}, X_0, Y_0} | \eta_{N_0, X_1, Y_1} | \cdots | \eta_{N_{n-1}, X_n, Y_n} | \end{aligned}$$

Therefore for all $n \in \mathbb{N}$, $\text{Stage}_n(\mathbf{X}, \mathbf{Y})$ is a singleton set. By Lemma D.5, $\mathcal{C}$ is productive; therefore by Lemma D.6 we have $\text{Stream}(\mathcal{C})$ is indiscrete. $\square$

Lemma 3.11. *If $\mathcal{C}$ is a SMC with monoidal zero morphisms, then $\text{Stream}(\mathcal{C})$ is indiscrete.*

Proof of Lemma 3.11. Fix $\mathbf{X}, \mathbf{Y} \in \text{ob}(\mathcal{C})^{\mathbb{N}}$. Take some $\langle f_0 | f_1 | \cdots | f_n | \in \text{Stage}_n(\mathbf{X}, \mathbf{Y})$ with witnesses

$f_i: M_{i-1} \otimes X_i \rightarrow M_i \otimes Y_i$ and $M_{-1} := I$. By dinaturality and the coherences for monoidal zero morphisms we have that

$$\begin{aligned} \langle f_0 | \cdots | f_{n-1} | f_n | &\approx \langle f_0 | \cdots | f_{n-1} | (0_{M_n, I} \otimes 1_{Y_n}) \circ f_n | \\ &= \langle f_0 | \cdots | f_{n-1} | 0_{M_n \otimes X_n, I \otimes Y_n} | \\ &\vdots \\ &= \langle 0_{X_0, Y_0} | \cdots | 0_{X_{n-1}, Y_{n-1}} | 0_{X_n, Y_n} | \end{aligned}$$

Therefore for all $n \in \mathbb{N}$, $\text{Stage}_n(\mathbf{X}, \mathbf{Y})$ is a singleton set. By Lemma D.5, $\mathcal{C}$ is productive; therefore by Lemma D.6, $\text{Stream}(\mathcal{C})$ is indiscrete. $\square$

E Proofs of Section 4

Lemma 4.3. *Approximation is a preorder.*

Proof of Lemma 4.3. We need to prove that:

- i. approximation is reflexive;
- ii. approximation is transitive.

i. Reflexivity. Immediate by taking $j' = j$.

ii. Transitivity. Suppose $\mathbf{f} \subseteq \mathbf{g}$ and $\mathbf{g} \subseteq \mathbf{h}$. Fix $j \in \mathcal{J}$. By $\mathbf{g} \subseteq \mathbf{h}$, there exists $j_0 \in \mathcal{J}$ with $j \subseteq j_0$ such that $(1_{Y_j} \otimes \top_{Y_{j_0 \setminus j}}) \circ g_{j_0} \subseteq h_j \otimes \top_{X_{j_0 \setminus j}}$. Moreover, since $\mathbf{f} \subseteq \mathbf{g}$, by restricting to the context j_0 , there exists some $j_1 \in \mathcal{J}$ with $j_0 \subseteq j_1$ such that $(1_{Y_{j_0}} \otimes \top_{Y_{j_1 \setminus j_0}}) \circ f_{j_1} \subseteq g_{j_0} \otimes \top_{X_{j_1 \setminus j_0}}$. Therefore $(1_{Y_j} \otimes \top_{Y_{j_1 \setminus j}}) \circ f_{j_1} \subseteq h_j \otimes \top_{X_{j_1 \setminus j}}$, so that $\mathbf{f} \subseteq \mathbf{h}$. $\square$

Proposition 4.5. $\mathbf{Mon}_{\mathcal{I}, \mathcal{J}}(\mathcal{C})$ is a discard bicategory.

Proof of Proposition 4.5. We need to show that:

- i. structural morphisms are monotone families;
- ii. composition of monotone families is monotone;
- iii. monoidal product of monotone families is monotone;
- iv. the discard bicategory axioms are satisfied.

i. Structural morphisms are monotone. Identities, symmetry, and discard are defined contextwise by tensoring the corresponding structural morphisms in $\mathcal{C}$:

$$(1_{\mathbf{X}})_j = 1_{X_j}, \quad (\top_{\mathbf{X}})_j = \bigotimes_{i \in j} \top_{X_i}, \quad \text{and} \quad (\sigma_{\mathbf{X}, \mathbf{Y}})_j = \bigotimes_{i \in j} \sigma_{X_i, Y_i}.$$

Take contexts $j \subseteq j'$. All structural morphisms f in $\mathcal{C}$ are total meaning that $\top \circ f = \top$; therefore, since all structural morphisms in $\mathbf{Mon}_{\mathcal{I}, \mathcal{J}}(\mathcal{C})$ are built from those in $\mathcal{C}$ we have that

$$(1_{Y_j} \otimes \top_{Y_{j' \setminus j}}) \circ f_{j'} = f_j \otimes \top_{X_{j' \setminus j}}.$$

Therefore all structural morphisms are causal (and thus monotone) families.

ii. Composition preserves monotonicity. Take two monotone $\mathbf{f}: \mathbf{X} \rightarrow \mathbf{Y}$ and $\mathbf{g}: \mathbf{Y} \rightarrow \mathbf{Z}$ families, and fix contexts $j \subseteq j'$ in $\mathcal{J}$. Using monotonicity of $\mathbf{f}$ and $\mathbf{g}$, it follows that

$$(1_{Z_j} \otimes \top_{Z_{j' \setminus j}}) \circ (g_{j'} \circ f_{j'}) = ((1_{Z_j} \otimes \top_{Z_{j' \setminus j}}) \circ g_{j'}) \circ f_{j'} \subseteq (g_j \otimes \top_{Y_{j' \setminus j}}) \circ f_{j'} \subseteq (g_j \circ f_j) \otimes \top_{X_{j' \setminus j}}.$$

Therefore $\mathbf{g} \circ \mathbf{f}$ is monotone.

iii. Monoidal product preserves monotonicity. Take two monotone families $\mathbf{f}: \mathbf{X} \rightarrow \mathbf{Y}, \mathbf{h}: \mathbf{X}' \rightarrow \mathbf{Y}'$, and fix contexts $j \subseteq j'$. It follows that

$$\begin{aligned} & (1_{(Y \otimes Y')_j} \otimes \top_{(Y \otimes Y')_{j' \setminus j}}) \circ (\zeta_{\mathbf{Y}, \mathbf{Y}'})_j \circ (f_{j'} \otimes h_{j'}) \circ (\zeta_{\mathbf{X}, \mathbf{X}'}^{-1})_j^{-1} \\ &= (\zeta_{\mathbf{Y}, \mathbf{Y}'})_j \circ (1_{Y_j} \otimes \top_{Y_{j' \setminus j}} \otimes 1_{Y'_j} \otimes \top_{Y'_{j' \setminus j}}) \circ (f_{j'} \otimes h_{j'}) \circ (\zeta_{\mathbf{X}, \mathbf{X}'}^{-1})_j^{-1} \\ &\subseteq (\zeta_{\mathbf{Y}, \mathbf{Y}'})_j \circ (f_j \otimes h_j) \circ (\zeta_{\mathbf{X}, \mathbf{X}'}^{-1})_j^{-1}. \end{aligned}$$

Therefore $\mathbf{f} \otimes \mathbf{h}$ is monotone.

iv. Discard bicategory axioms hold. i., ii., and iii. imply that approximation is a monoidal preorder enrichment; where all other discard-bicategory axioms, including lax naturality of the discard, hold contextwise. $\square$

Theorem 4.9. $\text{Obs}_{\mathcal{I}, \mathcal{J}}(\mathcal{C})$ is a poset-enriched discard bicategory.

Proof of Theorem 4.9. We need to show that observational equivalence is a congruence for:

- i. approximation;
- ii. composition;
- iii. monoidal product.

From which it follows that the discard bicategory structure of $\text{Mon}_{\mathcal{I}, \mathcal{J}}(\mathcal{C})$ from Proposition 4.5 descends to the quotient, where the preorder enrichment becomes a poset since mutually-approximating families are identified.

i. Congruence for approximation. This follow by transitivity.

ii. Congruence for composition. Take composable $\mathbf{f} \subseteq \mathbf{f}'$ and $\mathbf{g} \subseteq \mathbf{g}'$ and fix some $j \in \mathcal{J}$. Choose $j_f, j_g \in \mathcal{J}$ with $j \subseteq j_f$ and $j \subseteq j_g$ witnessing that $\mathbf{f}'$ approximates $\mathbf{f}$ and $\mathbf{g}'$ approximates $\mathbf{g}$. Using upward directedness of $\mathcal{J}$, let $j' \in \mathcal{J}$ be such that $j_f \cup j_g \subseteq j'$. Then:

$$(1_{Z_j} \otimes \top_{Z_{j' \setminus j}}) \circ g_{j'} \circ f_{j'} \subseteq g'_j \circ (1_{Y_j} \otimes \top_{Y_{j' \setminus j}}) \circ f_{j'} \subseteq (g'_j \circ f'_j) \otimes \top_{X_{j' \setminus j}}.$$

Thus $\mathbf{g} \circ \mathbf{f} \subseteq \mathbf{g}' \circ \mathbf{f}'$.

iii. Congruence for monoidal product. Take $\mathbf{f} \subseteq \mathbf{f}'$ and $\mathbf{h} \subseteq \mathbf{h}'$. Fix $j \in \mathcal{J}$. Choose $j_f, j_h \in \mathcal{J}$ with $j \subseteq j_f$ and $j \subseteq j_h$ witnessing that $\mathbf{f}'$ approximates $\mathbf{f}$ and $\mathbf{h}'$ approximates $\mathbf{h}$. Using upward directedness of $\mathcal{J}$, let $j' \in \mathcal{J}$ be such that $j_f \cup j_h \subseteq j'$. Then:

$$(1_{(Y \otimes Y')_j} \otimes \top_{(Y \otimes Y')_{j' \setminus j}}) \circ (\zeta_{\mathbf{Y}, \mathbf{Y}'})_j \circ (f_{j'} \otimes h_{j'}) \circ (\zeta_{\mathbf{X}, \mathbf{X}'}^{-1}_j \circ (f'_j \otimes h'_j) \circ (\zeta_{\mathbf{X}, \mathbf{X}'}^{-1}_{j'}) \subseteq (\zeta_{\mathbf{Y}, \mathbf{Y}'})_j \circ (f'_j \otimes h'_j) \circ (\zeta_{\mathbf{X}, \mathbf{X}'}^{-1}_{j'}) \circ (f_{j'} \otimes h_{j'}) \circ (\zeta_{\mathbf{X}, \mathbf{X}'}^{-1}_j \circ 1_{(Y \otimes Y')_{j' \setminus j}}).$$

Thus $\mathbf{f} \otimes \mathbf{h} \subseteq \mathbf{f}' \otimes \mathbf{h}'$. $\square$

Proposition 4.10. Let $F: \mathcal{C} \rightarrow \mathcal{D}$ be a discard bicategory functor. There is an induced discard bicategory functor $\text{Obs}_{\mathcal{I}, \mathcal{J}}(F): \text{Obs}_{\mathcal{I}, \mathcal{J}}(\mathcal{C}) \rightarrow \text{Obs}_{\mathcal{I}, \mathcal{J}}(\mathcal{D})$ defined contextwise.

Moreover, if F reflects the preorder then $\text{Obs}_{\mathcal{I}, \mathcal{J}}(F)$ is faithful. If F reflects the preorder and is full, then $\text{Obs}_{\mathcal{I}, \mathcal{J}}(F)$ is full and faithful.

Proof of Proposition 4.10. We need to show that $\text{Obs}_{\mathcal{I}, \mathcal{J}}(F)$:

- i. preserves the preorder on hom-sets;
- ii. is well-defined on observational equivalence classes;
- iii. is functorial and preserves the discard category structure;
- iv. is faithful when F reflects the preorder;
- v. is full when F reflects the preorder and is full.

i. Preservation of monotonicity. Fix contexts $j \subseteq j'$ in $\mathcal{J}$. By monotonicity of $\mathbf{f}$, we have

$$(1_{Y_j} \otimes \top_{Y_{j' \setminus j}}) \circ f_{j'} \subseteq f_j \otimes \top_{X_{j' \setminus j}}.$$

Using the fact that F preserves the preorder, it follows that

$$F((1_{Y_j} \otimes \top_{Y_{j' \setminus j}}) \circ f_{j'}) \subseteq F(f_j \otimes \top_{X_{j' \setminus j}}).$$

Since F is a discard functor and symmetric monoidal, it follows that

$$(1_{F(Y)_j} \otimes \top_{F(Y)_{j' \setminus j}}) \circ F(f_{j'}) \subseteq F(f_j) \otimes \top_{F(X)_{j' \setminus j}}.$$

Therefore $\{F(f_j)\}_{j \in \mathcal{J}}$ is a monotone net in $\mathcal{D}$.

ii. Well-definedness on equivalence class. Take two monotone families $\mathbf{f} \subseteq \mathbf{g}$ in $\text{Mon}_{\mathcal{I}, \mathcal{J}}(\mathcal{C})$, and fix some context $j \in \mathcal{J}$. By definition of approximation, there exists $j' \in \mathcal{J}$ with $j \subseteq j'$ such that $(1_{Y_j} \otimes \top_{Y_{j' \setminus j}}) \circ f_{j'} \subseteq g_j \otimes \top_{X_{j' \setminus j}}$. Therefore it follows that

$$(1_{F(Y)_j} \otimes \top_{F(Y)_{j' \setminus j}}) \circ F(f_{j'}) \subseteq F(g_j) \otimes \top_{F(X)_{j' \setminus j}}.$$

Thus $\text{Obs}_{\mathcal{I}, \mathcal{J}}(F)([\mathbf{f}]) \subseteq \text{Obs}_{\mathcal{I}, \mathcal{J}}(F)([\mathbf{g}])$; therefore, $\text{Obs}_{\mathcal{I}, \mathcal{J}}(F)$ is well-defined on observational equivalence classes.

iii. Functoriality. Functoriality holds contextwise.

iv. Faithfulness. Assume F reflects the preorder, and take two monotone families $\mathbf{f}$ and $\mathbf{g}$ such that $\text{Obs}_{\mathcal{I}, \mathcal{J}}(F)([\mathbf{f}]) = \text{Obs}_{\mathcal{I}, \mathcal{J}}(F)([\mathbf{g}])$. Then $\text{Obs}_{\mathcal{I}, \mathcal{J}}(F)([\mathbf{f}])$ and $\text{Obs}_{\mathcal{I}, \mathcal{J}}(F)([\mathbf{g}])$ approximate each other.

Next, take some context $j \in \mathcal{J}$. Then there exists some $j' \supseteq j$ such that

$$(1_{F(Y)_j} \otimes \top_{F(Y)_{j' \setminus j}}) \circ F(f_{j'}) \subseteq F(g_j) \otimes \top_{F(X)_{j' \setminus j}}.$$

Rewrite both sides as $F(u) \subseteq F(v)$. By reflection of the preorder, $u \subseteq v$, hence $\mathbf{f} \subseteq \mathbf{g}$. The reverse inclusion holds by symmetry, so that $[\mathbf{f}] = [\mathbf{g}]$. Thus $\text{Obs}_{\mathcal{I}, \mathcal{J}}(F)$ is faithful.

v. Fullness. Assume F reflects the preorder and is full.

Let $[\mathbf{h}]: \text{Obs}_{\mathcal{I}, \mathcal{J}}(F)(\mathbf{X}) \rightarrow \text{Obs}_{\mathcal{I}, \mathcal{J}}(F)(\mathbf{Y})$ be a morphism in $\text{Obs}_{\mathcal{I}, \mathcal{J}}(\mathcal{D})$. For each $j \in \mathcal{J}$, using fullness, choose some f_j in $\mathcal{C}$ such that $F(f_j) = h_j$, taking $\mathbf{f} := \{f_j\}_{j \in \mathcal{J}}$.

Monotonicity of $\mathbf{h}$ and the reflection of the preorder imply monotonicity of $\mathbf{f}$. Since $F(f_j) = h_j$ for all $j \in \mathcal{J}$, it follows that $\text{Obs}_{\mathcal{I}, \mathcal{J}}(F)([\mathbf{f}]) = [\mathbf{h}]$. $\square$

Proposition 4.11. *Take a discard bicategory $\mathcal{C}$ with monoidal zero morphisms, and upward-directed $\mathcal{J} \subseteq \mathcal{P}_f(\mathcal{I})$. Then $\text{Obs}_{\mathcal{I}, \mathcal{J}}(\mathcal{C})$ has monoidal zero morphisms. Moreover, if a monotone net $\mathbf{f}: \mathbf{X} \rightarrow \mathbf{Y}$ satisfies $f_{j_0} = 0$ for some $j_0 \in \mathcal{J}$, then $[\mathbf{f}] = [\{0\}_{j \in \mathcal{J}}]$ is a zero morphism.*

Proof of Proposition 4.11. Fix a contexts $j_0, j, j' \in \mathcal{J}$ with $j \cup j_0 \subseteq j'$. By monotonicity of $\mathbf{f}$, $f_{j'}$ factors through $f_{j_0} = 0$, hence $(1_{Y_j} \otimes \top_{Y_{j' \setminus j}}) \circ f_{j'} = 0$. This witnesses $\mathbf{f} \subseteq \{0\}$, where the converse is immediate. $\square$

Theorem 4.14. *If $\mathcal{C}$ is a discard bicategory then there is a symmetric monoidal functor $\text{St}([\mathbb{N}, \mathcal{C}], \text{Delay}) \rightarrow \text{Obs}_{\mathbb{N}}(\mathcal{C})$.*

Proof of Theorem 4.14. The functor $U: \text{St}([\mathbb{N}, \mathcal{C}], \text{Delay}) \rightarrow \text{Obs}_{\mathbb{N}}(\mathcal{C})$ is defined on a representative stateful morphism sequence by unrolling and discarding the output memory. By lax naturality, it is easy to show that this gives a monotone sequence and thus a valid initialised behaviour. To show that two representatives of the same equivalence class of $\text{St}([\mathbb{N}, \mathcal{C}], \text{Delay})$ yield the same initialised behaviour, the only non-trivial thing to prove is that the sliding equation holds. Let the grey circle denote the discard. We use lax naturality to prove the first inequality

$$\begin{array}{c}
U(\text{fbk}^{\mathbf{S}}((\mathbf{r} \otimes 1_{\mathbf{Y}}) \circ \mathbf{f}))_t \\
\parallel \\
\text{Diagram 1} \\
\text{Diagram 2} \\
\parallel \\
U(\text{fbk}^{\mathbf{S}}(\mathbf{f} \circ (\text{Delay}(\mathbf{r}) \otimes 1_{\mathbf{X}})))_t,
\end{array}$$

The diagrams are stateful morphism sequences. Diagram 1 shows a sequence of functions $f_0, f_1, f_2, \dots, f_t$ with intermediate results $r_0, r_1, r_2, \dots, r_t$ connected by wires. The final result r_t is connected to a grey circle (discard). Diagram 2 is identical to Diagram 1 but the discard is at the end of the sequence.

and similarly for the converse

$$\begin{array}{c}
(1_{Y_{[0,t]}} \otimes \top_{Y_{t+1}}) \circ (U(\text{fbk}^{\mathbf{S}}(\mathbf{f} \circ (\text{Delay}(\mathbf{r}) \otimes 1_{\mathbf{X}}))))_{t+1} \\
\parallel \\
\text{Diagram 3} \\
\text{Diagram 4} \\
\parallel \\
U(\text{fbk}^{\mathbf{S}}((\mathbf{r} \otimes 1_{\mathbf{Y}}) \circ \mathbf{f}))_t \otimes \top_{X_{t+1}}.
\end{array}$$

Diagram 3 shows a sequence of functions $f_0, f_1, f_2, \dots, f_t, f_{t+1}$ with intermediate results $r_0, r_1, r_2, \dots, r_t, r_{t+1}$. The final result r_{t+1} is connected to two grey circles (discards). Diagram 4 is identical to Diagram 3 but the discards are at the end of the sequence.

Note that the second inequality requires looking ahead to the context $[0, t+1] \supseteq [0, t]$. Therefore, both sides of the sliding equation are observationally equivalent and thus U is well-defined on equivalence classes. Moreover, symmetric monoidal functoriality follows immediately from the axiom $\top_X \otimes \top_Y = \top_{X \otimes Y}$. $\square$

Proposition 4.15. *For any discard bicategory $\mathcal{C}$, there is a monoidal natural transformation*

$$\delta: 1_{\text{Obs}_{\mathbb{N}}(\mathcal{C})} \Rightarrow \text{Delay}; \quad \delta_{\mathbf{X}} := \left[\left(\bigotimes_{k=0}^{t-1} 1_{X_k} \right) \otimes \top_{X_t} \right]_{t \in \mathbb{N}}.$$

Proof of Proposition 4.15. Naturality at any $\mathbf{f}: \mathbf{X} \rightarrow \mathbf{Y}$ amounts to the equation $\text{Delay}(\mathbf{f}) \circ \delta_{\mathbf{X}} = \delta_{\mathbf{Y}} \circ \mathbf{f}$, which follows by lax naturality of the discard upon looking ahead one context, exactly as in the sliding equation of Theorem 4.14. Monoidality follows from $\top_X \otimes \top_Y = \top_{X \otimes Y}$. $\square$

Theorem 4.18. *Given a compact-closed discard bicategory $\mathcal{C}$ with $\perp_X := (1_X \otimes \top_{X^*}) \circ \cup_X$, there is a monoidal natural isomorphism*

$$\delta: 1_{\text{Obs}_{\mathbb{Z}}(\mathcal{C})} \Rightarrow \mathbf{Shift}; \quad \delta_{\mathbf{X}} := [\{\perp_{X_{\ell-1}} \otimes (\bigotimes_{k=\ell}^{r-1} 1_{X_k}) \otimes \top_{X_r}\}_{[\ell, r] \in \mathbb{Z}_{\leq}^2}],$$

making $\text{Obs}_{\mathbb{Z}}(\mathcal{C})$ into a compact-closed feedback category.

Proof of Theorem 4.18. Using Proposition 3.5, it suffices to prove that:

- i. $\text{Obs}_{\mathbb{Z}}(\mathcal{C})$ is compact-closed;
- ii. the components of δ are monotone nets;
- iii. δ is a monoidal natural transformation.

i. Compact closure. For every object $\mathbf{X}$ of $\text{Obs}_{\mathbb{Z}}(\mathcal{C})$, we define the dual as $(\mathbf{X})^* := (X_k^*)_{k \in \mathbb{Z}}$. The cups and caps are defined component-wise by:

$$\cup_{\mathbf{X}} := \left[\left\{ (\zeta_{\mathbf{X}^*, \mathbf{X}})_{[\ell, r]} \circ \bigotimes_{k=\ell}^r \cup_{X_k} \right\}_{[\ell, r] \in \mathbb{Z}_{\leq}^2} \right] \quad \text{and} \quad \cap_{\mathbf{X}} := \left[\left\{ \bigotimes_{k=\ell}^r \cap_{X_k} \circ (\zeta_{\mathbf{X}, \mathbf{X}^*})_{[\ell, r]}^{-1} \right\}_{[\ell, r] \in \mathbb{Z}_{\leq}^2} \right].$$

Lax naturality ensures that $\cap_{X_k} \subseteq \top_{X_k^* \otimes X_k}$ and $\top_{X_k \otimes X_k^*} \circ \cup_{X_k} \subseteq 1_I = \top_I$. Therefore, it follows that $\cup_{\mathbf{X}}$ and $\cap_{\mathbf{X}}$ are monotone nets. The snake equations can be shown to hold component-wise.

ii. Monotonicity of delay. We start by showing monotonicity of each component of δ . Note that for any $[\ell, r] \in \mathbb{Z}_{\leq}^2$,

$$\begin{aligned} (\top_{X_{\ell-2}} \otimes 1_{X_{[\ell-1, r-1]}}) \circ (\delta_{\mathbf{X}})_{[\ell-1, r]} &= 1_{X_{[\ell-1, r-1]}} \otimes \top_{X_r} \subseteq \top_{X_{\ell-1}} \otimes (\perp_{X_{\ell-1}} \otimes 1_{X_{[\ell-1, r-1]}} \otimes \top_{X_r}) \\ &= \top_{X_{\ell-1}} \otimes (\delta_{\mathbf{X}})_{[\ell, r]}, \end{aligned}$$

where the inequality follows from the inequation $1_{X_{\ell-1}} \subseteq \perp_{X_{\ell-1}} \circ \top_{X_{\ell-1}}$. On the other hand we have:

$$(1_{X_{[\ell, r]}} \otimes \top_{X_{r+1}}) \circ (\delta_{\mathbf{X}})_{[\ell, r+1]} = (\delta_{\mathbf{X}})_{[\ell, r]} \otimes \top_{X_{r+1}}.$$

Therefore $\delta_{\mathbf{X}}$ is a monotone net.

iii. Monoidal naturality of delay. To show naturality, fix any $\mathbf{f}: \mathbf{X} \rightarrow \mathbf{Y}$ and consider the compositions $\mathbf{Shift}(\mathbf{f}) \circ \delta_{\mathbf{X}}: \mathbf{X} \rightarrow \mathbf{Shift}(\mathbf{Y})$ and $\delta_{\mathbf{Y}} \circ \mathbf{f}: \mathbf{X} \rightarrow \mathbf{Shift}(\mathbf{Y})$. Fix any $[\ell, r] \in \mathbb{Z}_{\leq}^2$.

First, to show $\mathbf{Shift}(\mathbf{f}) \circ \delta_{\mathbf{X}} \subseteq \delta_{\mathbf{Y}} \circ \mathbf{f}$ we look ahead to the context $[\ell, r+1] \supseteq [\ell, r]$:

$$\begin{aligned} (1_{Y_{[\ell-1, r-1]}} \otimes \top_{Y_r}) \circ (\mathbf{Shift}(\mathbf{f}) \circ \delta_{\mathbf{X}})_{[\ell, r+1]} &\subseteq ((\perp_{Y_{\ell-1}} \circ \top_{Y_{\ell-1}}) \otimes 1_{Y_{[\ell, r-1]}} \otimes \top_{Y_r}) \circ f_{[\ell-1, r]} \circ (\delta_{\mathbf{X}})_{[\ell, r+1]} \\ &\subseteq (\perp_{Y_{\ell-1}} \otimes 1_{Y_{[\ell, r-1]}} \otimes \top_{Y_r}) \circ f_{[\ell, r]} \circ (\top_{X_{\ell-1}} \otimes 1_{X_{[\ell, r]}}) \circ (\delta_{\mathbf{X}})_{[\ell, r+1]} \\ &\subseteq (\perp_{Y_{\ell-1}} \otimes 1_{Y_{[\ell, r-1]}} \otimes \top_{Y_r}) \circ f_{[\ell, r]} \circ (1_{X_{[\ell, r]}} \otimes \top_{X_{r+1}}) = (\delta_{\mathbf{Y}} \circ \mathbf{f})_{[\ell, r]} \otimes \top_{X_{r+1}}, \end{aligned}$$

where the first inequality uses $1_{Y_{\ell-1}} \subseteq \perp_{Y_{\ell-1}} \circ \top_{Y_{\ell-1}}$, the second is left monotonicity of f , and the third follows by definition of $\delta_{\mathbf{X}}$ and $\top_{X_{r+1}} \circ \perp_{X_{r+1}} \subseteq 1_I$. Similarly, for the other direction we look ahead to the context $[\ell-1, r] \supseteq [\ell, r]$:

$$\begin{aligned} (\top_{Y_{\ell-2}} \otimes 1_{Y_{[\ell-1, r-1]}}) (\delta_{\mathbf{Y}} \circ \mathbf{f})_{[\ell-1, r]} &\subseteq (1_{Y_{[\ell-1, r-1]}} \otimes \top_{Y_r}) \circ f_{[\ell-1, r]} \subseteq f_{[\ell-1, r-1]} \otimes \top_{X_r} \\ &\subseteq f_{[\ell-1, r-1]} \circ ((\perp_{X_{\ell-1}} \circ \top_{X_{\ell-1}}) \otimes 1_{X_{[\ell, r-1]}} \otimes \top_{X_r}) \\ &= \top_{X_{\ell-1}} \otimes (f_{[\ell-1, r-1]} \circ (\delta_{\mathbf{X}})_{[\ell, r]}) = \top_{X_{\ell-1}} \otimes [\mathbf{Shift}(\mathbf{f}) \circ \delta_{\mathbf{X}}]_{[\ell, r]}. \end{aligned}$$

Therefore we have shown that the two sides of naturality are observationally equal.

Finally, coherence of the delay natural transformation with respect to the unit and the monoidal product follows from the coherence axioms of $\top$. Since $\text{Obs}_{\mathbb{Z}}(\mathcal{C})$ is compact closed, Proposition 3.5 ensures that δ is moreover a natural isomorphism. $\square$

Corollary 4.19. *Given a compact-closed discard bicategory $\mathcal{C}$, there are feedback functors $\text{St}(\mathcal{C}) \rightarrow \text{St}([\mathbb{Z}, \mathcal{C}], \mathbf{Shift}) \rightarrow \text{Obs}_{\mathbb{Z}}(\mathcal{C})$.*

Proof of Corollary 4.19. The first functor is given by Lemma 3.7.

By Theorem 4.18, $\text{Obs}_{\mathbb{Z}}(\mathcal{C})$ is a compact-closed feedback category, with delay endofunctor

$$\mathbf{Shift}: \text{Obs}_{\mathbb{Z}}(\mathcal{C}) \rightarrow \text{Obs}_{\mathbb{Z}}(\mathcal{C}).$$

Define a symmetric monoidal functor $J: [\mathbb{Z}, \mathcal{C}] \rightarrow \text{Obs}_{\mathbb{Z}}(\mathcal{C})$ acting as the identity on objects and on morphisms by:

$$J(f) := \left[\left\{ \bigotimes_{k=\ell}^r f_k \right\}_{[\ell, r] \in \mathbb{Z}_{\leq}^2} \right].$$

Fix a sequence $f_k: X_k \rightarrow Y_k$ and a context $[\ell, r] \in \mathbb{Z}_{\leq}^2$. Using monoidality of $\top$ and $\perp$, the right monotonicity inequality for $J(f)$ reduces to $\top_Y \circ f_k \subseteq \top_X$, and the left monotonicity inequality reduces to $f_k \circ \perp_X \subseteq \perp_Y$, both of which hold by lax naturality.

Moreover, it is immediate that J preserves the $\mathbf{Shift}$ endofunctor, i.e. $\mathbf{Shift} \circ J = J \circ \mathbf{Shift}$.

Invoking Proposition D.2 with $\Gamma = 1_{[\mathbb{Z}, \mathcal{C}]}$ and $\Delta = \mathbf{Shift}$, there is a unique feedback functor $F: \text{St}([\mathbb{Z}, \mathcal{C}]) \rightarrow \text{Obs}_{\mathbb{Z}}(\mathcal{C})$ such that $F \circ \eta = J$ as desired.

Finally, F is a feedback functor because both source and target are compact-closed feedback categories with trivial guard, therefore by Proposition 3.5, the feedback operator and the delay determine one another. Therefore, the preservation of feedback forces that the delay and induced compact-closed structure are preserved. $\square$

F Proofs of Section 5

Lemma 5.3. *Given an I -indexed set of compact Hausdorff spaces $\mathbf{X} = \{X_k \in \text{ob}(\mathbf{KHaus})\}_{k \in I}$, and a monotone net of closed subspaces $\mathbf{C} := \{C_j \subseteq \prod_{k \in j} X_k\}_{j \in \mathcal{J}}$, then there is a compact, closed subspace*

$$\text{lim}(\mathbf{C}) := \bigcap_{j \in \mathcal{J}} (\pi_j^\infty)^{-1}(C_j) \subseteq \prod_{k \in I} X_k.$$

Proof of Lemma 5.3. The product $\prod_{k \in I} X_k$ is compact with the product topology by Tychonoff's theorem and any intersection of closed subsets of a compact space is closed and compact. $\square$

Theorem 5.5 (Compactness). *Given a set $\mathcal{I}$ with cofinal $\mathcal{J} \subseteq \mathcal{P}_f(\mathcal{I})$, then lim induces a poset-enriched discard functor $\text{Lim}: \text{Obs}_{\mathcal{I}, \mathcal{J}}(\text{Rel}(\mathbf{KHaus})) \rightarrow \text{Rel}(\mathbf{KHaus})$.*

Proof of Theorem 5.5. Define Lim on objects by

$$\mathbf{X} = \{X_k \in \text{ob}(\mathbf{KHaus})\}_{k \in \mathcal{I}} \mapsto \prod_{k \in \mathcal{I}} X_k,$$

and on morphisms $[\mathbf{S}]: \mathbf{X} \rightarrow \mathbf{Y}$ by

$$\text{Lim}([\mathbf{S}]) = \zeta_{\mathbf{X}, \mathbf{Y}}(\text{lim}(\mathbf{S})) \subseteq \left(\prod_{i \in \mathbb{Z}} X_i \right) \times \left(\prod_{i \in \mathbb{Z}} Y_i \right),$$

recalling the canonical shuffling isomorphism

$$\zeta_{\mathbf{X}, \mathbf{Y}}: \prod_{k \in \mathcal{I}} X_k \otimes Y_k \cong \prod_{k \in \mathcal{I}} X_k \times \prod_{k \in \mathcal{I}} Y_k.$$

We show that Lim is a discard functor, i.e. that it preserves:

- i. the monoidal unit;
- ii. discards;

- iii. identities;
- iv. the poset structure (and hence is well-defined on equivalence classes);
- v. monoidal products;
- vi. composition laxly ($\text{Lim}(\mathbf{T}) \circ \text{Lim}(\mathbf{S}) \subseteq \text{Lim}(\mathbf{T} \circ \mathbf{S})$);
- vii. composition oplaxly ($\text{Lim}(\mathbf{T} \circ \mathbf{S}) \subseteq \text{Lim}(\mathbf{T}) \circ \text{Lim}(\mathbf{S})$).

i. Preservation of monoidal unit. Lim preserves the monoidal unit since $\prod_{k \in \mathcal{I}} \{*\} \cong \{*\}$.

ii. Preservation of discards. Lim preserves discards, since

$$\text{lim}(\top_{\mathbf{X}}) = \bigcap_{j \in \mathcal{J}} (\pi_j^\infty)^{-1}(X_j) = \bigcap_{j \in \mathcal{J}} \prod_{k \in \mathcal{I}} X_k = \prod_{k \in \mathcal{I}} X_k.$$

iii. Preservation of identities. We want to show $1_{\prod_{k \in \mathcal{I}} X_k} = \text{Lim}(1_{\mathbf{X}})$.

$1_{\prod_{k \in \mathcal{I}} X_k} \subseteq \text{Lim}(1_{\mathbf{X}})$. First, for any j we have $\pi_j^\infty 1_{\prod_{k \in \mathcal{I}} X_k} \subseteq 1_{X_j}$ and therefore

$$1_{\prod_{k \in \mathcal{I}} X_k} \subseteq \bigcap_{j \in \mathcal{J}} (\pi_j^\infty)^{-1}(1_{X_j}) = \text{Lim}(1_{\mathbf{X}}).$$

$\text{Lim}(1_{\mathbf{X}}) \subseteq 1_{\prod_{k \in \mathcal{I}} X_k}$. For the other direction, suppose $(x, y) \in \text{Lim}(1_{\mathbf{X}})$. Therefore $\pi_j^\infty(x, y) \in 1_{X_j}$, so that for all j , $\pi_j^\infty x = \pi_j^\infty y$. Using the fact that $\mathcal{J}$ is a *cover*, for any $k \in \mathcal{I}$ there exists some $j \ni k$ and projecting out the k^{th} component of X_j we get $x_k = y_k$. Therefore $x = y$ and we have shown $\text{Lim}(1_{\mathbf{X}}) \subseteq 1_{\prod_{k \in \mathcal{I}} X_k}$.

iv. Preservation of poset and well-definedness on equivalence classes. Suppose $\mathbf{A} \subseteq \mathbf{B}$, then there is a function f that to any context j associates a larger context $j' = f(j) \supseteq j$ such that $\pi_j^{j'} A_{j'} \subseteq B_j$. Then since $\pi_j^\infty = \pi_j^{j'} \circ \pi_{j'}^\infty$ we have $(\pi_j^\infty)^{-1} = (\pi_{j'}^\infty)^{-1} \circ (\pi_j^{j'})^{-1}$ and so $(\pi_{j'}^\infty)^{-1}(A_{j'}) \subseteq (\pi_j^\infty)^{-1}(B_j)$. Therefore

$$\text{lim}(\mathbf{A}) = \bigcap_{j \in \mathcal{J}} (\pi_j^\infty)^{-1}(A_j) \subseteq \bigcap_{j \in \mathcal{J}} (\pi_{f(j)}^\infty)^{-1}(A_{f(j)}) \subseteq \bigcap_{j \in \mathcal{J}} (\pi_j^\infty)^{-1}(B_j) = \text{lim}(\mathbf{B}),$$

so that Lim preserves the poset structure on monotone nets, which implies that Lim is well-defined on observational equivalence classes.

v. Preservation of monoidal product. Let $\mathbf{S}$ and $\mathbf{T}$ be two observational nets. Without loss of generality assume that $S_j \subseteq A_j$ and $T_j \subseteq B_j$ for some nets $\mathbf{A}, \mathbf{B} \in \text{ob}(\mathcal{C})^{\mathcal{I}}$. Then $\pi_{j'}^\infty(a, b) = (\pi_{j'}^{\infty, A} a, \pi_{j'}^{\infty, B} b)$ for the projections $\pi_{j'}^{\infty, X}: \prod_{k \in \mathcal{I}} X_k \rightarrow \prod_{k \in \mathcal{I}} X_k$. Therefore:

$$\begin{aligned} \text{lim}(\mathbf{S} \otimes \mathbf{T}) &= \bigcap_{j \in \mathcal{J}} ((\pi_{j'}^{\infty, A})^{-1}(S_j) \times (\pi_{j'}^{\infty, B})^{-1}(T_j)) = \bigcap_{j \in \mathcal{J}} ((\pi_{j'}^{\infty, A})^{-1}(S_j)) \times \bigcap_{j \in \mathcal{J}} ((\pi_{j'}^{\infty, B})^{-1}(T_j)) \\ &\cong \text{lim}(\mathbf{S}) \times \text{lim}(\mathbf{T}), \end{aligned}$$

from which it follows that Lim preserves monoidal products.

vi. Preservation of composition: laxness. Take composable nets $\mathbf{S}$ and $\mathbf{T}$. We start by proving that Lim laxly preserves composition, meaning that $\text{Lim}(\mathbf{T}) \circ \text{Lim}(\mathbf{S}) \subseteq \text{Lim}(\mathbf{T} \circ \mathbf{S})$.

Suppose that $(x, z) \in \text{Lim}(\mathbf{T}) \circ \text{Lim}(\mathbf{S})$ then there exists some y such that $(x, y) \in \text{Lim}(\mathbf{S})$ and $(y, z) \in \text{Lim}(\mathbf{T})$. Note that for any $j_0 \in \mathcal{J}$, $\pi_{j_0}^\infty(\mathbf{C}) \subseteq C_{j_0}$ because

$$\pi_{j_0}^\infty(\text{lim}(\mathbf{C})) = \pi_{j_0}^\infty\left(\bigcap_{j \in \mathbb{Z}_{\leq}^\infty} (\pi_j^\infty)^{-1}(C_j)\right) \subseteq \pi_{j_0}^\infty((\pi_{j_0}^\infty)^{-1}(C_{j_0})) = C_{j_0}.$$

Therefore, for any j , $\pi_j^\infty(x, y) = (\pi_j^\infty x, \pi_j^\infty y) \in S_j$ and $(\pi_j^\infty y, \pi_j^\infty z) \in T_j$ which implies that $(\pi_j^\infty x, \pi_j^\infty z) \in T_j \circ S_j = (T \circ S)_j$. Therefore, $(x, z) \in \text{Lim}(\mathbf{T} \circ \mathbf{S})$.

vii. Preservation of composition: oplaxness. Now we prove the harder direction, that Lim oplaxly preserves composition, meaning that $\text{Lim}(\mathbf{T} \circ \mathbf{S}) \subseteq \text{Lim}(\mathbf{T}) \circ \text{Lim}(\mathbf{S})$.

Suppose $(x, z) \in \text{Lim}(\mathbf{T} \circ \mathbf{S})$ then for all j , $(\pi_j^\infty x, \pi_j^\infty z) \in (T \circ S)_j = T_j \circ S_j$ there exists some y_j such that $(\pi_j^\infty x, y_j) \in S_j$ and $(y_j, \pi_j^\infty z) \in T_j$. Therefore the following sets are non-empty:

$$W_j = \left\{ y_j \in \prod_{i \in \mathcal{I}} Y_i \mid (\pi_j^\infty x, y_j) \in S_j, (y_j, \pi_j^\infty z) \in T_j \right\}.$$

They are moreover closed, since W_j is the intersection of the image of $\{\pi_j^\infty x\}$ under S_j and the preimage of $\{\pi_j^\infty z\}$ under T_j and both are closed relations.

Now take any $j' \supseteq j$. It follows that

$$\begin{aligned} \pi_{j'}^{j'}(W_{j'}) &= \left\{ y_j \in \prod_{i \in \mathcal{I}} Y_i \mid \exists y_{j'} \in \prod_{i \in \mathcal{I}'} Y_i : \begin{array}{l} \pi_{j'}^{j'}(y_{j'}) = y_j, \\ (\pi_{j'}^\infty x, y_{j'}) \in S_{j'}, \\ (y_{j'}, \pi_{j'}^\infty z) \in T_{j'} \end{array} \right\} \\ &\subseteq \left\{ y_j \in \prod_{i \in \mathcal{I}} Y_i \mid \begin{array}{l} (\pi_j^\infty x, y_j) \in S_j, \\ (y_j, \pi_j^\infty z) \in T_j \end{array} \right\} = W_j. \end{aligned}$$

Therefore $(\pi_{j'}^{j'})^{-1}(W_{j'}) \supseteq W_j$.

Now for each $j \in \mathcal{J}$, define:

$$C_j := (\pi_j^\infty)^{-1}(W_j) \subseteq \prod_{i \in \mathbb{Z}} Y_i.$$

Whenever $j \subseteq j'$, $\pi_j^\infty = \pi_{j'}^{j'} \circ \pi_{j'}^\infty$, we have that $(\pi_j^\infty)^{-1} = (\pi_{j'}^\infty)^{-1} \circ (\pi_{j'}^{j'})^{-1}$, therefore

$$C_j = (\pi_j^\infty)^{-1}(W_j) = (\pi_{j'}^\infty)^{-1} \circ (\pi_{j'}^{j'})^{-1}(W_j) \supseteq (\pi_{j'}^\infty)^{-1}(W_{j'}) = C_{j'}.$$

Since each W_j is closed and the projections π_j^∞ are continuous, C_j is also closed. Moreover, since each W_j is non-empty so is C_j .

We show that, since $\mathcal{J}$ is *directed*, the C_j satisfy the finite intersection property. Fix any n and take arbitrary contexts $\{j_1, \dots, j_n\}$. By directedness there exists some $j^* \in \mathcal{J}$ such that $j^* \supseteq j_i$ for all i . Therefore $C_{j^*} \subseteq \bigcap_{i=1}^n C_{j_i}$ and since C_{j^*} is non-empty, the finite intersection property holds. The intersection of closed subsets of a compact space satisfying the finite intersection property is nonempty, so that:

$$\bigcap_{j \in \mathcal{J}} (\pi_j^\infty)^{-1}(W_j) = \bigcap_{j \in \mathcal{J}} C_j \neq \emptyset.$$

Finally, take $y \in \bigcap_{j \in \mathcal{J}} C_j$, then for every j , $\pi_j^\infty(x, y) \in S_j$ and $\pi_j^\infty(y, z) \in T_j$. Therefore $(x, y) \in \text{Lim}(\mathbf{S})$ and $(y, z) \in \text{Lim}(\mathbf{T})$ and it follows that $\text{Lim}(\mathbf{T} \circ \mathbf{S}) \subseteq \text{Lim}(\mathbf{T}) \circ \text{Lim}(\mathbf{S})$. $\square$

Definition F.1. A poset $(P, \leq)$ is **Artinian** in case for every sequence $(x_k)_{k \in \mathbb{N}}$ in P with $x_{k+1} \leq x_k$ for all $k \in \mathbb{N}$, there exists $d \in \mathbb{N}$ such that $x_k = x_d$ for all $k \geq d$.

Importantly, the example that we care about, $\text{Rel}(\mathbf{FSet})$, is enriched in Artinian posets since every strictly descending chain of subsets of a finite set terminates.

Lemma F.2. Let $\mathcal{C}$ be an Artinian poset-enriched compact-closed discard category satisfying $1_X \subseteq \perp_X \circ \top_X$ and $\top_X \circ \perp_X \subseteq 1_I$. Let $\mathcal{J} \subseteq \mathcal{P}_f(\mathcal{I})$ be upward-directed. Then every monotone net $\mathbf{S}$ of morphisms in $\mathcal{C}$ is observationally equivalent to a unique causal net $\mathbf{fix}(\mathbf{S})$.

Proof. Fix $j \in \mathcal{J}$. For each $j' \in \mathcal{J}$ with $j \subseteq j'$, set

$$B_j^{j'} := (1_{Y_j} \otimes \top_{Y_{j' \setminus j}}) \circ S_{j'} \circ (1_{X_j} \otimes \perp_{X_{j' \setminus j}}) : X_j \rightarrow Y_j.$$

By monotonicity of $\mathbf{S}$, enlarging j' can only refine this morphism, so the family $\{B_j^{j'} \mid j' \in \mathcal{J}, j \subseteq j'\}$ stabilises at some $\rho(\mathbf{S}, j) \in \mathcal{J}$ with $j \subseteq \rho(\mathbf{S}, j)$. Define $\mathbf{fix}(\mathbf{S}) := \{B_j^{\rho(\mathbf{S}, j)}\}_{j \in \mathcal{J}}$.

We need to show that $\mathbf{fix}(\mathbf{S})$ is:

- i. causal;
- ii. observationally equivalent to $\mathbf{S}$;
- iii. the unique such causal net.

i. Causality. Let $j \subseteq j'$. Then $(1_{Y_j} \otimes \top_{Y_{j' \setminus j}}) \circ \mathbf{fix}(\mathbf{S})_{j'} = (1_{Y_j} \otimes \top_{Y_{j' \setminus j}}) \circ B_{j'}^{\rho(\mathbf{S}, j')} = B_j^{\rho(\mathbf{S}, j')}$. Since $\rho(\mathbf{S}, j')$ contains j' , it also contains j , and by stabilisation of the family for j we have $B_j^{\rho(\mathbf{S}, j')} = B_j^{\rho(\mathbf{S}, j)} = \mathbf{fix}(\mathbf{S})_j$. Thus $\mathbf{fix}(\mathbf{S})$ is causal.

ii. Observational equivalence. For each j , monotonicity at $j \subseteq \rho(\mathbf{S}, j)$ yields

$$(1_{Y_j} \otimes \top_{Y_{\rho(\mathbf{S}, j) \setminus j}}) \circ S_{\rho(\mathbf{S}, j)} \subseteq S_j \otimes \top_{X_{\rho(\mathbf{S}, j) \setminus j}}.$$

Precomposing $1_{X_j} \otimes \perp_{X_{\rho(\mathbf{S}, j) \setminus j}}$ and using $\top \circ \perp \subseteq 1_I$ yields $\mathbf{fix}(\mathbf{S})_j \subseteq S_j$, and thus $\mathbf{fix}(\mathbf{S}) \subseteq \mathbf{S}$.

For the converse, using $1 \subseteq \perp \circ \top$ on the extra input factors gives

$$(1_{Y_j} \otimes \top_{Y_{\rho(\mathbf{S}, j) \setminus j}}) \circ S_{\rho(\mathbf{S}, j)} \subseteq \mathbf{fix}(\mathbf{S})_j \otimes \top_{X_{\rho(\mathbf{S}, j) \setminus j}},$$

so that $\mathbf{S} \subseteq \mathbf{fix}(\mathbf{S})$. Therefore $\mathbf{S}$ and $\mathbf{fix}(\mathbf{S})$ are observationally equivalent.

iii. Uniqueness. Let $\mathbf{T}$ be a causal net observationally equivalent to $\mathbf{S}$. Then $\mathbf{T}$ is observationally equivalent to $\mathbf{fix}(\mathbf{S})$. Since $\mathbf{T}$ and $\mathbf{fix}(\mathbf{S})$ are causal, approximation implies contextwise comparison, hence $\mathbf{T} = \mathbf{fix}(\mathbf{S})$ contextwise. $\square$

Lemma F.3. Given a set $\mathcal{I}$ and cofinal $\mathcal{J} \subseteq \mathcal{P}_f(\mathcal{I})$, and any causal monotone net $\mathbf{S} = \{S_j\}_{j \in \mathcal{J}}$ in $\text{Rel}(\mathbf{FSet})$, then for all $j \in \mathcal{J}$, we have that $\pi_j^\infty(\mathbf{lim}(\mathbf{S})) = S_j$.

Proof. The inclusion $\pi_j^\infty(\mathbf{lim}(\mathbf{S})) \subseteq S_j$ is immediate since $\mathbf{lim}(\mathbf{S}) \subseteq (\pi_j^\infty)^{-1}(S_j)$.

For the reverse inclusion, fix $j_0 \in \mathcal{J}$. If $S_{j_0} = \emptyset$ there is nothing to show. Otherwise, choose $x_{j_0} \in S_{j_0}$. For each $k \in \mathcal{J}$ with $k \supseteq j_0$, set

$$C_k := (\pi_k^\infty)^{-1}(S_k) \cap (\pi_{j_0}^\infty)^{-1}(\{x_{j_0}\}) \subseteq \prod_{i \in \mathcal{I}} X_i.$$

Since $\mathbf{S}$ is causal, for every $k \supseteq j_0$ we have $\pi_{j_0}^k(S_k) = S_{j_0}$, so each C_k is nonempty.

Fix $n \in \mathbb{N}$ and a family $(k_m)_{1 \leq m \leq n}$ in $\mathcal{J}$ with $k_m \supseteq j_0$ for all m . Choose $k^* \in \mathcal{J}$ with $k^* \supseteq \bigcup_{1 \leq m \leq n} k_m$. Then $C_{k^*} \subseteq \bigcap_{1 \leq m \leq n} C_{k_m}$, so $\{C_k\}_{k \supseteq j_0}$ has the finite intersection property. Since $\prod_{i \in \mathcal{I}} X_i$ is compact Hausdorff, $\bigcap_{k \supseteq j_0} C_k \neq \emptyset$. Choose y in this intersection. Then $y \in \mathbf{lim}(\mathbf{S})$ and $\pi_{j_0}^\infty(y) = x_{j_0}$, so $x_{j_0} \in \pi_{j_0}^\infty(\mathbf{lim}(\mathbf{S}))$. $\square$

From these two lemmas it follows immediately that:

G Proofs of Section 6

Theorem 6.2. *There is an oplax normal feedback functor $\mathcal{Z}^*: \mathbf{AR}_{\mathbb{K}(\mathbb{Z})}^{\text{fd}} \rightarrow \mathbf{Obs}_{\mathbb{Z}}(\mathbf{AR}_{\mathbb{K}}^{\text{fd}})$ given on objects by $n \mapsto n^{\mathbb{Z}}$ and on morphisms $S: n \rightarrow m$ by*

$$\mathcal{Z}^*(S) := \left[\left\{ \left\{ (\zeta_j^n(\mathbf{x}), \zeta_j^m(\mathbf{y})) \mid (\mathbf{x}, \mathbf{y}) \in S \right\} \right\}_{j \in \mathbb{Z}_{\leq}^2} \right];$$

where oplax means that $\mathcal{Z}^*(T \circ S) \subseteq \mathcal{Z}^*(T) \circ \mathcal{Z}^*(S)$ and normal means that $\mathcal{Z}^*(1_n) = 1_{n^{\mathbb{Z}}}$.

Proof of Theorem 6.2. We need to show that $\mathcal{Z}^*$:

- i. produces a monotone net of affine relations;
- ii. is oplax;
- iii. is normal and preserves the relevant structure.

i. Well-definedness. We show that $\mathcal{Z}^*$ produces a monotone net of affine relations. First note that for all $j \in \mathbb{Z}_{\leq}^2$, $(\mathcal{Z}^*(S))_j$ is an $\mathbb{K}$ -affine subspace; moreover for all $n, m \in \mathbb{N}$, ζ_j^n and ζ_j^m are $\mathbb{K}$ -linear maps. Furthermore, each $(\mathcal{Z}^*(S))_j$ is defined via the projection of the same infinite affine subspace, so $\mathcal{Z}^*(S)$ produces a causal, and thus monotone, net of affine relations.

ii. Oplaxness. Now we prove that $\mathcal{Z}^*$ is an oplax functor. Fix $j \in \mathbb{Z}_{\leq}^2$. If $T \circ S = \emptyset$ then $(\mathcal{Z}^*(T \circ S))_j = \emptyset$ by definition, so the inclusion is vacuous.

Otherwise, take some $(u, w) \in (\mathcal{Z}^*(T \circ S))_j$. By definition of $(\mathcal{Z}^*(T \circ S))_j$, there exists some $(\mathbf{x}, \mathbf{z}) \in T \circ S$ such that $u = \zeta_j^n(\mathbf{x})$ and $w = \zeta_j^p(\mathbf{z})$. Since $(\mathbf{x}, \mathbf{z}) \in T \circ S$, by definition of relational composition in $\mathbf{AR}_{\mathbb{K}(\mathbb{Z})}^{\text{fd}}$ there exists some $\mathbf{y} \in \mathbb{K}(\mathbb{Z})^m$ such that $(\mathbf{x}, \mathbf{y}) \in S$ and $(\mathbf{y}, \mathbf{z}) \in T$. Since $(\mathbf{x}, \mathbf{y}) \in S$, by definition of $(\mathcal{Z}^*(S))_j$ we have

$$(u, \zeta_j^m(\mathbf{y})) = (\zeta_j^n(\mathbf{x}), \zeta_j^m(\mathbf{y})) \in (\mathcal{Z}^*(S))_j.$$

Similarly

$$(\zeta_j^m(\mathbf{y}), w) = (\zeta_j^m(\mathbf{y}), \zeta_j^p(\mathbf{z})) \in (\mathcal{Z}^*(T))_j.$$

Thus, by definition of relational composition in $\mathbf{AR}_{\mathbb{K}}^{\text{fd}}$,

$$(u, w) \in (\mathcal{Z}^*(T))_j \circ (\mathcal{Z}^*(S))_j.$$

Therefore, $\mathcal{Z}^*$ is oplax.

iii. Symmetric monoidal and feedback structure. $\mathcal{Z}^*$ preserves the identity, as well as all of the other symmetric monoidal and feedback structure, since they are defined contextwise. $\square$

Proposition 6.3. *Given a finite field $\mathbb{F}_q$, and a cofinal set $\mathcal{J} \subseteq \mathcal{P}_f(\mathcal{I})$ there is a faithful poset-enriched discard functor $\text{Lim}: \mathbf{Obs}_{\mathcal{I}, \mathcal{J}}(\mathbf{AR}_{\mathbb{F}_q}^{\text{fd}}) \rightarrow \mathbf{AR}_{\mathbb{F}_q}$.*

Proof of Proposition 6.3. By Theorem 5.5 there is a poset-enriched discard functor $\text{Lim}: \mathbf{Obs}_{\mathcal{I}, \mathcal{J}}(\mathbf{Rel}(\mathbf{KHaus})) \rightarrow \mathbf{Rel}(\mathbf{KHaus})$. Since $\mathbb{F}_q$ is a finite field, any vector space over $\mathbb{F}_q$ is compact and Hausdorff with respect to the product topology of discrete topologies. Moreover since the projections are linear maps, inverse images of linear maps are linear maps, and intersections of linear subspaces are linear subspaces. Therefore, it follows that there is an induced poset-enriched discard functor $\mathbf{Obs}_{\mathcal{I}, \mathcal{J}}(\mathbf{AR}_{\mathbb{F}_q}^{\text{fd}}) \rightarrow \mathbf{AR}_{\mathbb{F}_q}$. Moreover, because the objects of $\mathbf{AR}_{\mathbb{F}_q}^{\text{fd}}$ are compact, by Corollary 5.6, it follows that this functor is moreover faithful. $\square$

Theorem 6.5. Let $\mathbb{F}_q$ be a finite field and $S \subseteq \mathbb{F}_q(z)^\ell$ be a finite-dimensional $\mathbb{F}_q(z)$ -affine subspace. Then $(\text{Lim} \circ \mathcal{Z}^*)(S) = \text{beh}(S) \cap \mathcal{Z}_\ell^\gamma(\mathbb{F}_q(z)^\ell)$.

Proof of Theorem 6.5. We prove the claim for affine subspaces $S \subseteq \mathbb{F}_q(z)^\ell$. The analogous statement for affine relations $R \subseteq \mathbb{F}_q(z)^n \times \mathbb{F}_q(z)^m$ follows immediately by viewing R as an affine subspace of $\mathbb{F}_q(z)^{n+m}$ and transporting along the canonical shuffling isomorphism

$$\zeta_{n,m}: (\mathbb{F}_q^n)^\mathbb{Z} \times (\mathbb{F}_q^m)^\mathbb{Z} \cong (\mathbb{F}_q^{n+m})^\mathbb{Z}.$$

First, take $S = \emptyset$, then $(\text{Lim} \circ \mathcal{Z}^*)(S) = \emptyset = \text{beh}(S)$, so the claim follows immediately.

Assume otherwise, taking some $\mathbf{a} \in S$. Define $R := S - \mathbf{a}$. Therefore, $R \subseteq \mathbb{F}_q(z)^\ell$ is an $\mathbb{F}_q(z)$ -linear subspace and $S = R + \mathbf{a}$. By definition of beh on affine subspaces,

$$\text{beh}(S) = \text{beh}(R) + \mathcal{Z}_\ell^\gamma(\mathbf{a}).$$

Since $\text{Lim} \circ \mathcal{Z}^*$ is given on morphisms by $\mathcal{Z}^\gamma$, and $\mathcal{Z}^\gamma$ is $\mathbb{F}_q[z, 1/z]$ -linear, we also have

$$(\text{Lim} \circ \mathcal{Z}^*)(S) = (\text{Lim} \circ \mathcal{Z}^*)(R) + \mathcal{Z}_\ell^\gamma(\mathbf{a}),$$

and $\mathcal{Z}_\ell^\gamma(\mathbf{a}) \in \mathcal{Z}_\ell^\gamma(\mathbb{F}_q(z)^\ell)$. Thus it suffices to prove the following for the linear subspace R :

- i. for all $u \in \mathbb{F}_q((z))^\ell$, $\mathcal{Z}_r^\gamma(B(z)u) = B(\Delta) \mathcal{Z}_\ell^\gamma(u)$;
- ii. $(\text{Lim} \circ \mathcal{Z}^*)(R) \subseteq \text{beh}(R)$;
- iii. $\text{beh}(R) \cap \mathcal{Z}_\ell^\gamma(\mathbb{F}_q(z)^\ell) \subseteq (\text{Lim} \circ \mathcal{Z}^*)(R)$.

Choose some $r \in \mathbb{N}$ and an $\mathbb{F}_q(z)$ -linear map $A: \mathbb{F}_q(z)^\ell \rightarrow \mathbb{F}_q(z)^r$ with $R = \ker A$. Pick $d(z) \in \mathbb{F}_q[z, 1/z] \setminus \{0\}$ clearing denominators and set $B(z) := d(z)A \in \mathbb{F}_q[z, 1/z]^{r \times \ell}$. Then $\text{beh}(R) = \ker(B(\Delta)) \subseteq (\mathbb{F}_q^\mathbb{Z})^\ell$.

i. Linearity. Since $\mathcal{Z}^\gamma$ is $\mathbb{F}_q[z, 1/z]$ -linear, for each entry $b(z) \in \mathbb{F}_q[z, 1/z]$ and $x \in \mathbb{F}_q((z))$ we have $\mathcal{Z}_1^\gamma(b(z)x) = b(\Delta) \mathcal{Z}_1^\gamma(x)$, and applying this entrywise to $B(z)$ and componentwise to vectors yields $\mathcal{Z}_r^\gamma(B(z)u) = B(\Delta) \mathcal{Z}_\ell^\gamma(u)$.

ii. Forward inclusion. Take $u \in (\text{Lim} \circ \mathcal{Z}^*)(R)$. Then there exists some $x \in R$ such that $u = \mathcal{Z}_\ell^\gamma(x)$. Since $x \in R = \ker(B(z))$, we have that $B(z)x = 0$; thus,

$$0 = \mathcal{Z}_r^\gamma(B(z)x) = B(\Delta) \mathcal{Z}_\ell^\gamma(x) = B(\Delta)u,$$

therefore $u \in \ker(B(\Delta)) = \text{beh}(R)$.

iii. Backward inclusion. Take $u \in \text{beh}(R) \cap \mathcal{Z}_\ell^\gamma(\mathbb{F}_q(z)^\ell)$. Choose $x \in \mathbb{F}_q(z)^\ell$ with $u = \mathcal{Z}_\ell^\gamma(x)$. Since $u \in \text{beh}(R) = \ker(B(\Delta))$, we have that $B(\Delta)u = 0$, hence $0 = B(\Delta) \mathcal{Z}_\ell^\gamma(x) = \mathcal{Z}_r^\gamma(B(z)x)$. By injectivity of $\mathcal{Z}_r^\gamma$, $B(z)x = 0$, so $x \in \ker(B(z)) = R$. Thus $u = \mathcal{Z}_\ell^\gamma(x) \in (\text{Lim} \circ \mathcal{Z}^*)(R)$.

Therefore $(\text{Lim} \circ \mathcal{Z}^*)(R) = \text{beh}(R) \cap \mathcal{Z}_\ell^\gamma(\mathbb{F}_q(z)^\ell)$. $\square$

Lemma 6.7. There is a full feedback functor $\text{St}_{\text{LR}_{\mathbb{K}}^{\text{fd}}}(\text{AR}_{\mathbb{K}}^{\text{fd}}) \rightarrow \text{AR}_{\mathbb{K}(z)}^{\text{fd}}$, given by identifying the delays and extending scalars along $\mathbb{K}(z)/\mathbb{K}$, and extending scalars along the geometric series embedding $\mathbb{K}((z))/\mathbb{K}(z)$ induces a feedback functor $\text{Ext}: \text{St}_{\text{LR}_{\mathbb{K}}^{\text{fd}}}(\text{AR}_{\mathbb{K}}^{\text{fd}}) \rightarrow \text{AR}_{\mathbb{K}(z)}^{\text{fd}}$.

Proof of Lemma 6.7. By Proposition 3.5, $\text{St}_{\text{LR}_{\mathbb{K}}^{\text{fd}}}(\text{AR}_{\mathbb{K}}^{\text{fd}})$ is presented by freely adding a “discrete-time delay” automorphism to $\text{AR}_{\mathbb{K}}^{\text{fd}}$ which is natural on the subcategory $\text{LR}_{\mathbb{K}}^{\text{fd}}$. $\square$

Lemma 6.8. *Let $\eta: \text{AR}_{\mathbb{K}}^{\text{fd}} \rightarrow \text{St}_{\text{LR}_{\mathbb{K}}^{\text{fd}}}(\text{AR}_{\mathbb{K}}^{\text{fd}})$ be the canonical functor. Define $J: \text{AR}_{\mathbb{K}}^{\text{fd}} \rightarrow \text{Obs}_{\mathbb{Z}}(\text{AR}_{\mathbb{K}}^{\text{fd}})$ by $J(V)_t := V$ for all $t \in \mathbb{Z}$, and on morphisms by $J(\emptyset) := [\{\emptyset\}_{j \in \mathbb{Z}_{\leq}^2}]$ and*

$$J(A + \mathbf{a}) := \left[\left\{ \bigoplus_{t=\ell}^r A + \mathbf{a} \quad \text{if } t = 0, \right. \right. \\ \left. \left. A \quad \text{if } t \neq 0 \right\}_{[\ell, r] \in \mathbb{Z}_{\leq}^2} \right].$$

There is a unique feedback functor $\text{Unroll}: \text{St}_{\text{LR}_{\mathbb{K}}^{\text{fd}}}(\text{AR}_{\mathbb{K}}^{\text{fd}}) \rightarrow \text{Obs}_{\mathbb{Z}}(\text{AR}_{\mathbb{K}}^{\text{fd}})$ satisfying $\text{Unroll} \circ \eta = J$.

Proof of Lemma 6.8. We proceed by using the universal property of the state construction (see Proposition D.2).

By inspection, J produces monotone nets and preserves identities. The only nontrivial part is to show that composition is preserved. Take $f \in \text{AR}_{\mathbb{K}}^{\text{fd}}(\mathbb{K}^n, \mathbb{K}^m)$ and $g \in \text{AR}_{\mathbb{K}}^{\text{fd}}(\mathbb{K}^m, \mathbb{K}^p)$. If f or g is empty then $J(g \circ f) = [\{\emptyset\}_{j \in \mathbb{Z}_{\leq}^2}] = J(g) \circ J(f)$. Similarly if $g \circ f$ is empty, then $J(g \circ f)$ is empty at every context $j \in \mathbb{Z}_{\leq}^2$, hence by Proposition 4.11, $J(g \circ f) = [\{\emptyset\}_{j \in \mathbb{Z}_{\leq}^2}] = J(g) \circ J(f)$. Otherwise, suppose that we have nonempty affine relations $A + \mathbf{a} \in \text{AR}_{\mathbb{K}}^{\text{fd}}(\mathbb{K}^n, \mathbb{K}^m)$ and $B + \mathbf{b} \in \text{AR}_{\mathbb{K}}^{\text{fd}}(\mathbb{K}^m, \mathbb{K}^p)$, with nonempty composite $C + \mathbf{c} \in \text{AR}_{\mathbb{K}}^{\text{fd}}(\mathbb{K}^n, \mathbb{K}^p)$. Then:

$$\begin{aligned} & J(C + \mathbf{c}) \\ &= \left[\left\{ \bigoplus_{k=\ell}^r \begin{pmatrix} C + \mathbf{c} & \text{if } k = 0 \\ C & \text{if } k \neq 0 \end{pmatrix} \right\}_{[\ell, r] \in \mathbb{Z}_{\leq}^2} \right] \\ &= \left[\left\{ \bigoplus_{k=\ell}^r \begin{pmatrix} C + \mathbf{c} & \text{if } k = 0 \\ B \circ A & \text{if } k \neq 0 \end{pmatrix} \right\}_{[\ell, r] \in \mathbb{Z}_{\leq}^2} \right] \\ &= \left[\left\{ \bigoplus_{k=\ell}^r \begin{pmatrix} B + \mathbf{b} & \text{if } k = 0 \\ B & \text{if } k \neq 0 \end{pmatrix} \circ \begin{pmatrix} A + \mathbf{a} & \text{if } k = 0 \\ A & \text{if } k \neq 0 \end{pmatrix} \right\}_{[\ell, r] \in \mathbb{Z}_{\leq}^2} \right] \\ &= \left[\left\{ \bigoplus_{k=\ell}^r \begin{pmatrix} B + \mathbf{b} & \text{if } k = 0 \\ B & \text{if } k \neq 0 \end{pmatrix} \right\}_{[\ell, r] \in \mathbb{Z}_{\leq}^2} \right] \\ &\quad \circ \left[\left\{ \bigoplus_{k=\ell}^r \begin{pmatrix} A + \mathbf{a} & \text{if } k = 0 \\ A & \text{if } k \neq 0 \end{pmatrix} \right\}_{[\ell, r] \in \mathbb{Z}_{\leq}^2} \right] \\ &= J(B + \mathbf{b}) \circ J(A + \mathbf{a}). \end{aligned}$$

Let $J|_{\text{LR}_{\mathbb{K}}^{\text{fd}}}$ denote the restriction of J along the inclusion $\text{LR}_{\mathbb{K}}^{\text{fd}} \hookrightarrow \text{AR}_{\mathbb{K}}^{\text{fd}}$. For shift-invariance, take $L \in \text{LR}_{\mathbb{K}}^{\text{fd}}(V, W)$ and $[\ell, r] \in \mathbb{Z}_{\leq}^2$; then

$$(\text{Shift}(J|_{\text{LR}_{\mathbb{K}}^{\text{fd}}}(L)))_{[\ell, r]} = (J(L))_{[\ell-1, r-1]} = \bigoplus_{t=\ell-1}^{r-1} L = \bigoplus_{t=\ell}^r L = (J(L))_{[\ell, r]}.$$

Now apply Proposition D.2 with

$$\begin{aligned} \Gamma: \text{LR}_{\mathbb{K}}^{\text{fd}} &\hookrightarrow \text{AR}_{\mathbb{K}}^{\text{fd}}, & \Delta &= 1_{\text{LR}_{\mathbb{K}}^{\text{fd}}}, & \Gamma' &= 1_{\text{Obs}_{\mathbb{Z}}(\text{AR}_{\mathbb{K}}^{\text{fd}})}, \\ \Delta' &= \text{Shift}, & F &= J, & F|_{\text{LR}_{\mathbb{K}}^{\text{fd}}} &= J|_{\text{LR}_{\mathbb{K}}^{\text{fd}}}. \end{aligned}$$

The condition $\Gamma' \circ F|_{\text{LR}_{\mathbb{K}}^{\text{fd}}} = F \circ \Gamma$ is the statement that $J|_{\text{LR}_{\mathbb{K}}^{\text{fd}}}$ is the restriction of J , and the condition $\Delta' \circ F|_{\text{LR}_{\mathbb{K}}^{\text{fd}}} = F|_{\text{LR}_{\mathbb{K}}^{\text{fd}}} \circ \Delta$ is exactly shift-invariance of $J|_{\text{LR}_{\mathbb{K}}^{\text{fd}}}$. Hence there exists a unique feedback functor

$$\text{Unroll}: \text{St}_{\text{LR}_{\mathbb{K}}^{\text{fd}}}(\text{AR}_{\mathbb{K}}^{\text{fd}}) \rightarrow \text{Obs}_{\mathbb{Z}}(\text{AR}_{\mathbb{K}}^{\text{fd}})$$

such that $\text{Unroll} \circ \eta = J$. By construction, Unroll agrees with J as defined. $\square$

Proposition 6.9. *Given any polynomial $z + a \in \mathbb{K}[z]$ for $a \neq 0$:*

$$\text{Unroll}\left(\begin{array}{c} \text{---} a_0 \text{---} \\ \text{---} a_1 \text{---} \vdots \partial \\ \text{---} a_n \text{---} \partial^n \end{array}\right) = \text{Unroll}\left(\begin{array}{c} \text{---} \text{---} \end{array}\right) \not\subseteq \text{Unroll}\left(\begin{array}{c} \text{---} a_0 \text{---} \\ \text{---} a_1 \text{---} \vdots \partial \\ \text{---} a_n \text{---} \partial^n \end{array}\right)$$

Proof of Proposition 6.9. Given a polynomial $a(z) \in \mathbb{K}[z]$, define the following morphism in $\mathrm{St}_{\mathrm{LR}_{\mathbb{K}}}^{\mathrm{fd}}(\mathrm{AR}_{\mathbb{K}}^{\mathrm{fd}})$:

$$\mu_a(z) := \text{Diagram with nodes } a_0, a_1, \dots, a_n \text{ and } \partial, \partial^n$$

Take some nonzero $a \in \mathbb{K}$. We need to show that $\mathcal{Z}^*(\text{AR}_{\mathbb{K}\langle\langle z \rangle\rangle}^{\text{fd}})$:

- i. satisfies Equation (7);
- ii. only satisfies Equation (6) oploxly.

i. **Left inverse.** First, observe that

Diagram illustrating the unrolling of a quantum circuit. The left side shows a circuit with multiple layers of gates labeled 'a' and a vertical ellipsis indicating repetition. The right side shows the unrolled version of the same circuit, where the gates are explicitly shown in sequence.

Therefore

$$(\text{Unroll}(\mu_{z+a}) \circ \text{Unroll}(\mu_{z+a}^*)_{[\ell,r]})$$

From which it follows that

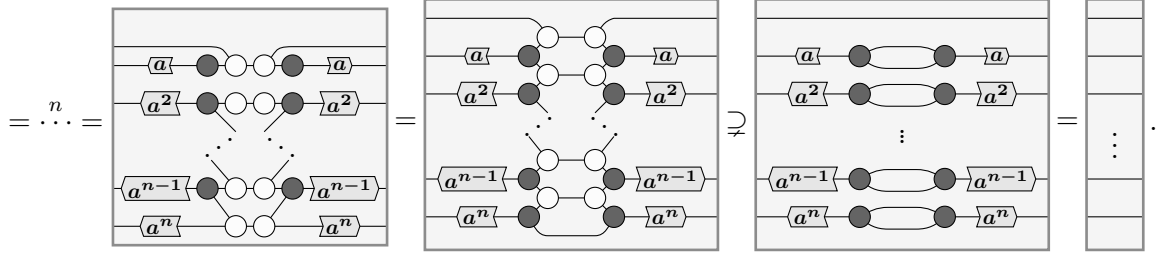
$$(\text{Unroll}(\mu_{z+a}) \circ \text{Unroll}(\mu_{z+a})^*)_{[\ell, r]} \circ (\top_1 \oplus 1_{r-\ell}) = \top_1 \oplus 1_{r-\ell},$$

so that

$$\text{Unroll}(\mu_{z+a}) \circ \text{Unroll}(\mu_{z+a})^* = 1_{1\mathbb{Z}}.$$

ii. **Right inverse.** On the other hand, fixing $n := r - \ell$,

$$(\text{Unroll}(\mu_{z+a})^* \circ \text{Unroll}(\mu_{z+a}))_{[l,r]}$$



Notice that the final inclusion is strict, which can be checked semantically as the bottom diagram has half the dimension of the one on top.

Therefore $\text{Unroll}(\mu_{z+a})^* \circ \text{Unroll}(\mu_{z+a}) \supsetneq 1_{1^{\mathbb{Z}}}$. $\square$

Proposition 6.10. *For any map f in $\text{St}_{\text{LR}_{\mathbb{K}}^{\text{fd}}}(\text{AR}_{\mathbb{K}}^{\text{fd}})$, we have that $\mathcal{Z}^*(\text{Ext}(f)) \subseteq \text{Unroll}(f)$.*

Proof of Proposition 6.10. A simple diagram chase shows that the diagram commutes strictly on affine displacements, linear relations, and the delay. Therefore, by oplaxness and normality, the desired inclusion holds. $\square$